Broadcast Technology UG1

Signals

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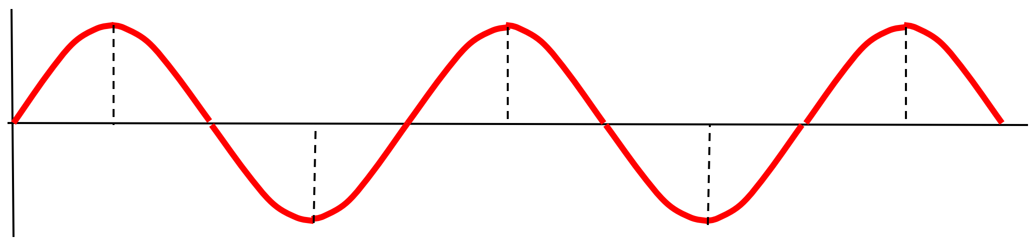
Why <u>Sine</u> Waves ?
The Signal
The Channel
From Analogue to Digital
Bandwidth
Pulse Transmission, Interference (ISI)
Noise
Signal Distortion
IRE Levels

Why Sine Waves ???

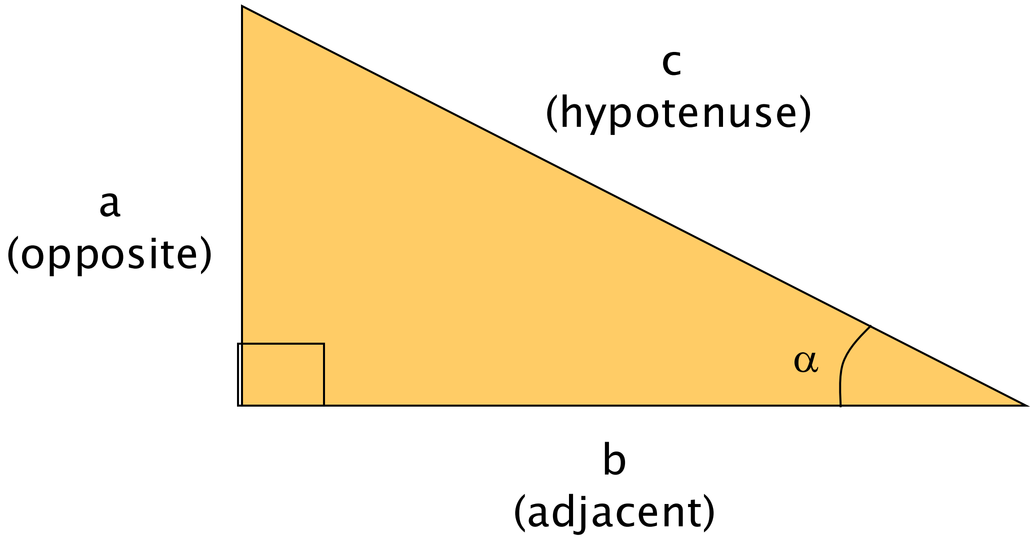


Sine waves are naturally produced by rotating machinery, like generators in power plants.

Smooth Transmission: Sine waves allow for efficient transmission over long distances. Their smooth, periodic nature minimizes energy loss when stepped up or down using transformers, which are essential for reducing power loss in transmission lines.

Why sine (sin) Waves?

sine (sin):



$$\sin \alpha = \frac{a}{c}$$

i.e. ***a ratio*** of the opposite side and hypotenuse of a right-angled triangle

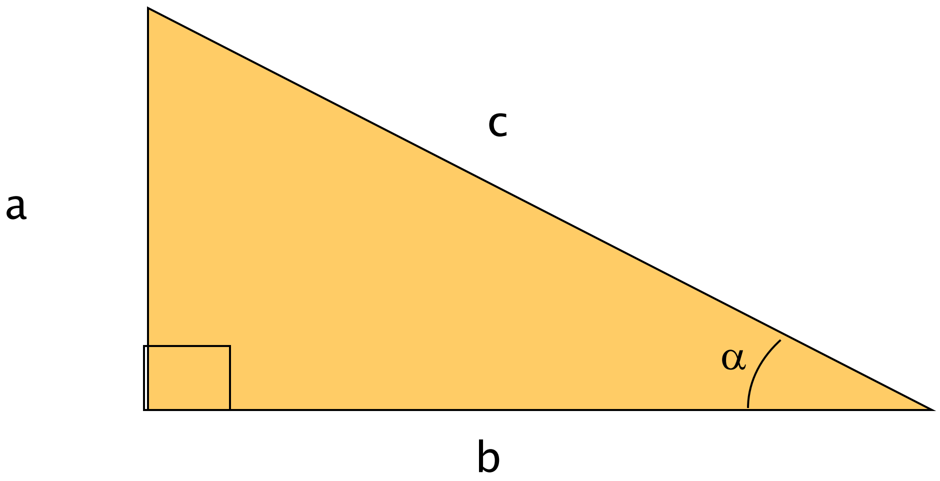
for $\alpha = 0° \Rightarrow 90°$

$\sin \alpha = 0 \Rightarrow 1.00$

notable value:

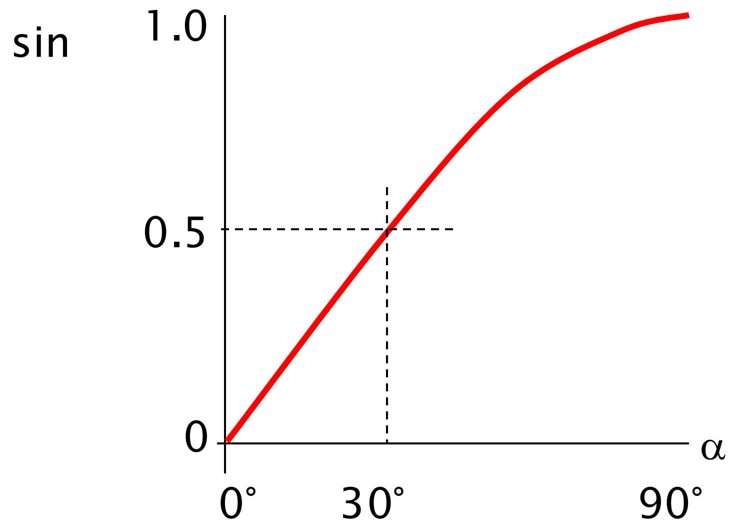
$\sin 30° = 0.500$

sine (sin)

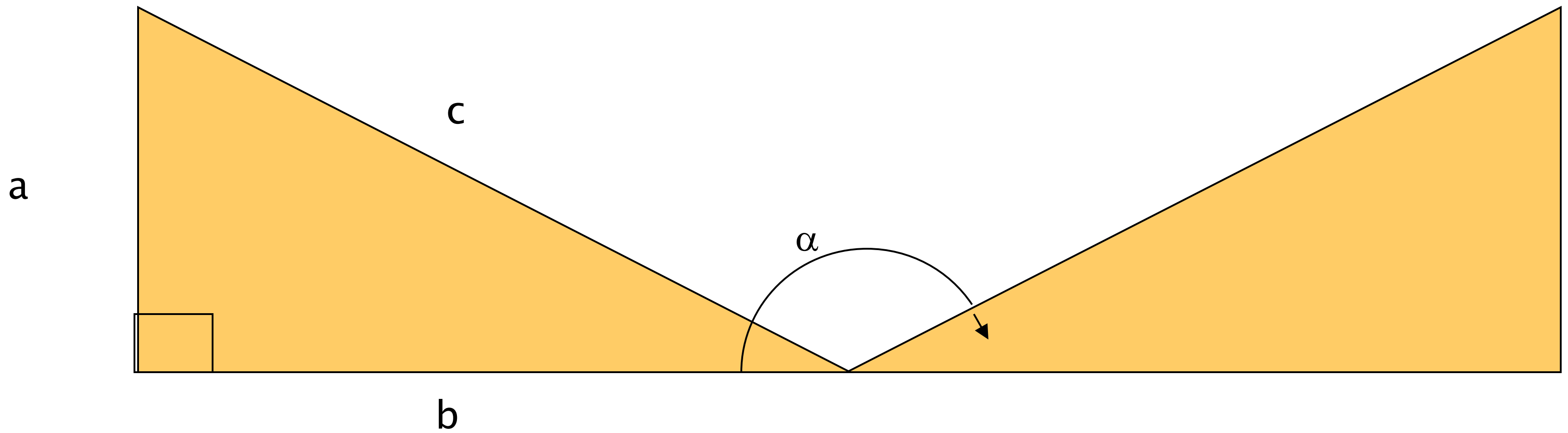


$$\sin \alpha = \frac{a}{c}$$

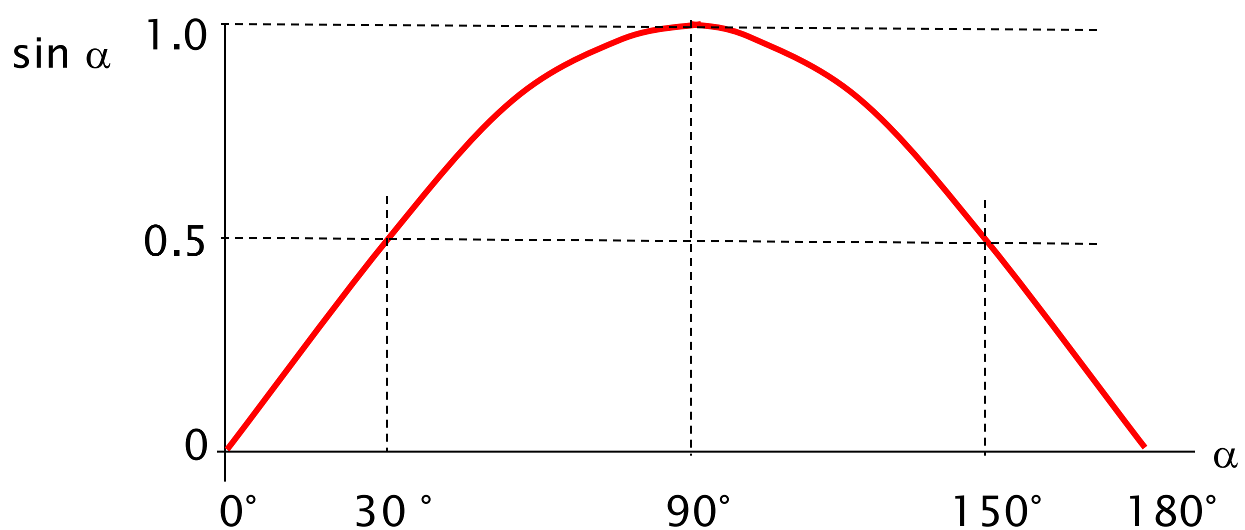
Plotting values of sin α between 0° and 90° gives a graph ....



sine (sin)

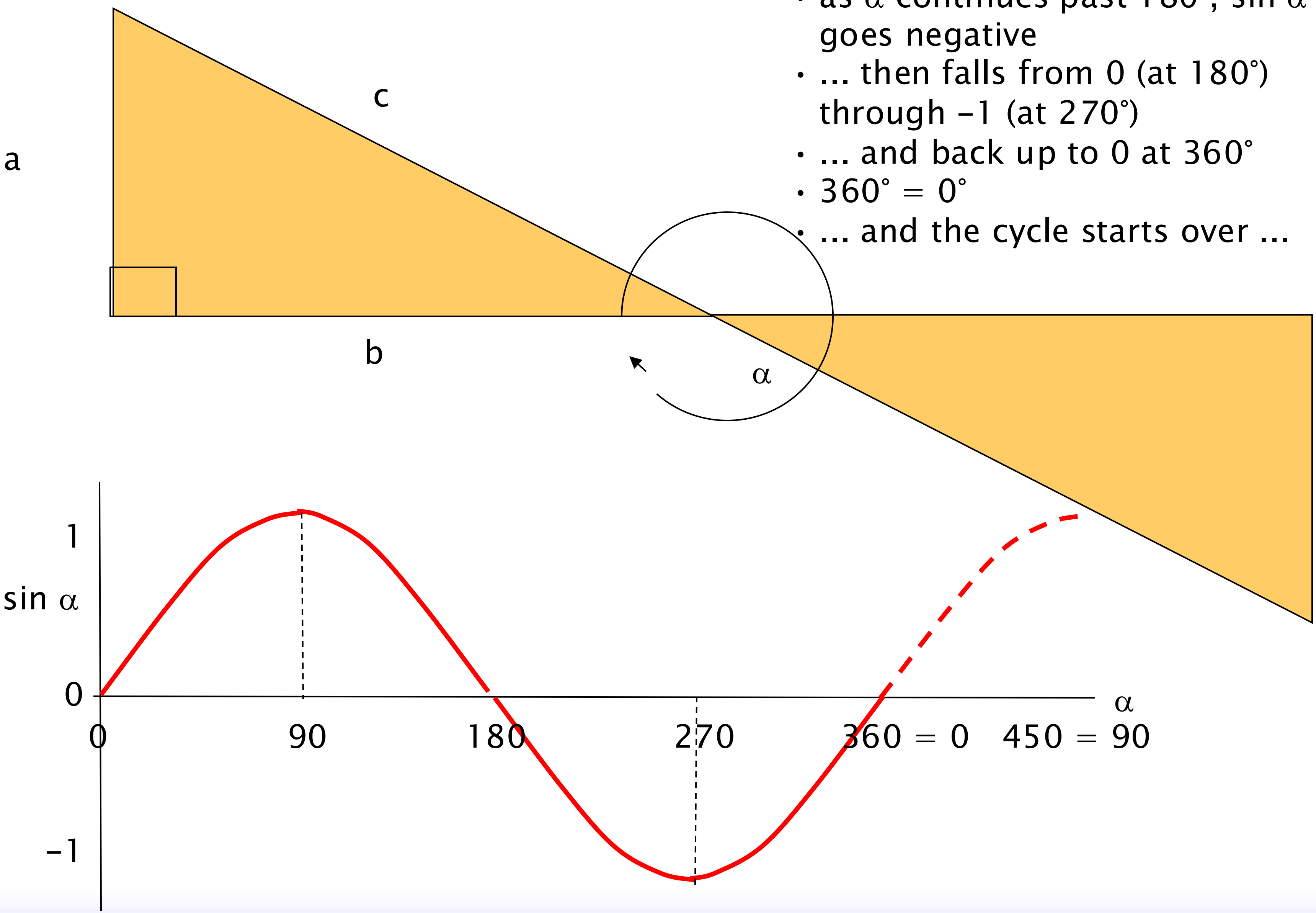


as α continues past 90° and towards 180°, the value sinα falls back to 0



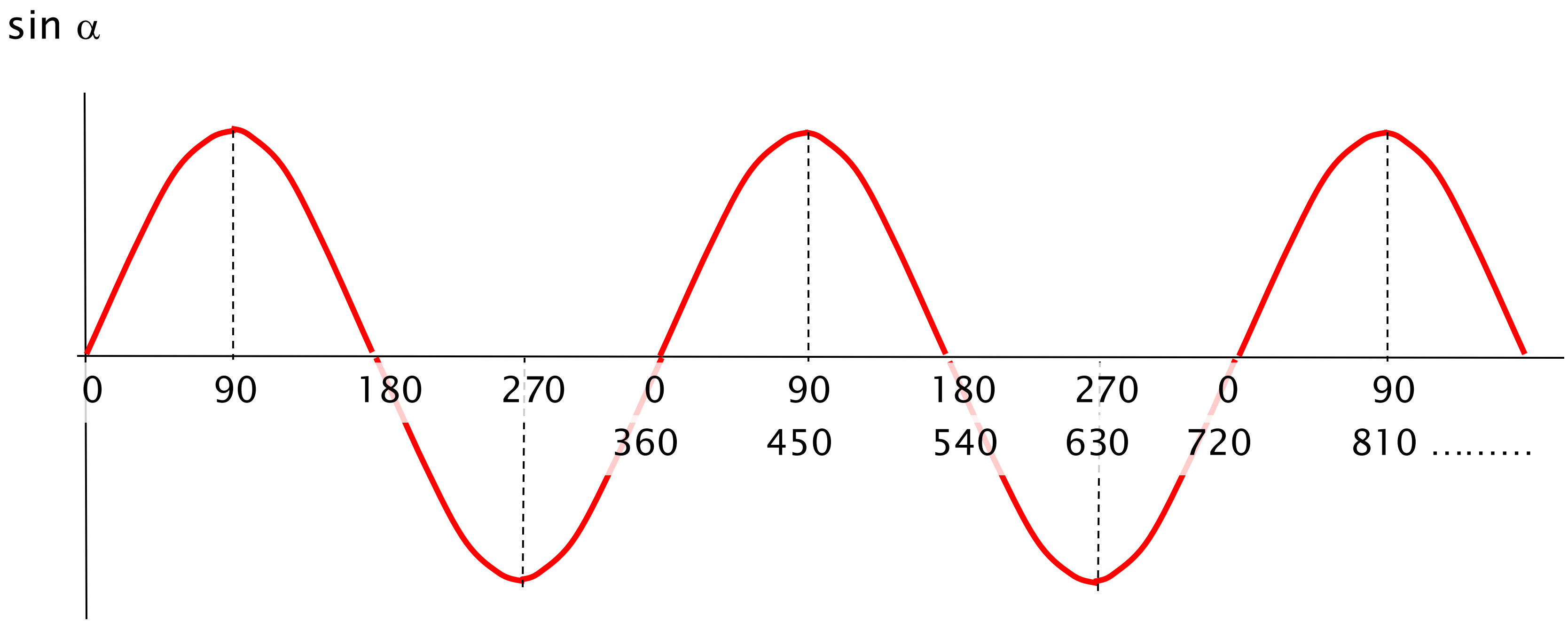
sine (sin)

- as α continues past 180°, sin α goes negative
- ... then falls from 0 (at 180°) through –1 (at 270°)
- ... and back up to 0 at 360°
- 360° = 0°
- ... and the cycle starts over ...

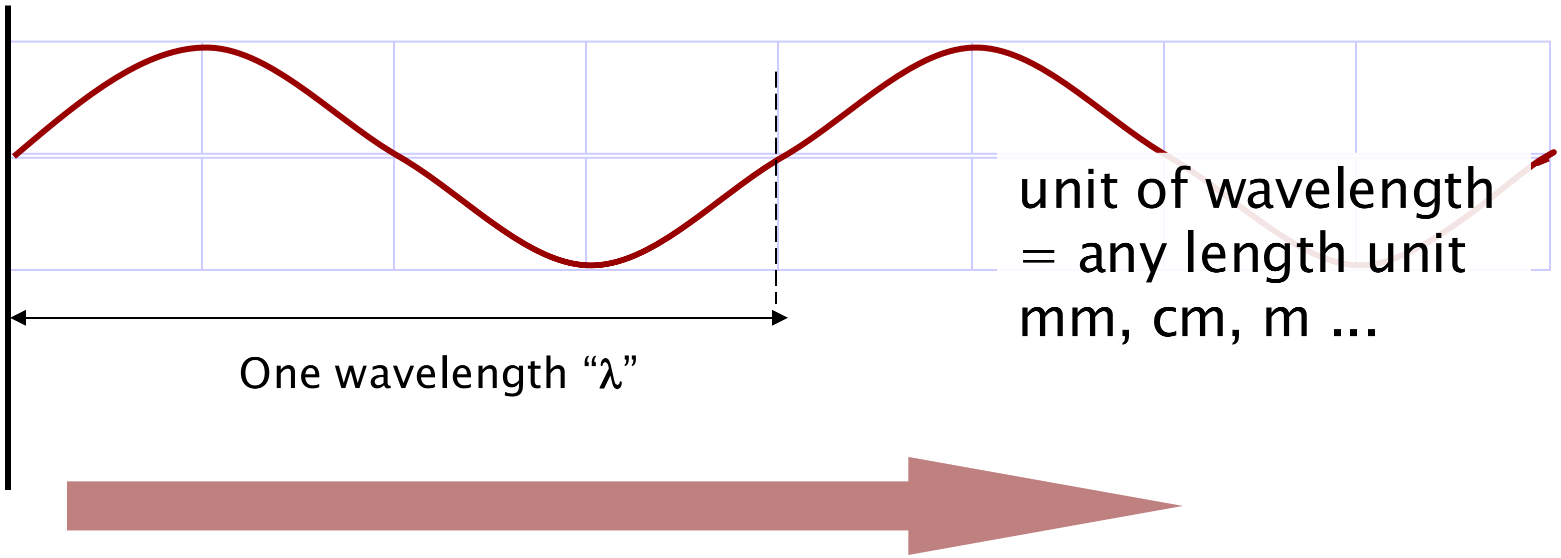


sine (sin)

As α continues, the curve of sin α forms a <u>*sin wave*</u>



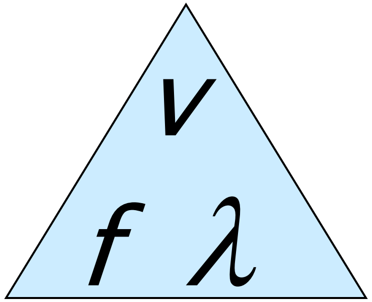
Waveforms: Wavelength, Frequency and Period



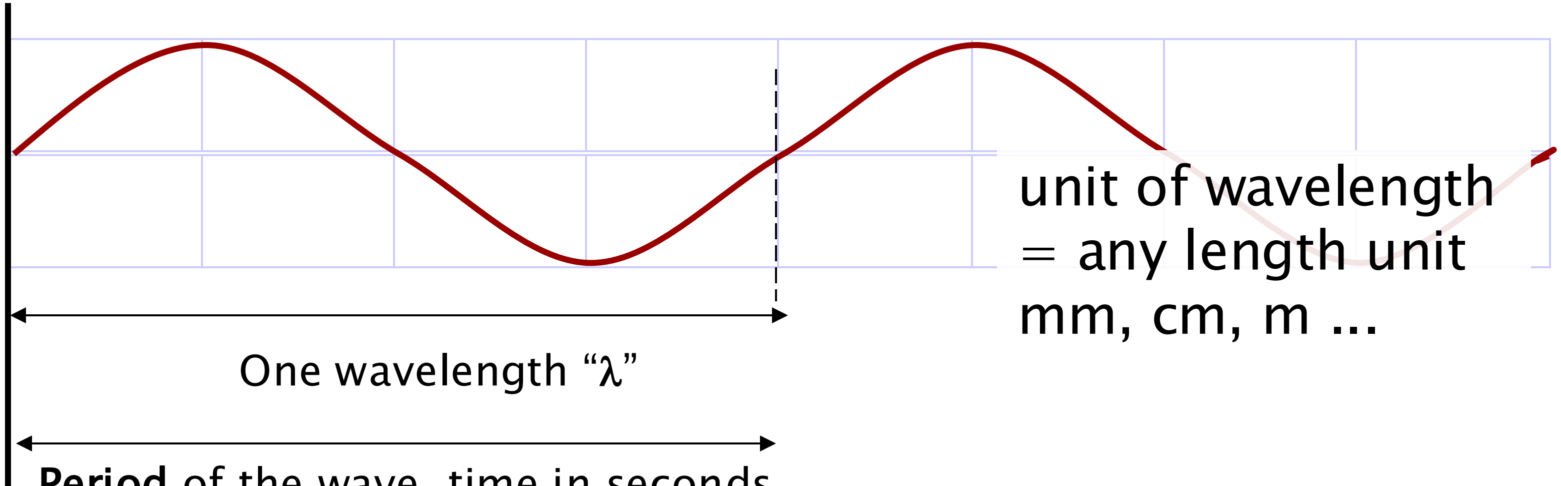
unit of wavelength
= any length unit
mm, cm, m ...

frequency = number of wavelengths / unit of time (number of **cycles per second**)
i.e. velocity = frequency x wavelength
Wavelength= **Physical distance** between two identical points on consecutive cycles of a wave

$$v = f\lambda \qquad f = \frac{v}{\lambda} \qquad \lambda = \frac{v}{f}$$



Waveforms: Wavelength, Frequency and Period

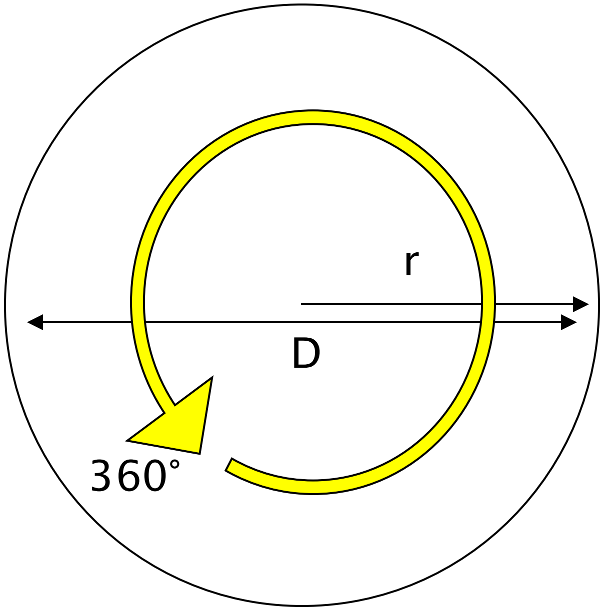


unit of wavelength
= any length unit
mm, cm, m ...

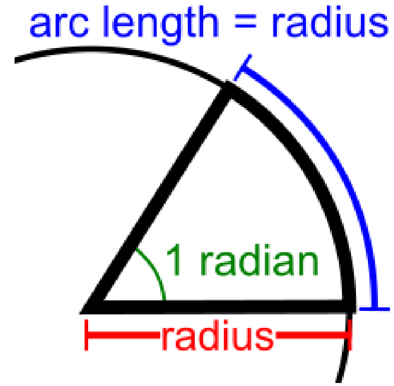
The period of the wave = T = the length of time for one wavelength
(**time taken** for one complete cycle of the wave.)

$$f = \frac{1}{T} \qquad T = \frac{1}{f} \qquad f\,T = 1 \text{ second}$$

Radians



Circumference
$= \pi D$
$= 2\pi r$



Radian
= the angle where the arc swept = the radius of the circle
i.e. 1 Rad = $180/\pi^{\circ} \approx 57.3^{\circ}$
i.e. 2π Rad = 360°

So if we take an arc around the circle the same length as the radius (r)
It will trace an angle of 57.3˚
… call this an angle = 1 *radian*
But consider – how many radius lengths are there around the circumference?
$= 2\pi r \div r$
$= 2\pi$

The Signal

Shure SM58 Specifications

Product Specifications

SM58® Cardioid Dynamic Microphone

Overview

The legendary SM58® is an industry-standard, highly versatile cardioid dynamic vocal microphone that is consistently the first choice of vocal performers around the globe. Even in extreme conditions, the SM58 is tailored to target the main sound source while minimizing background noise, delivering warm and clear vocal reproduction.

Features

- Frequency response tailored for vocals, with brightened midrange and bass rolloff
- Uniform cardioid pickup pattern isolates the main sound source and minimizes background noise
- Pneumatic shock-mount system cuts down handling noise
- Effective, built-in spherical wind and pop filter
- Supplied with break-resistant stand adapter which rotates 180 degrees
- Legendary Shure quality, ruggedness and reliability
- Cardioid (unidirectional) dynamic
- Frequency response: 50 to 15,000 Hz

Available Models

SM58-LC	Includes Stand Adapter and Zippered Pouch
SM58-CN	Includes 7.6 m (25 ft) XLR-Male to XLR-Female Cable, Swivel Adapter and a Zippered Pouch
SM58S	Includes Integrated On/Off Switch, Swivel Adapter and a Zippered Pouch

SM58

http://www.shure.com/idc/groups/public/documents/webcontent/us_pro_sm58_specsheet.pdf

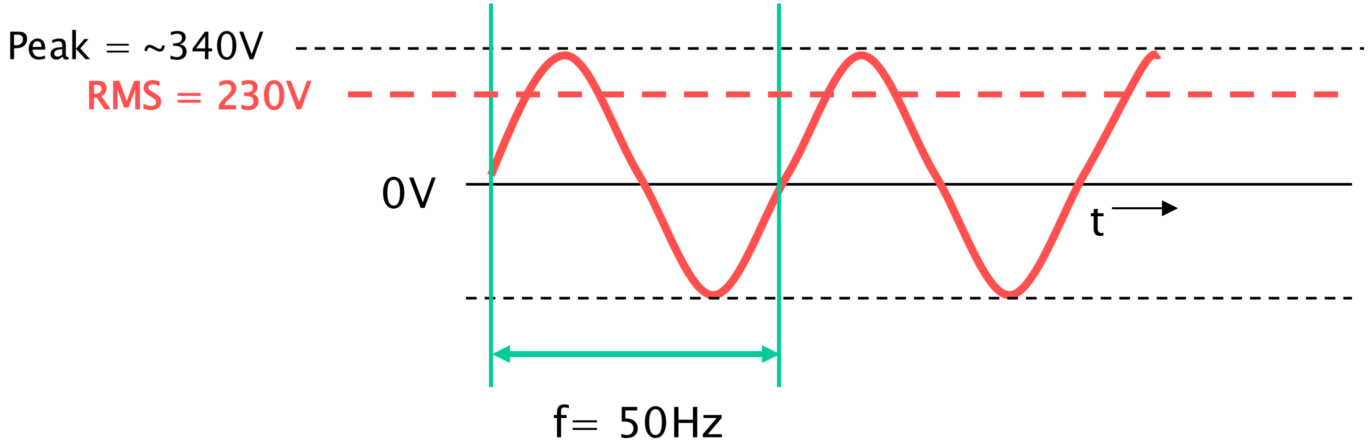
Specifications

Type	Dynamic
Frequency Response	50 to 15,000 Hz
Polar Pattern	Cardioid
Sensitivity (at 1,000 Hz Open Circuit Voltage)	–54.5 dBV/Pa (1.85 mV) 1 Pa = 94 dB SPL
Impedance	Rated impedance is 150Ω (300Ω actual) for connection to microphone inputs rated low impedance
Polarity	Positive pressure on diaphragm produces positive voltage on pin 2 with respect to pin 3.
Case	Dark gray, enamel-painted, die cast metal; matte-finished, silver colored, spherical steel mesh grille
Connector	Three-pin professional audio connector (male XLR type)
Connector	Three-pin professional audio connector (male XLR type)
Net Weight	298 grams (10.5 oz)
Dimensions	162 mm (6-3/8 in.) L x 51 mm (2 in.) W

Output voltage ~ 0.19mV $= \frac{19}{1000}$ V

RMS Voltage, e.g. Mains Supply

AC Power supply in UK (and the rest of Europe) is 230V AC
- AC = "alternating current"
- … alternating at a rate of 50 Hz (cycles per second)
- North America = 120V @ 60Hz
- Varies quite a bit around the world



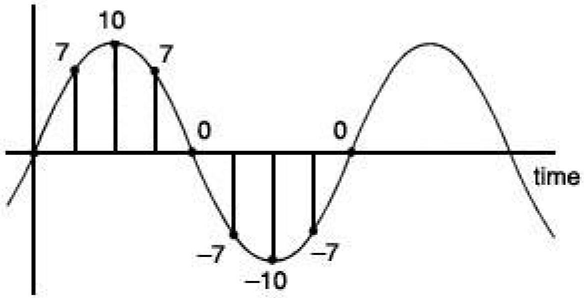
<u>Problem</u>: What is the "average" value of the sinusoidal waveform?
<u>Answer</u>: Zero … but obviously this is misleading

<u>Mathematically</u> this is expressed as RMS … Root Mean Square
<u>Practically</u>, could consider the RMS voltage as the voltage required to produce the equivalent power (e.g. *heating effect*) as a DC voltage

Root Mean Square

For example: suppose the time samples are as shown in the diagram:

(note: eight samples)



Values	0	7	10	7	0	-7	-10	-7
Squares	0	49	100	49	0	49	100	49

Sum of squares = 396
Average of squares = 396/8 = almost 50
Square root = approx 7

With more intervals the rms average turns out to be
= (peak value) / $\sqrt{2}$
= peak value/1.41
= 0.707peak value

RMS value is calculated by:
1. Squaring the function (to make all values positive).
2. Finding the mean (average) over one cycle.
3. Taking the square root of that mean.

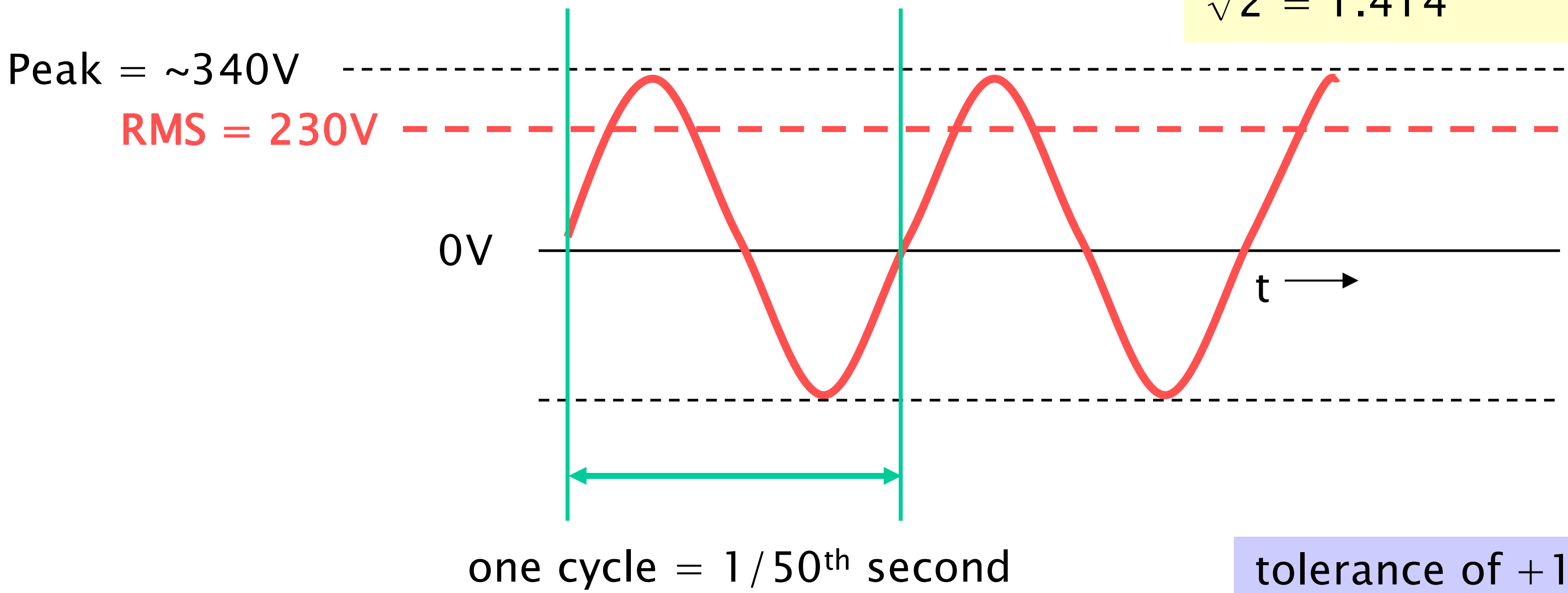
http://www.practicalphysics.org/go/Guidance_107.html

Root Mean Square – the Power Supply

<u>240V RMS, 50Hz:</u>

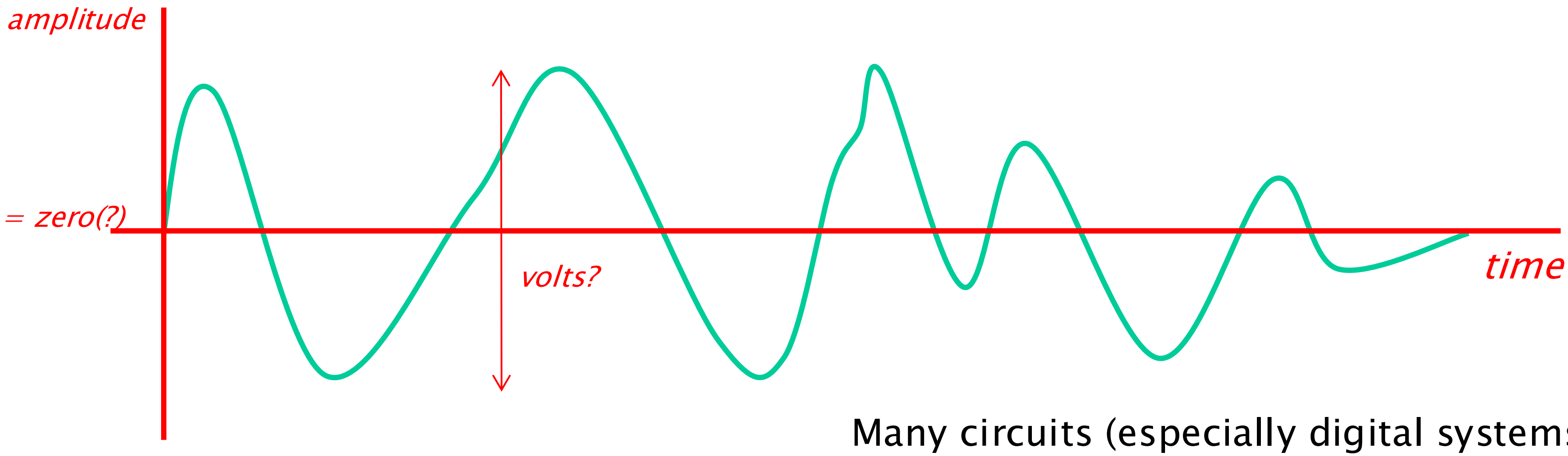
$V_{peak} = \sqrt{2} \times V_{rms}$

$\sqrt{2} = 1.414$

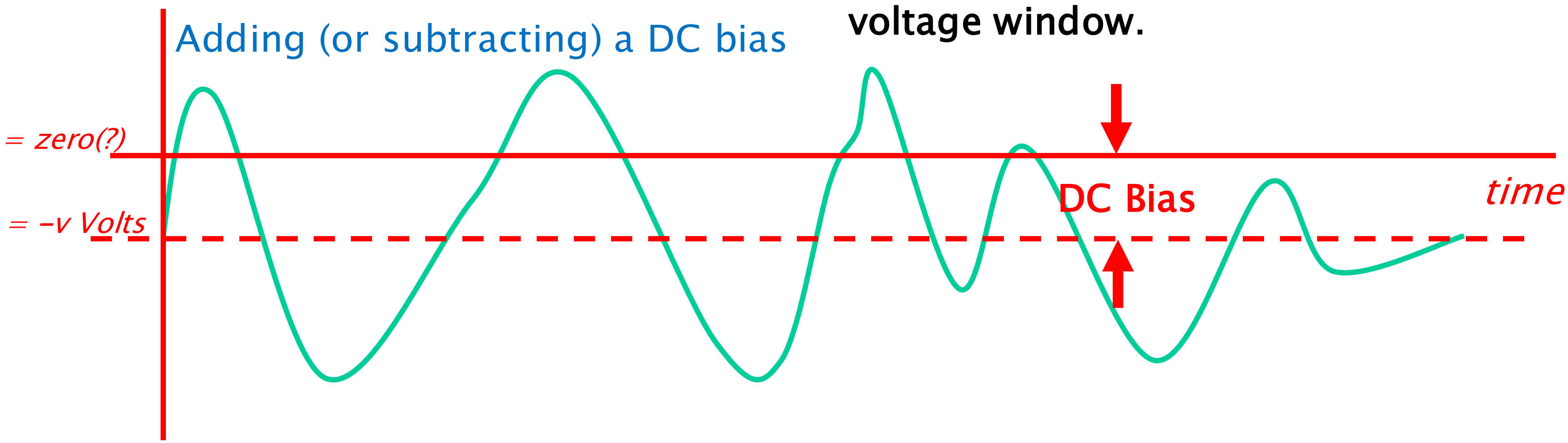


tolerance of +10%, –15%

Analogue Signal – *for example …*

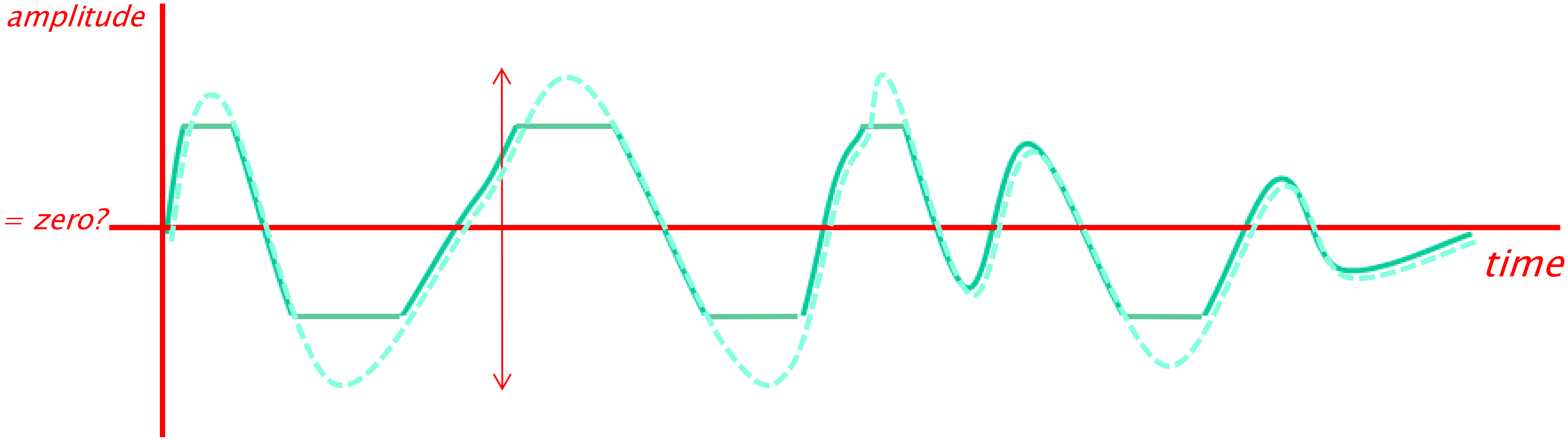


Many circuits (especially digital systems or op-amps) can't handle negative voltages. It ensures the **waveform sits in the right voltage window.**



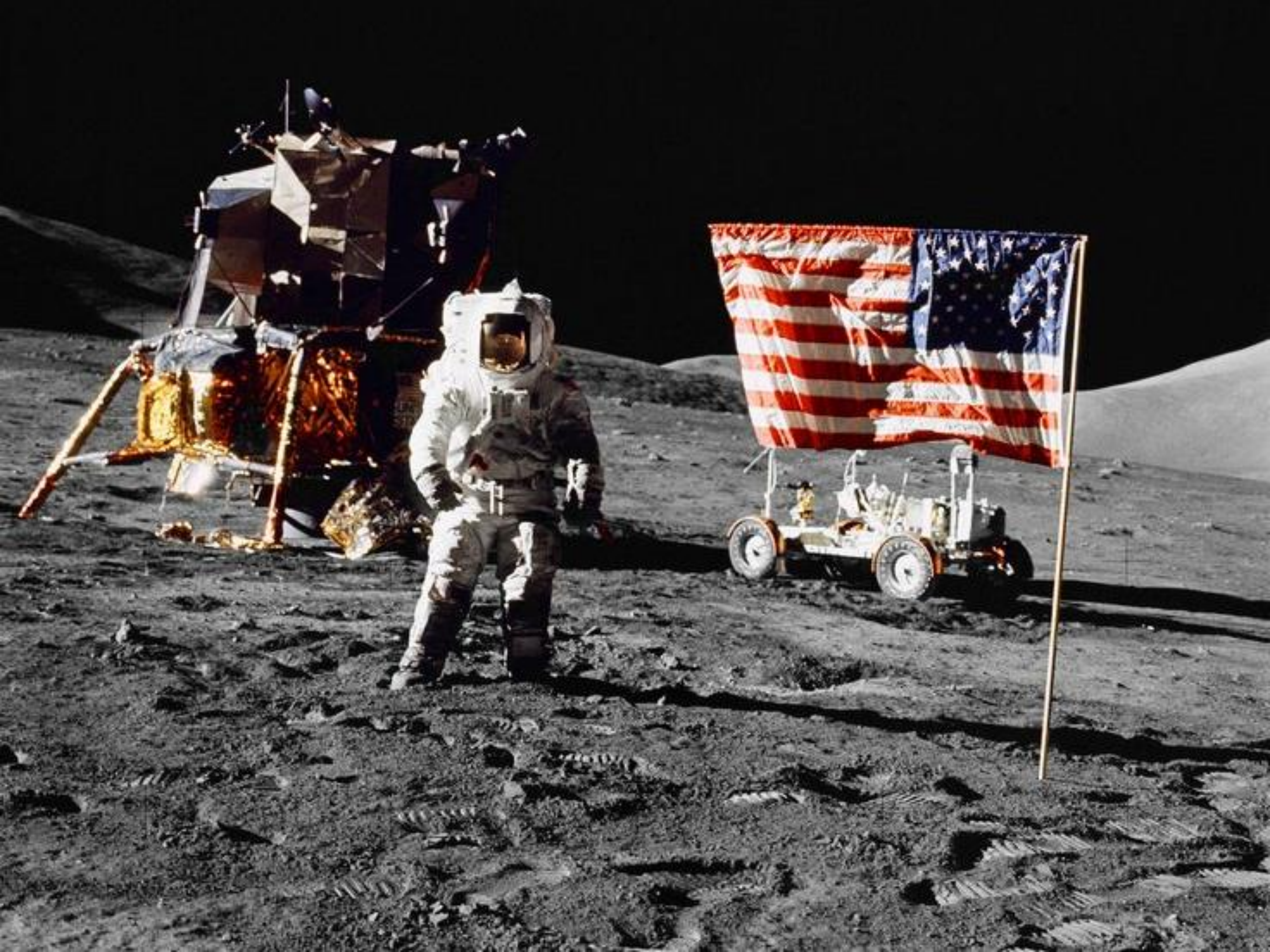
Analogue Signal – *for example …*

If signal is amplified to beyond DC availability in the circuit
… clipping of the signal

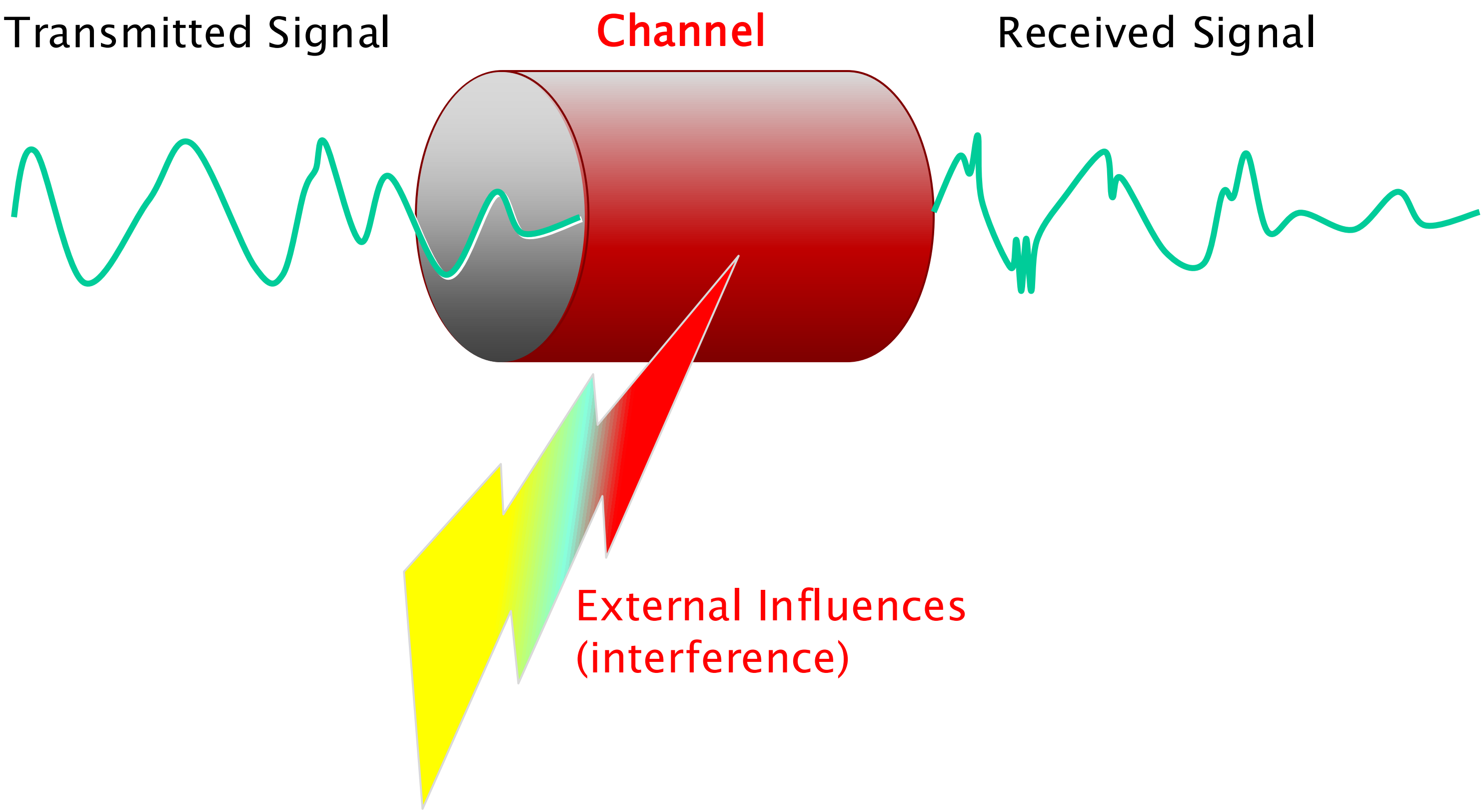


Clipping happens when an amplifier tries to output a signal that's **beyond its power supply limits.** Since it physically *can't* go higher or lower than its supply rails, the signal gets “flattened” or **cut off** at those limits.

The Channel

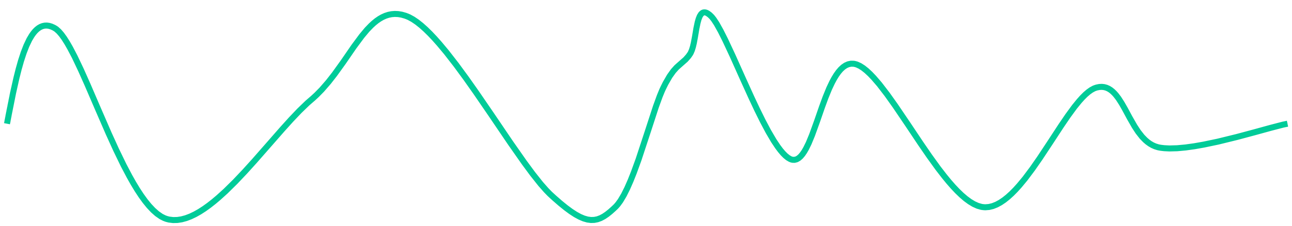


Signal and Channel

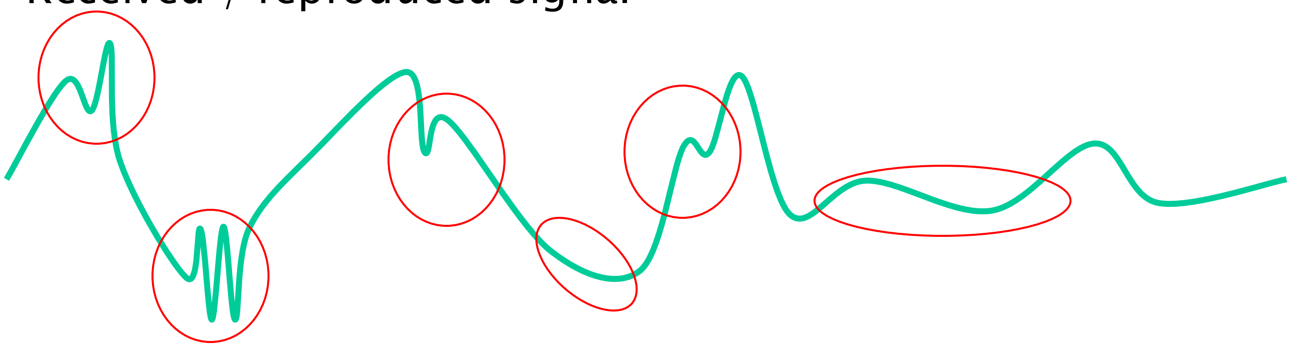


Analogue Signal Problems In The Reproduction Process

Original signal

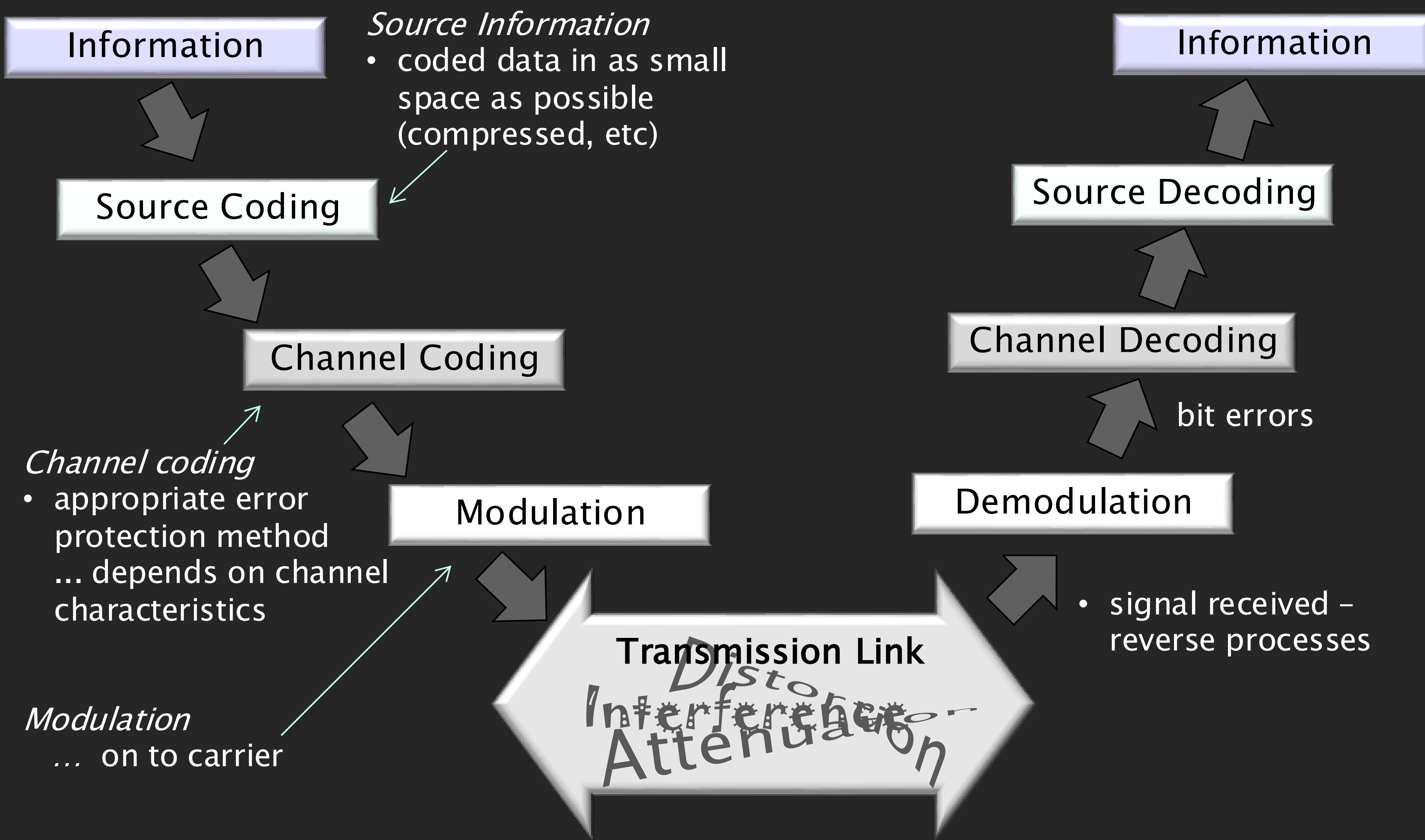


Received / reproduced signal



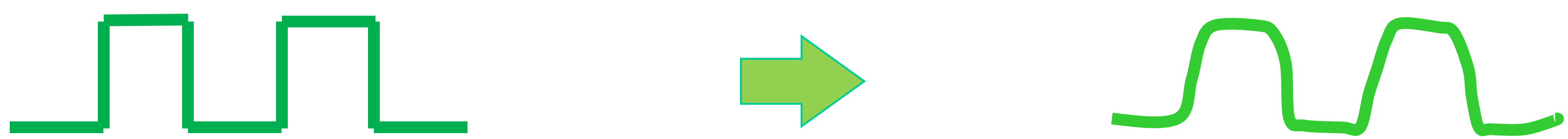
- phase effects: Phase distortion occurs when different frequency components of a signal are delayed by different amounts.
- Interference: Unwanted external signals
- other signals crosstalk: Leakage of one signal into another nearby channel.
- various intermodulation products: When **two or more signals mix non-linearly**, creating new frequencies
- mechanical transducer ineffective / inefficient: Microphones, speakers, or pickups not perfectly converting energy between **electrical** ↔ **mechanical** forms.

Information Transmission, Signal Encoding



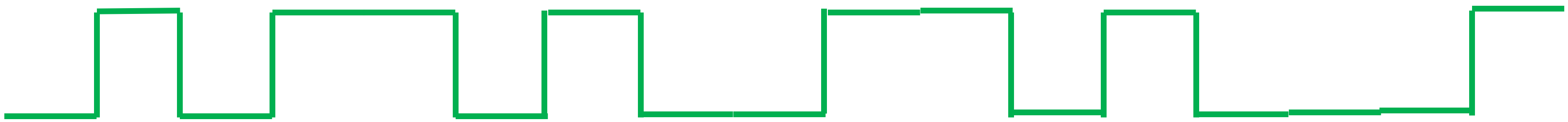
- transmission is subject to noise interference
- aim – to approach the Shannon Limit

Pulse Transmission, Interference (ISI) Eye Diagrams

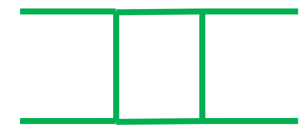


Original generated signal → transmission system → received signal

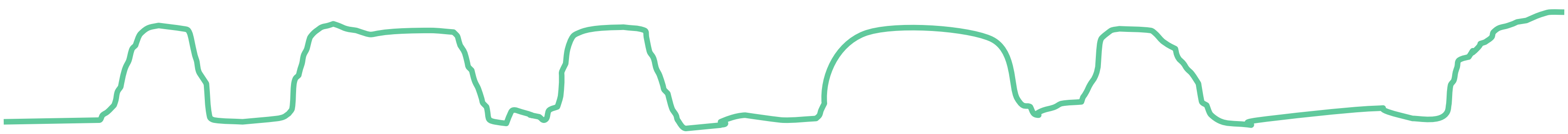
Quality of the Transmission



Theoretically perfect random pulse train



Pulse train: two or three pulse periods successively sampled, overlayed on each other.
Pulse train is a powerful visualisation technique used in signal analysis, especially to assess **signal stability**, **jitter**, and **timing consistency.**

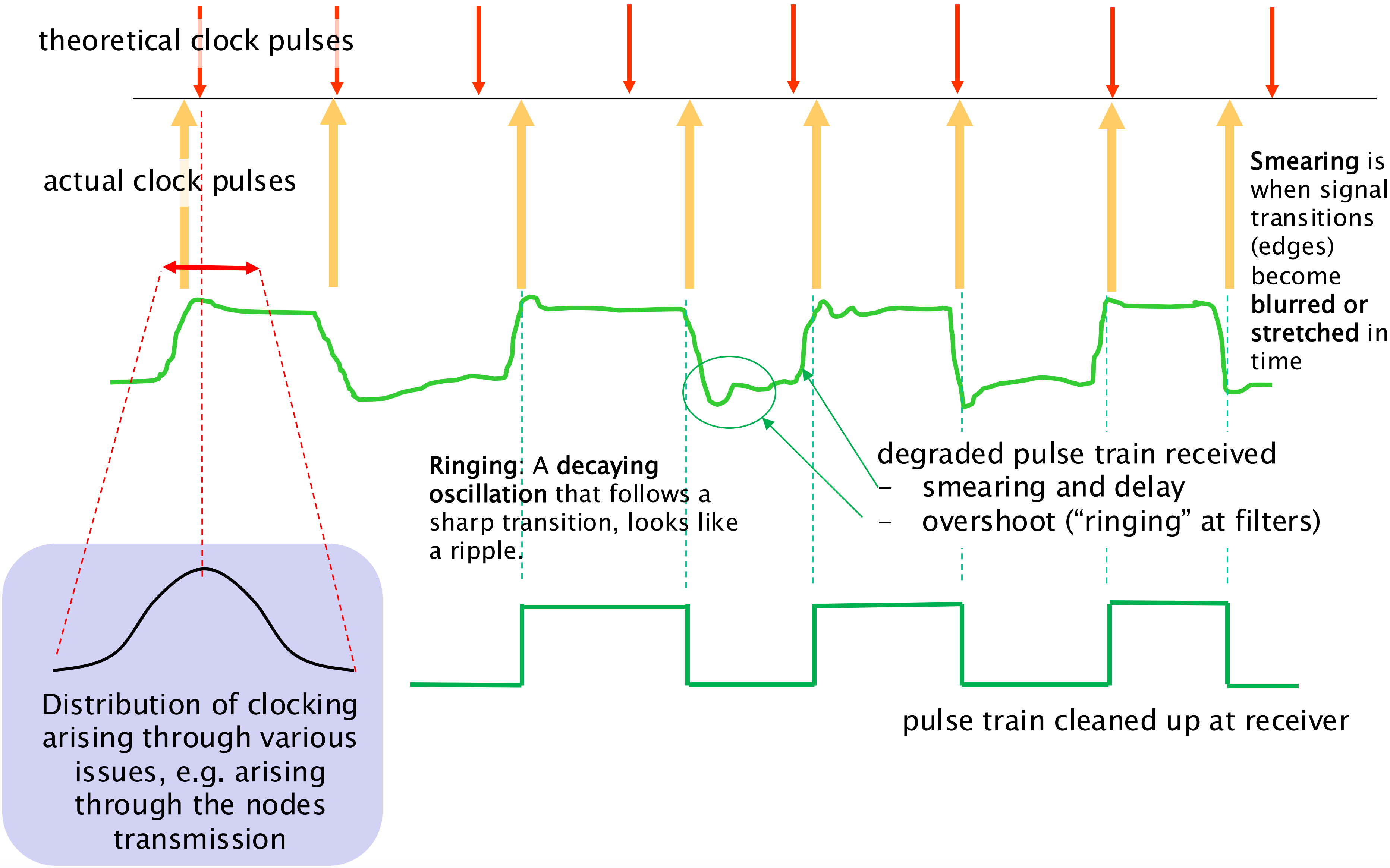


Random pulse train affected by imperfect transmission medias

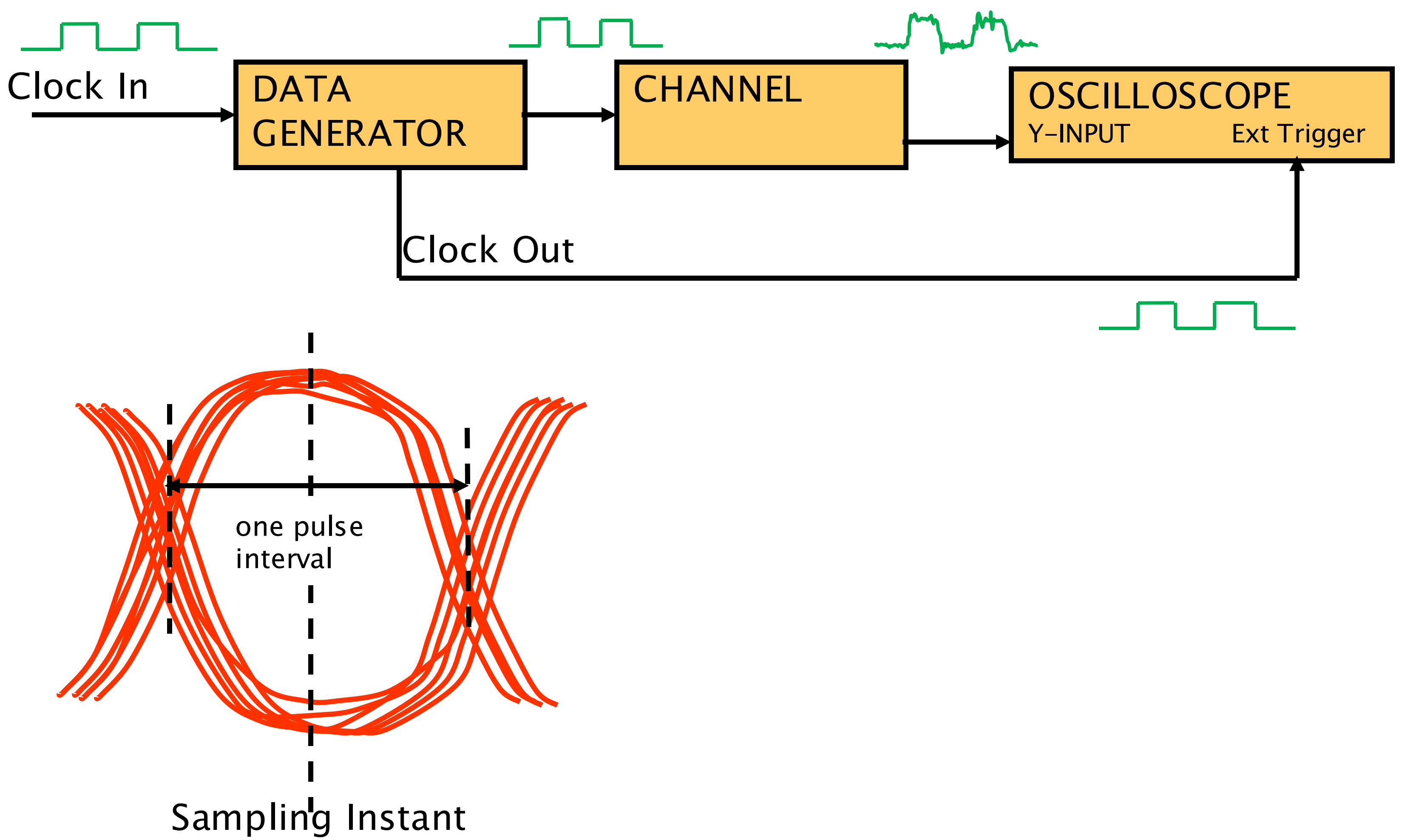
Practical pulse train: successively sampled, overlayed

Eye diagram shows the quality of the signal path. The wider and more open the eye, the cleaner the signal.

Jitter refers to **timing variations** in the edges of a digital signal—pulses **don't arrive exactly when they should.**

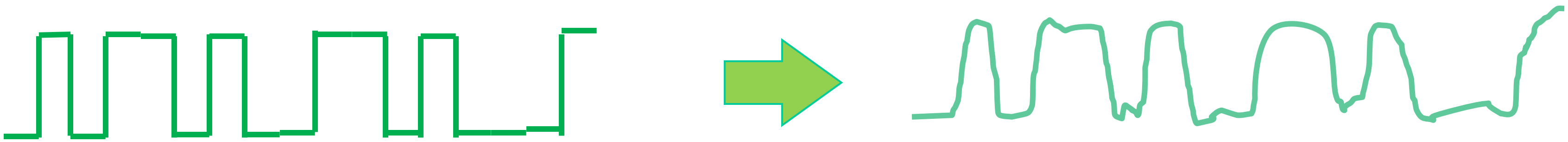


Eye Diagram – test set-up

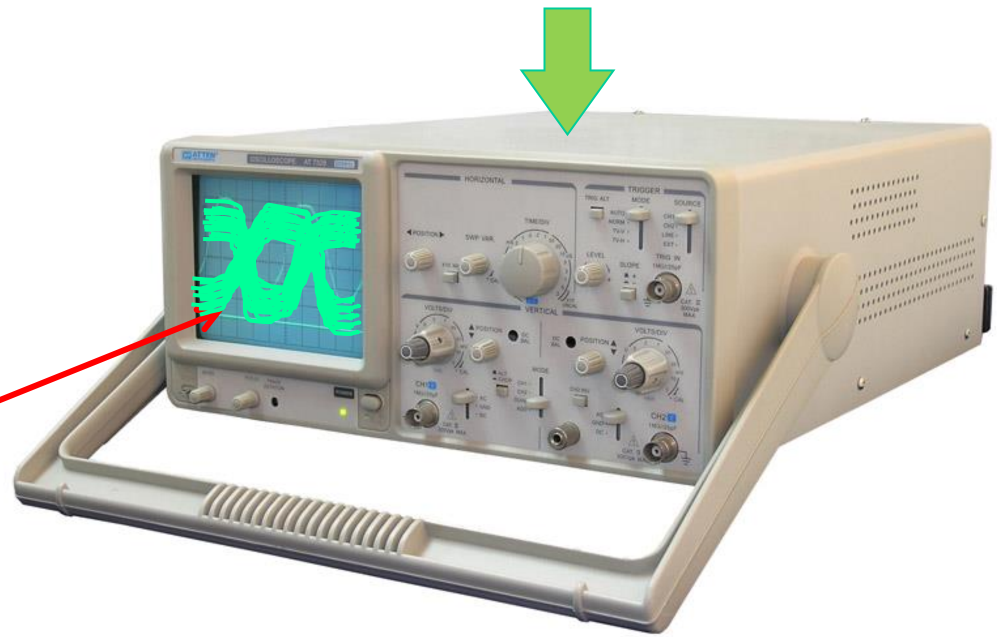


ISI – Eye Diagram

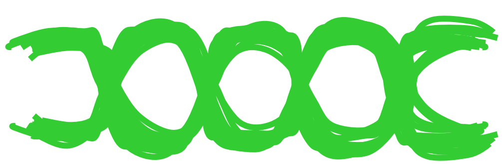
Demonstrated by “eye pattern” analysis of impulse response
It indicates the performance of a transmission system



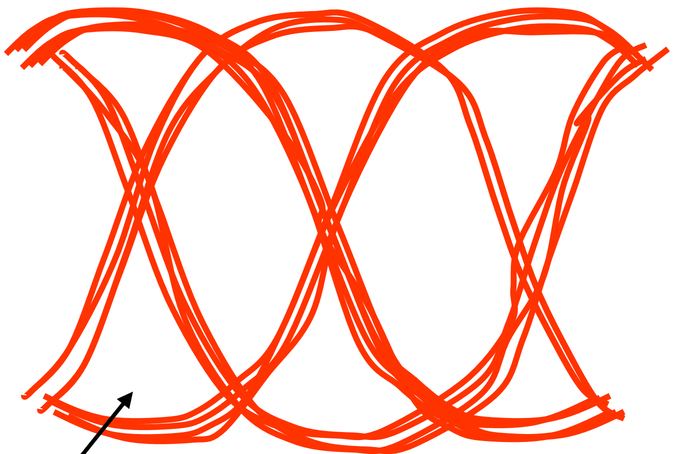
original generated signal → transmission system → received signal



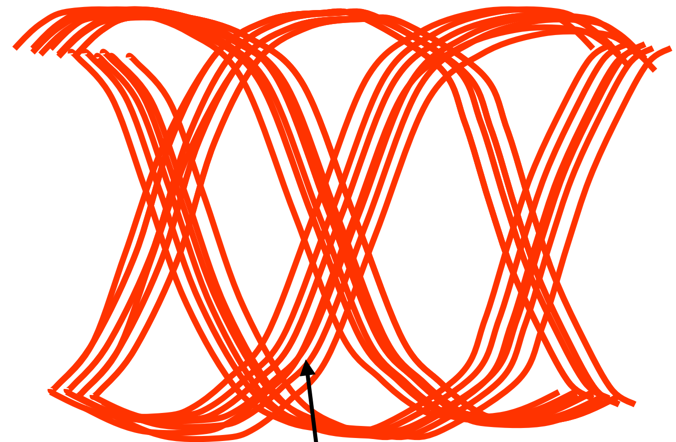
a run of pulses are fed into a scope – locked on to one or two pulse widths superimposed 1’s and 0’s



Different Eye Patterns



Signal only –
various small
individual random
distortions (but
likely OK)
– eye is “open”



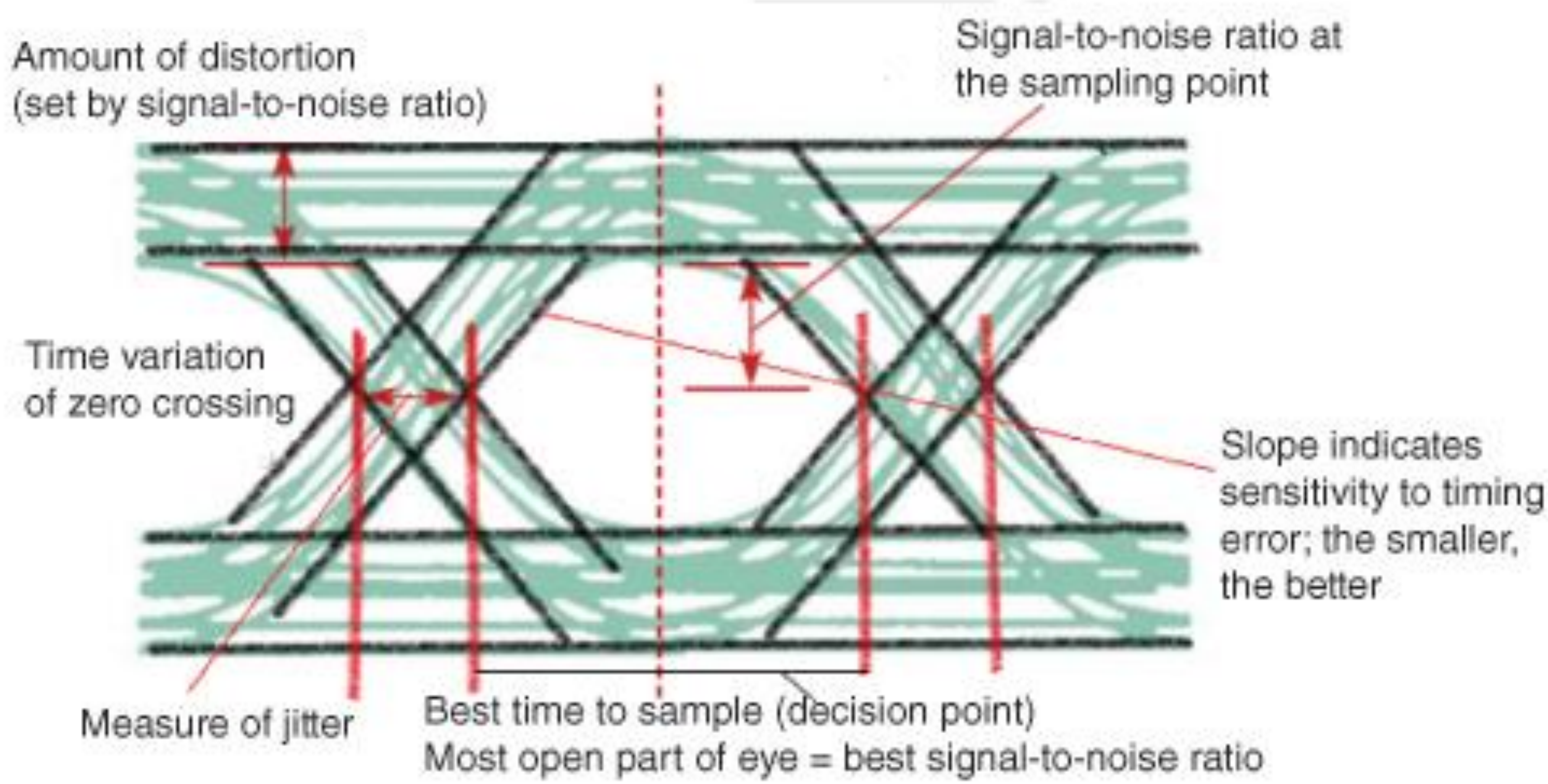
Signal + timing errors
– eye closing

Signal + noise – eye
closing (amount
depends on noise)

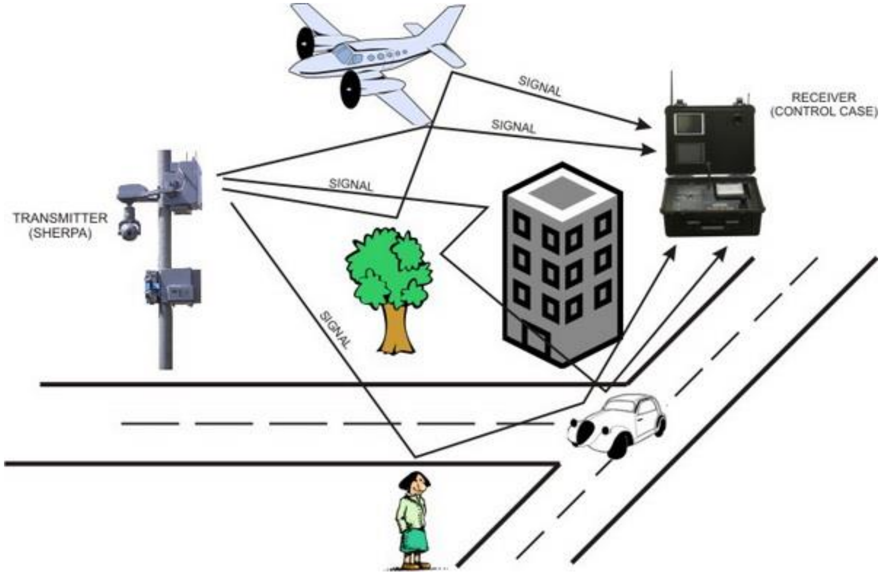
ISI – Eye Diagram

Can reveal important information

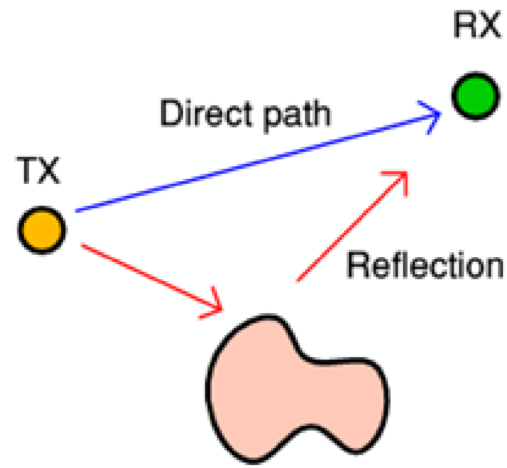
- best point for sampling
- SNR at the sampling point
- amount of jitter and distortion
- time variation at zero crossing – jitter

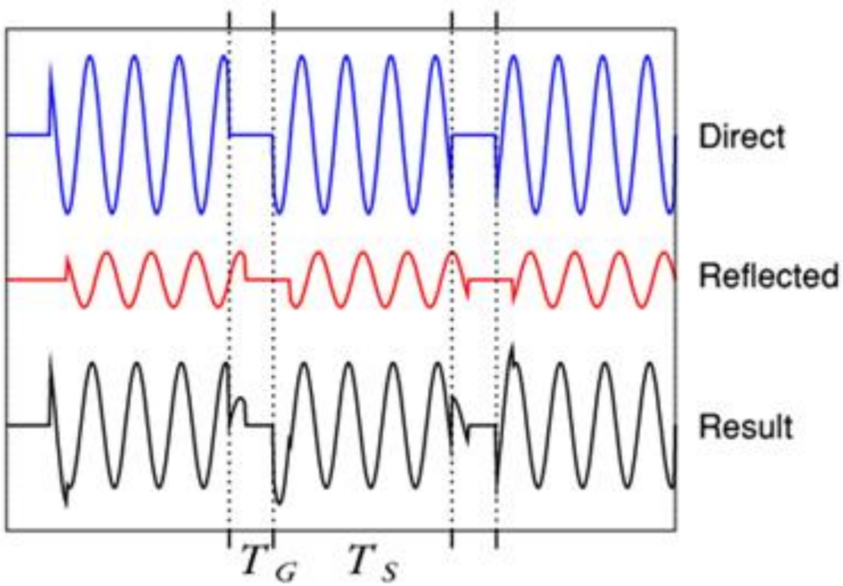


Multipath ISI



- an undesirable effect – reflections of radio waves hitting various objects in the environment
- unavoidable





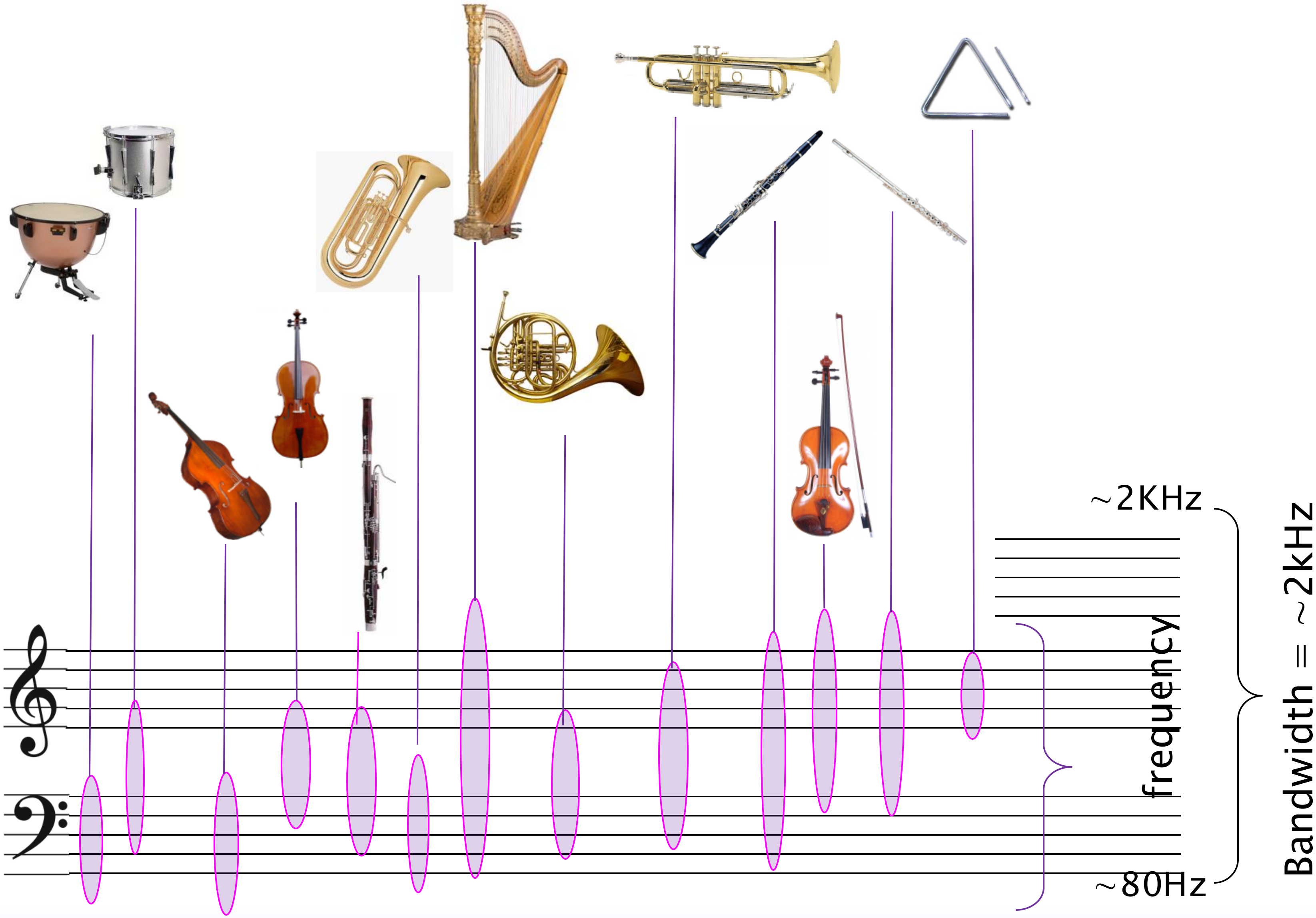
http://www.st-andrews.ac.uk/~www_pa/Scots_Guide/RadCom/part19/page3.html

- i.e. interference between a symbol and its own reflection … another form of ISI

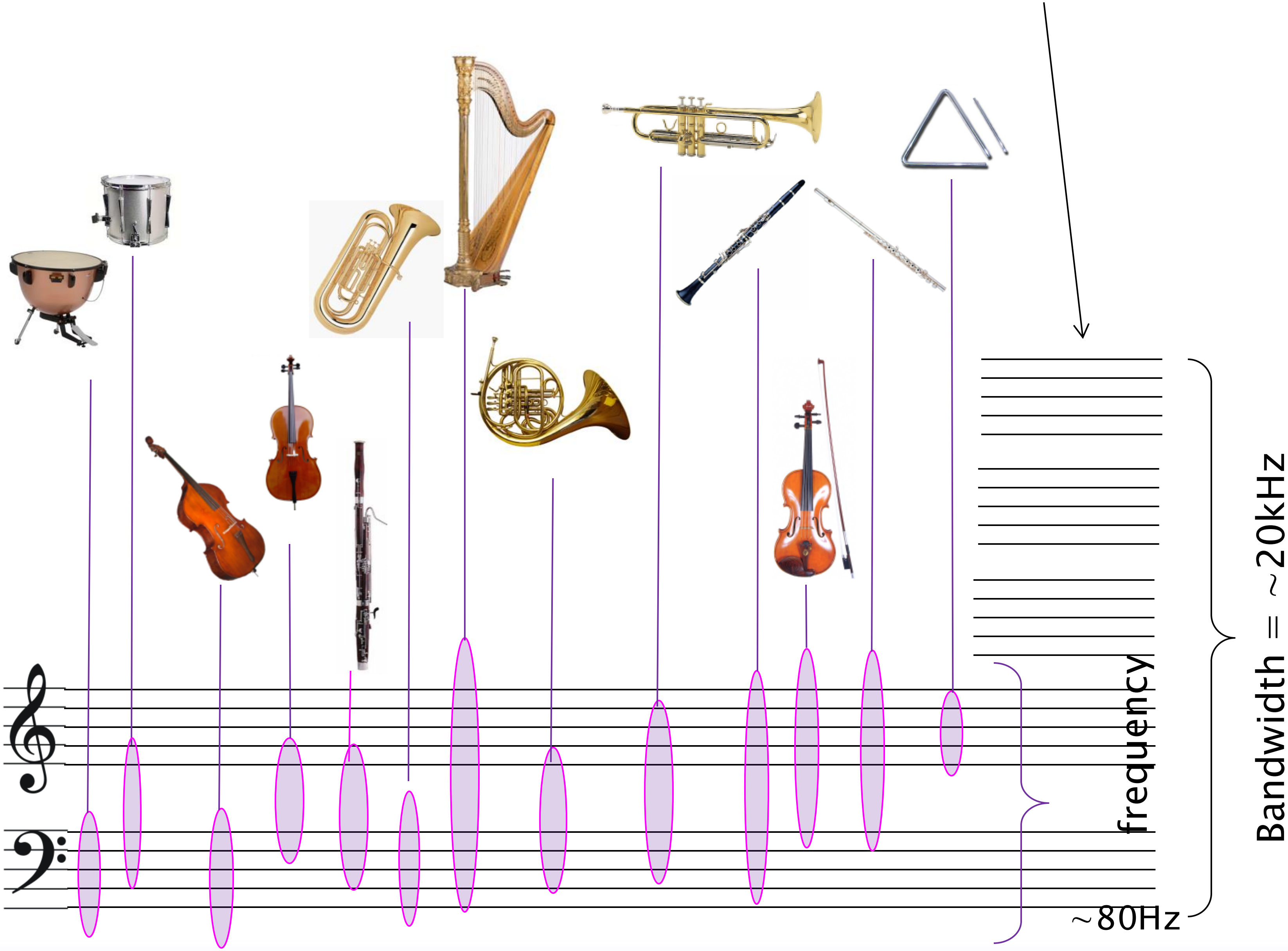

BBC HD
1080 lines

Bandwidth

Frequencies, Bandwidth – fundamental freqs



Frequencies, Bandwidth – add the harmonics



Bandwidth – Analogue Signals

Bandwidth is the range of frequencies an analogue signal occupies or requires for accurate transmission and reproduction.

Bandwidth = 20KHz

freq

~20Hz

~20kHz

Audio from 20 Hz to 20 kHz → needs ~20 kHz **analogue bandwidth.**

Baseband signals – the range of frequencies of the original signal

… unmodulated, not frequency shifted

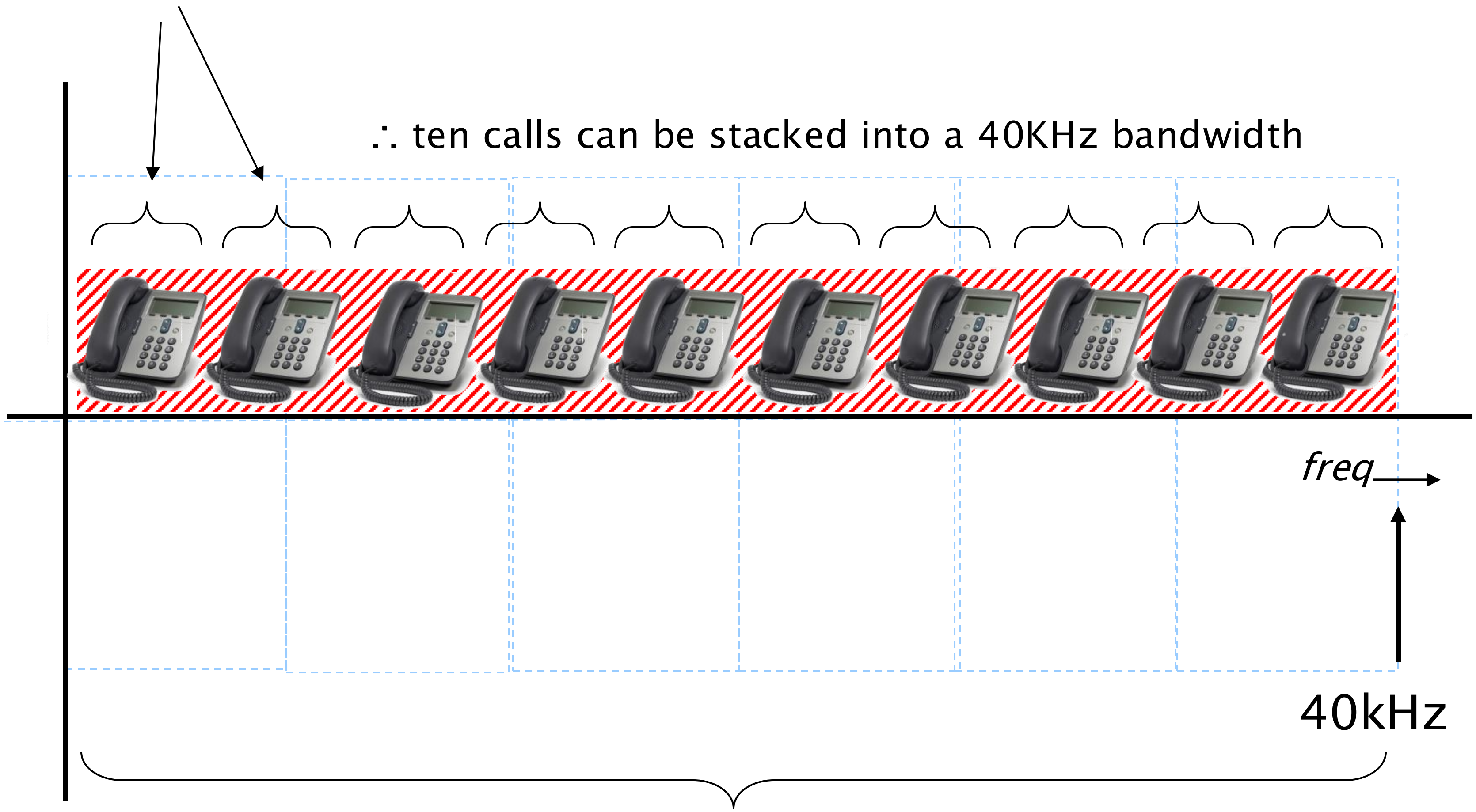
… It contains the **raw frequency content** of the source

Sampled Bandwidth – Analogue Signals

"Traditional" telephone network – using Frequency Division Multiplexing

Telephone bandwidth = 4KHz

∴ ten calls can be stacked into a 40KHz bandwidth



40KHz communications channel

Digital Signals, Bandwidth and Spectra

Does Analogue Bandwidth relate to Digital Bandwidth?

... it depends ... but not an accurate relationship

Digital Bandwidth: Refers to the **data rate** needed to transmit or store digital representations of signals.

Data rate in bits/second

= bandwidth of the communications channel

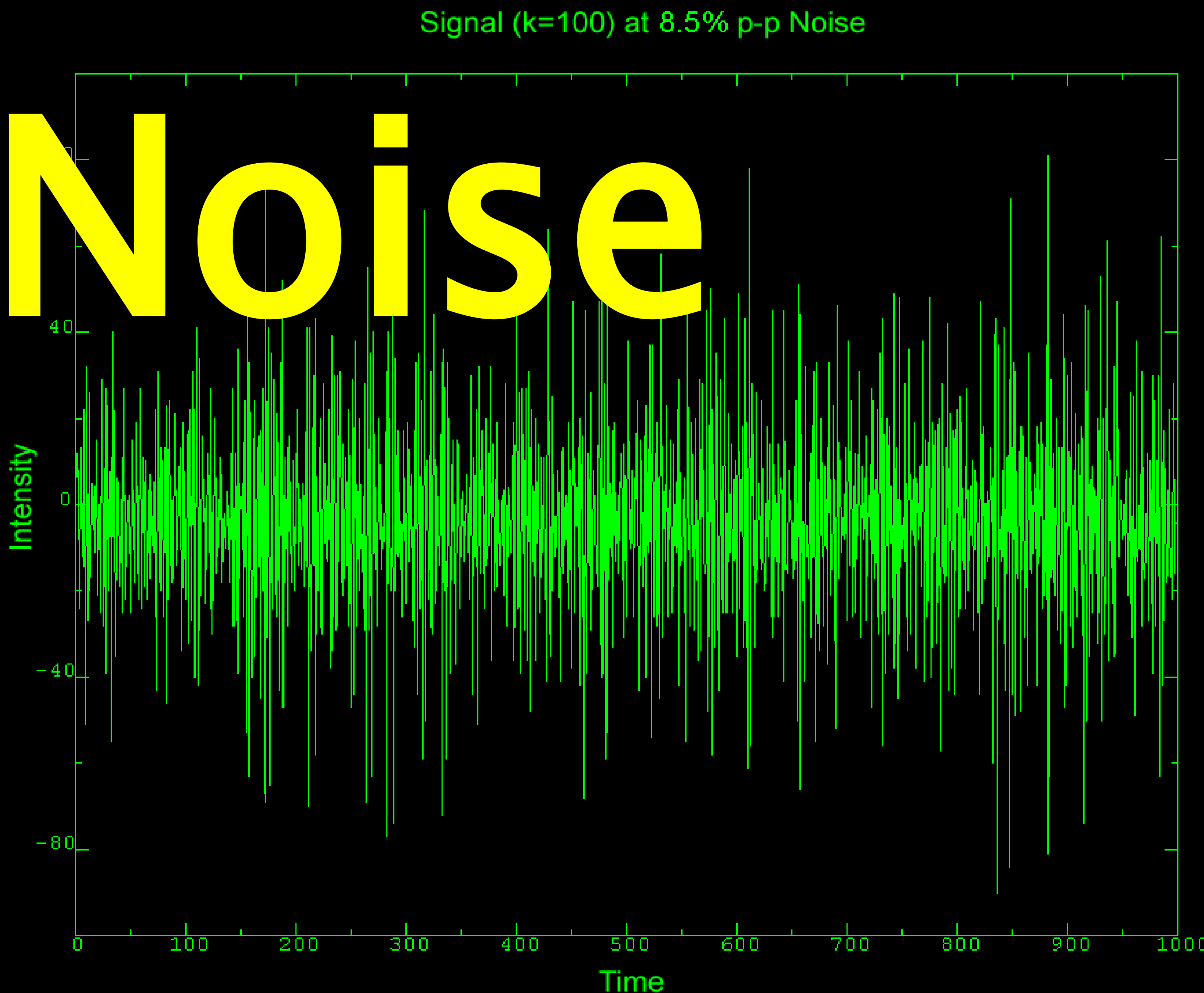
- e.g. 20kHz signal, sampled at 44.1kHz
  with 16bit coding (= 65536 quantisation levels) = 44100 samples/sec x 16bits/sample
  = 705600 bits/s = 705.6 kb/s
  … for stereo Bitrate= 705.6 x2 = 1411.2kb/s=1.4112Mb/s

- compress to mp3 @320kb/s; bandwidth = 320kb/s
- compress this to mp3 (or aac) @128kb/s; bandwidth = 128kb/s

- typical SD TV channel, avc =>3Mb/s
- typical HD TV channel, avc = >8Mb/s

- depends on no of bits, compression, etc
- *but consider the practical transmission channel … the modulation level; channel S/N*

Noise


Signal (k=100) at 8.5% p-p Noise
Intensity
40
0
−40
−80
0
100
200
300
400
500
600
700
800
900
1000
Time

Noise

IMPORTANCE:

- for audio or video:

  … noise is an obvious disturbance to the original noiseless signal

- for data transfer

  … noise can destroy data; can reduce data rates; require retransmissions, etc

Noise

Nature of noise

- typically random – a stochastic process

Noise is manifested as a voltage

- … alongside the voltage of the original signal

Various causes of noise

- (coming next …)

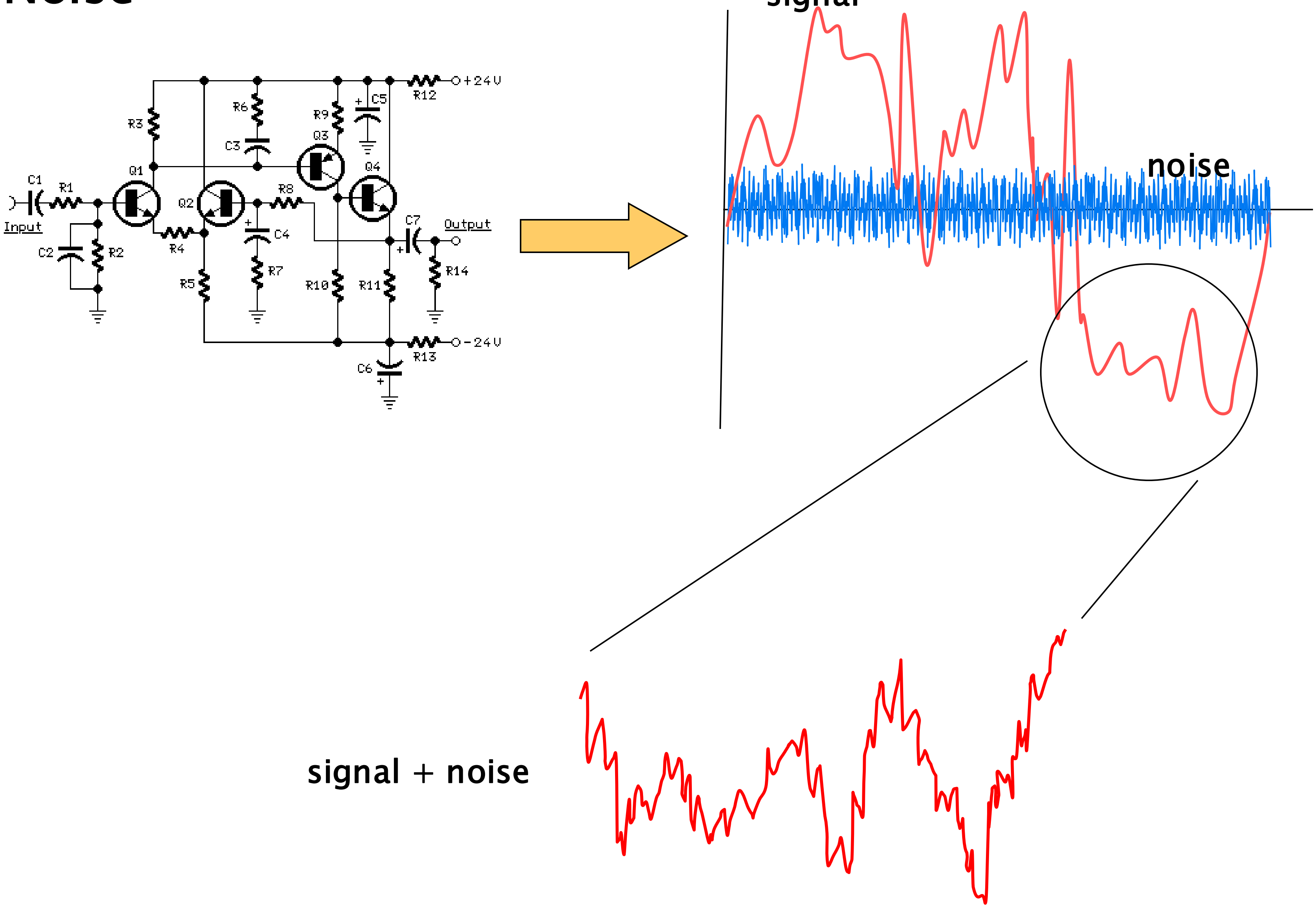
Need ways to measure and represent noise

- signal / noise ratio

Useful noise (sometimes)

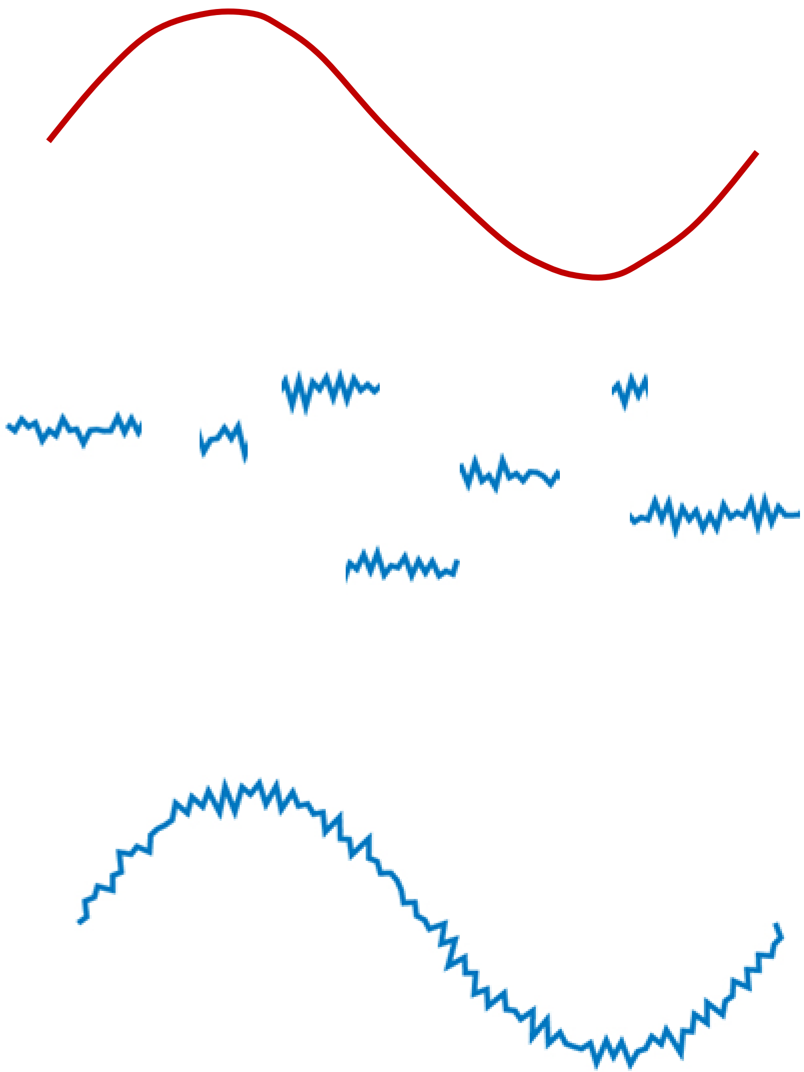
- white and pink noise

Noise



signal
noise
signal + noise
C1
R1
Input
C2
R2
R3
Q1
R4
R5
Q2
R6
C3
C4
R7
R8
R9
Q3
Q4
C5
R10
R11
R12
+24V
C7
Output
R14
R13
-24V
C6

Channel Noise

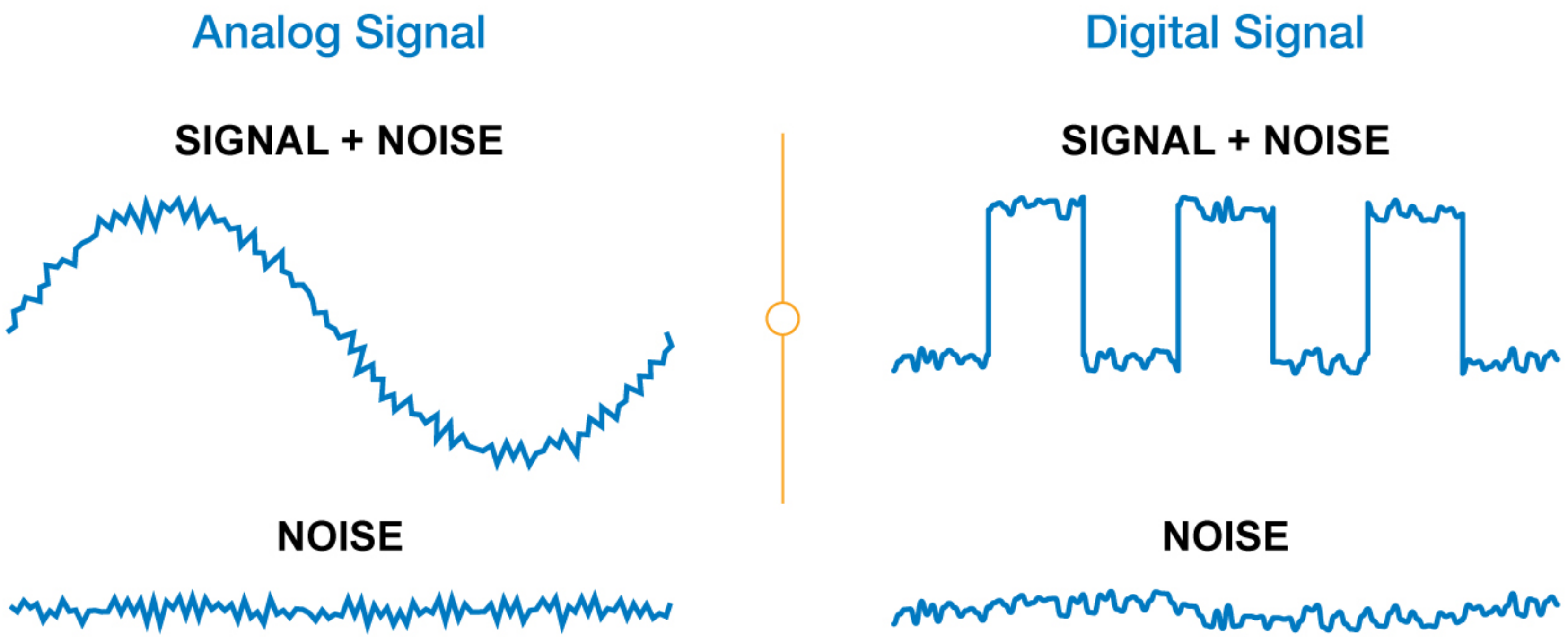


clean signal

channel noise

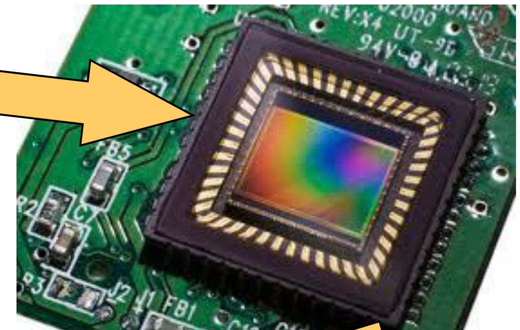
- Electromagnetic interference: This type of noise is caused by external **electromagnetic radiation** from various sources.
- **Additive noise** refers to any external noise that is added to the desired signal. The term "additive" means that the noise simply **adds** to the signal, resulting in distortion.

noise affected signal

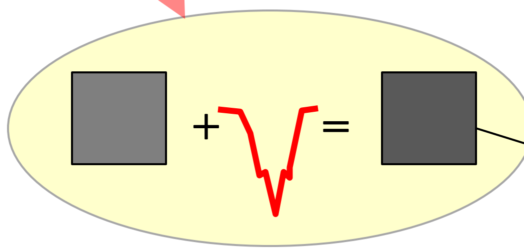

Analog Signal
SIGNAL + NOISE
NOISE
Digital Signal
SIGNAL + NOISE
NOISE

Sensor Noise

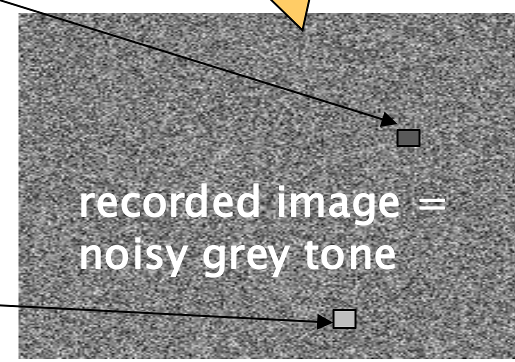
any *active* circuit (and many passive circuits) including camera sensors will produce noise ...

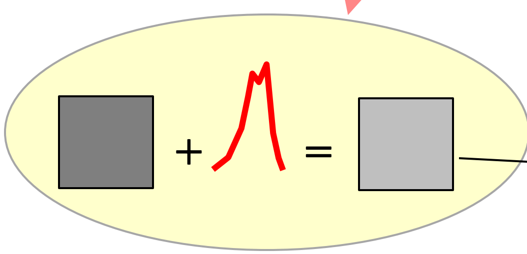


At the instant of image capture:



photocell charge plus negative noise spike = darker pixel





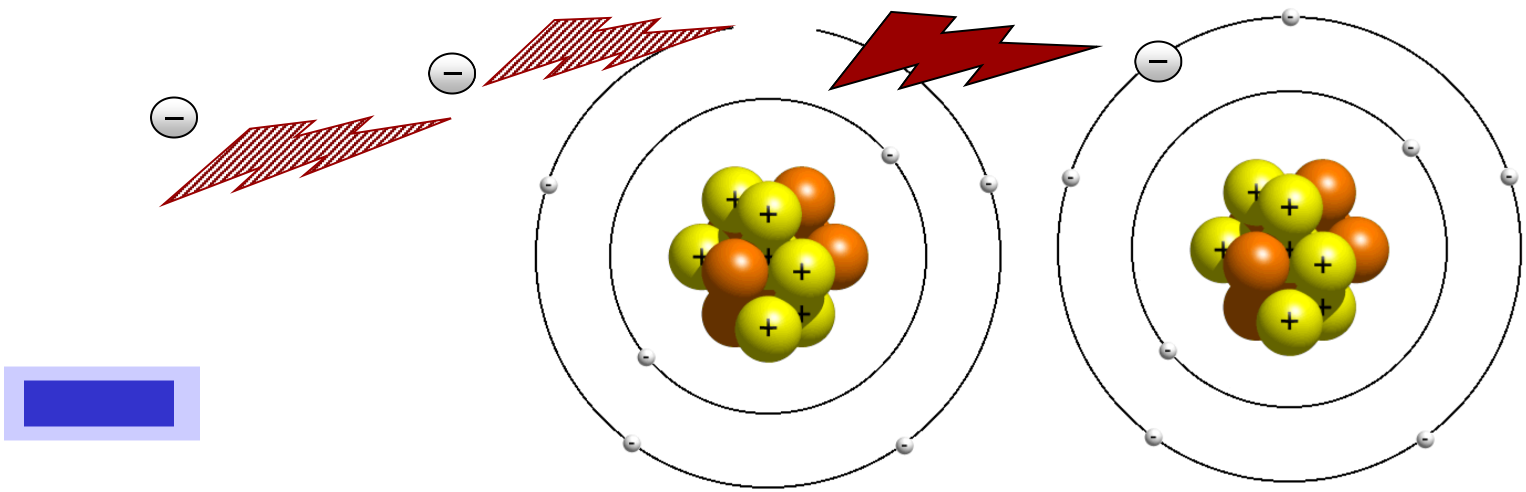
photocell charge plus positive noise spike = brighter pixel

Conduction

Movement of electrons in ONE direction gives rise to ELECTRIC current
... is really a movement of CHARGE

No field – random movement
... net current flow is zero

Conductors (metals, e.g. Copper) – have loose bindings to their outer electrons

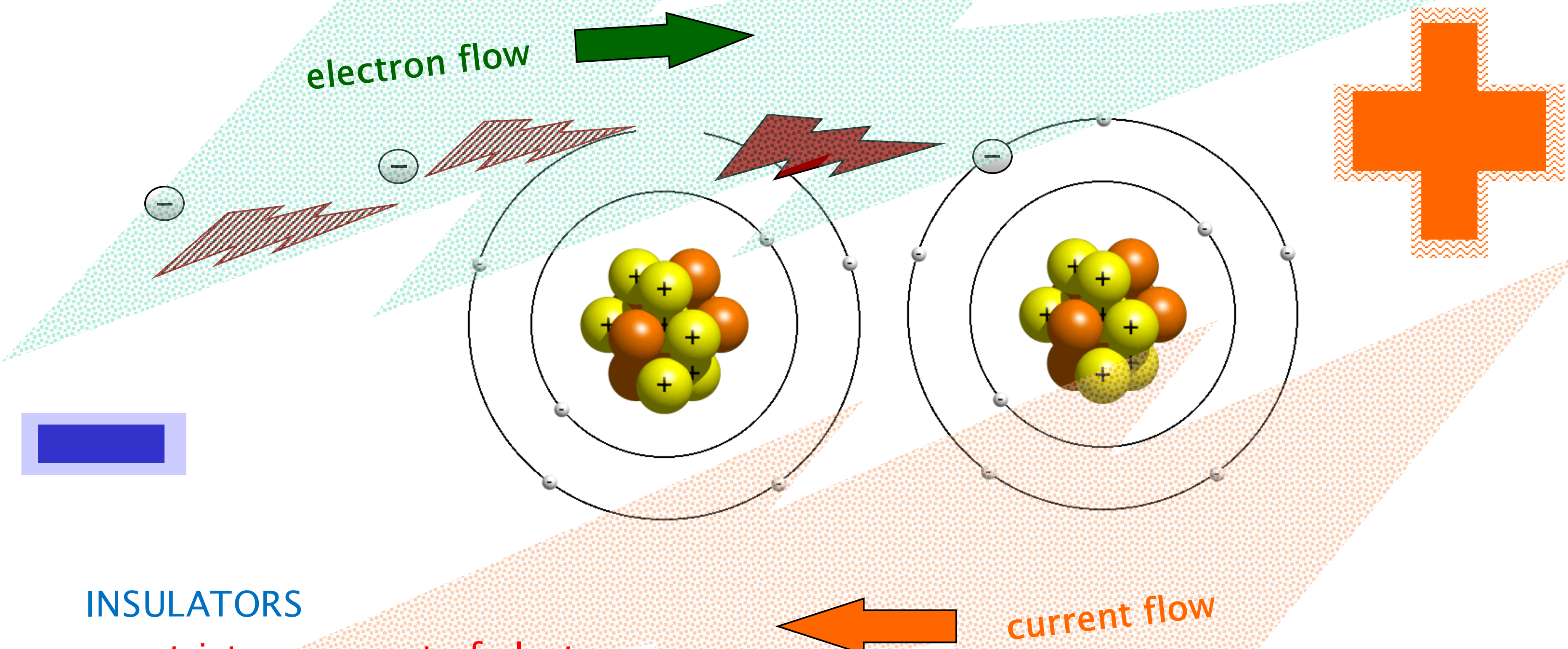


- electrons move **from the negative terminal** (lower potential) to the **positive terminal** (higher potential), which is opposite to the direction of conventional current.

Conductors

CONDUCTORS

- allow electrons to flow
- electrons easily freed and move through lattice of atoms when electric field is applied
- e.g. – gold, copper, tin, carbon

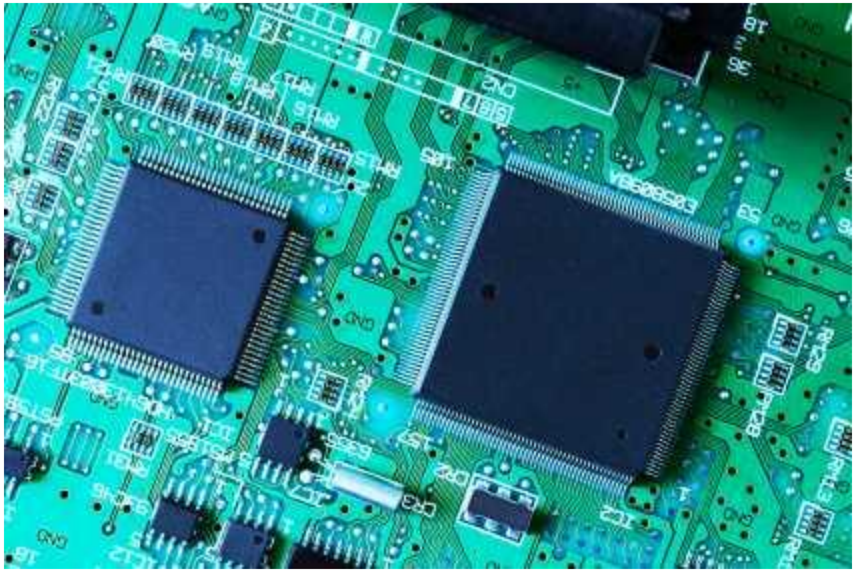


INSULATORS

- restrict movement of electrons
- electrons firmly bound to atom

Semiconductors

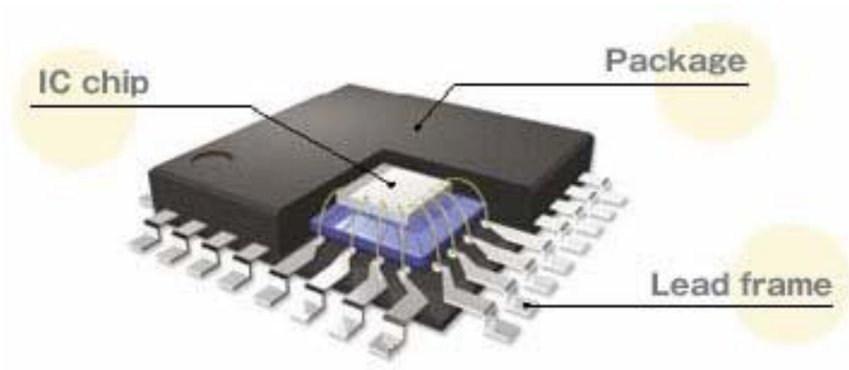
Operate as both/either conductors or insulators under controlled circumstances
… i.e. within a circuit



Applications

- diodes, transistors – CPUs (millions of transistors), sensors

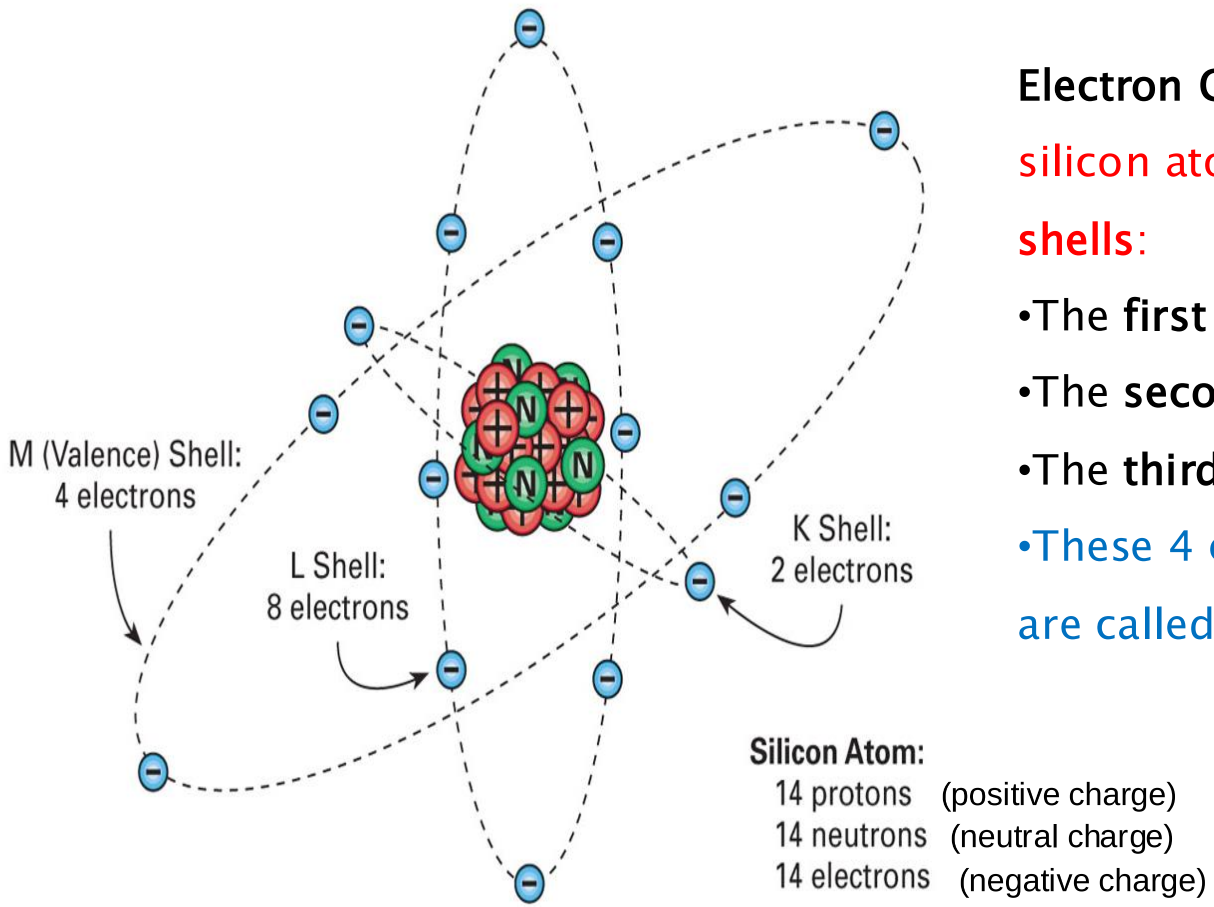
Some semiconductors are intrinsic
... but usually doped to assist the semiconductor action



Semiconductor materials

- Germanium (Ge), Silicon (Si) [intrinsic]
- Impurities (doping) are added to their crystal lattice structures to make n-type or p-type semiconductors

Semiconductor – Silicon atom

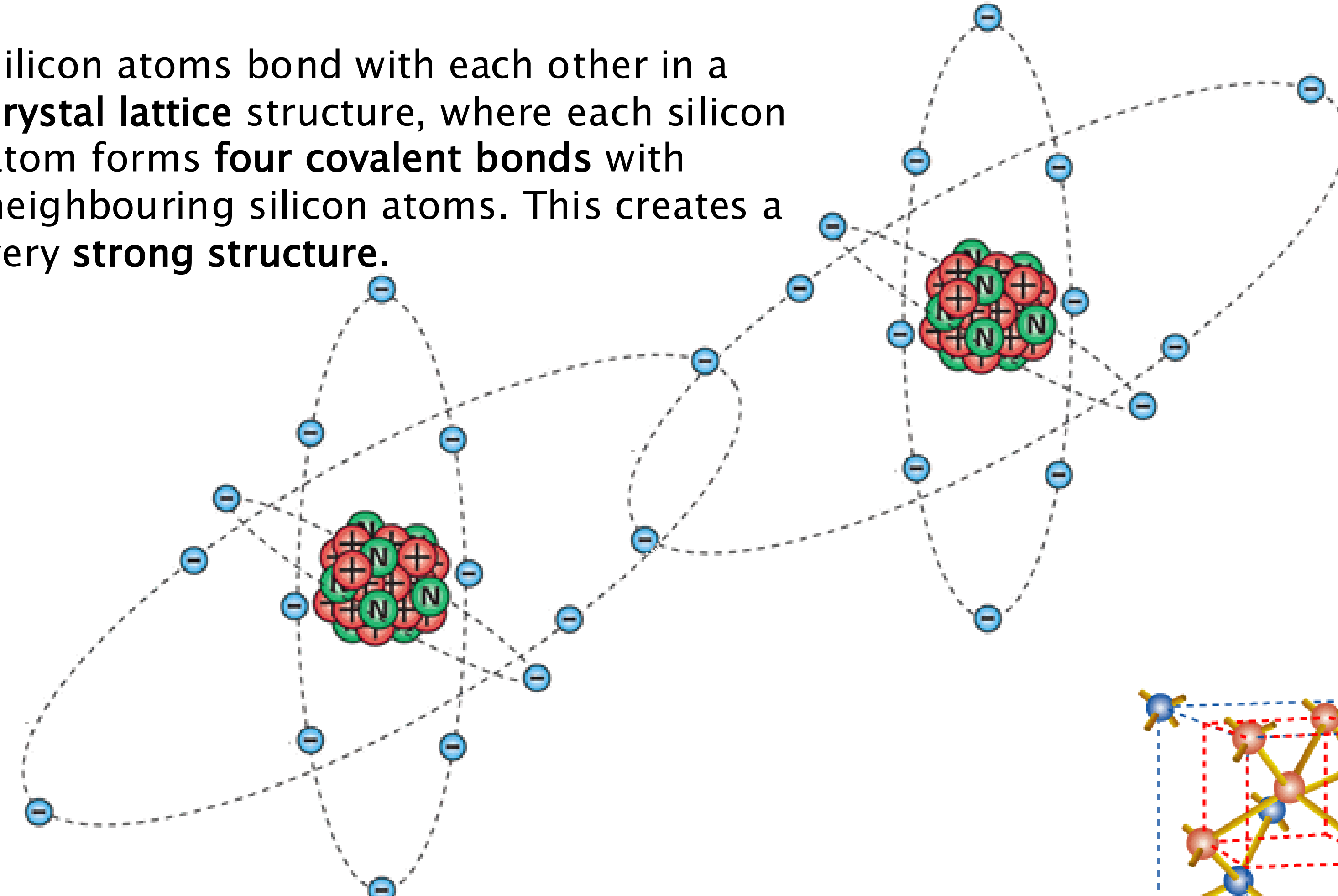


Electron Configuration: The electrons in a silicon atom are arranged in **three electron shells**:

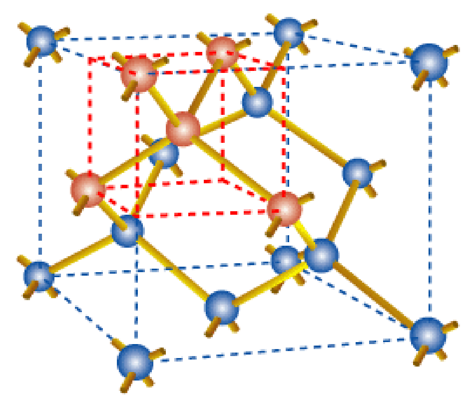
- The **first shell** has 2 electrons (K shell).
- The **second shell** has 8 electrons (L shell).
- The **third shell** has 4 electrons (M shell).
- These 4 electrons in the outermost shell are called **valence electrons**.

Semiconductor – Silicon atom

Silicon atoms bond with each other in a **crystal lattice** structure, where each silicon atom forms **four covalent bonds** with neighbouring silicon atoms. This creates a very **strong structure.**



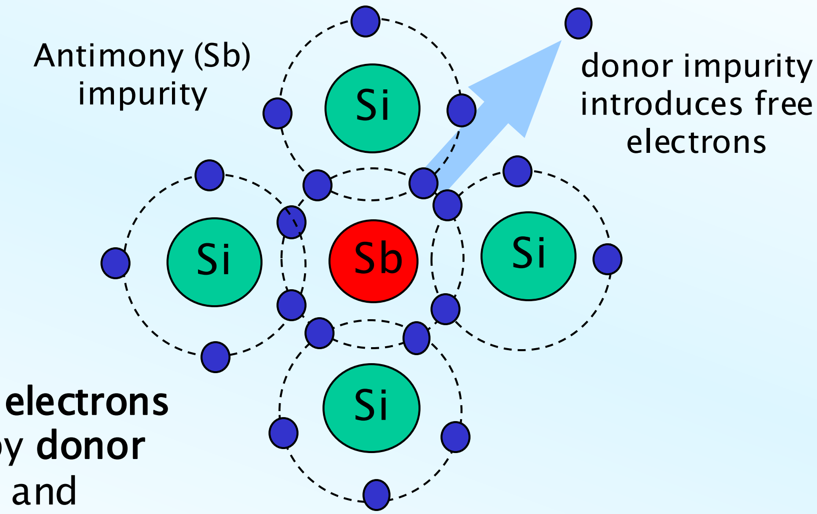
… into a lattice structure



Semiconductors (simplified 2D view of the lattice)

for n-type (negative type)

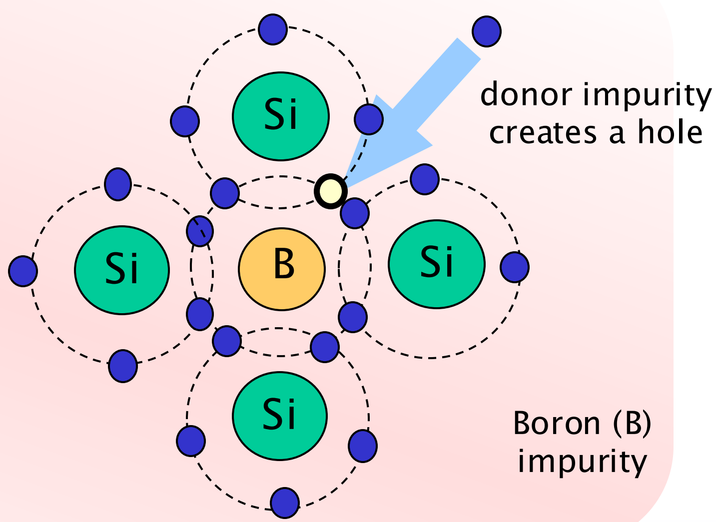
- "Donor" impurities
  - Antimony
  - Arsenic
  - Phosphorus
- introduces "free" electrons



•**N-type semiconductors** have **extra electrons** (negative charge carriers) donated by **donor impurities** like phosphorus, arsenic, and antimony.

for p-type

- "Acceptor" impurities
  - Boron
  - Aluminium
  - Gallium
- introduces "holes" – room for free electrons

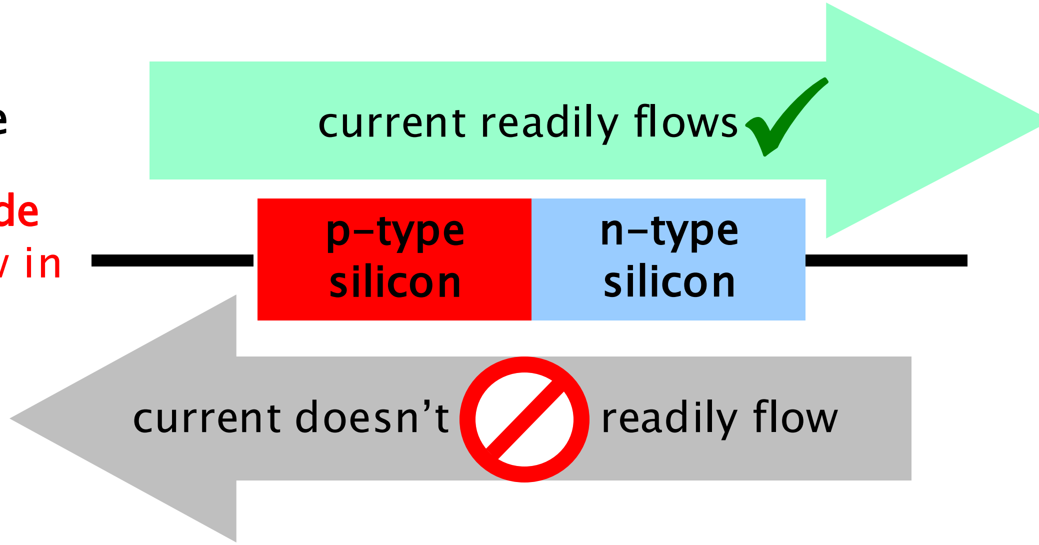


Conversely, introducing an atom like **boron** with 3 valence electrons creates a **hole** (a missing electron) in the crystal structure. This creates a **p-type semiconductor**, where the missing electron behaves as a **positive charge carrier** (hole).

P–N Junctions

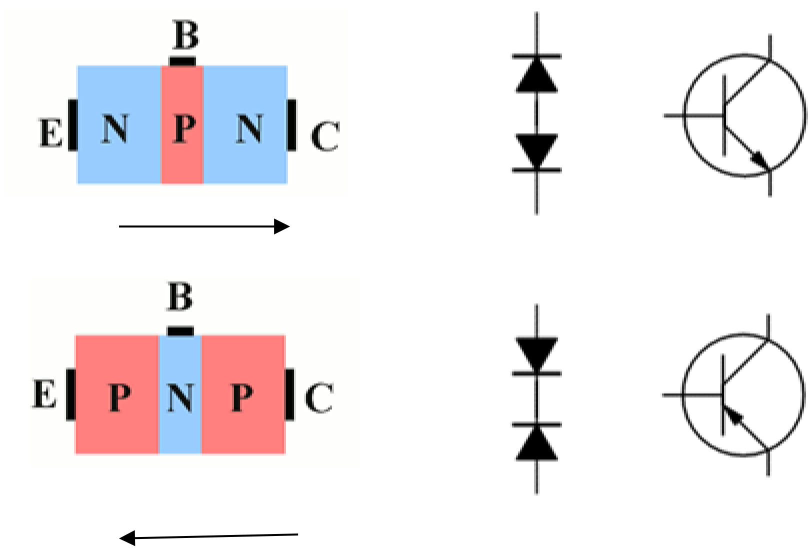
Semiconductor Diode

A **semiconductor diode** allows current to flow in one direction only.



Transistors: a semiconductor device used to amplify or switch electronic signals.

- here – Bipolar junction
  - top – NPN transistor
  - bottom – PNP transistor

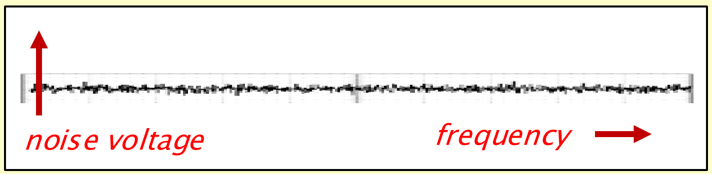


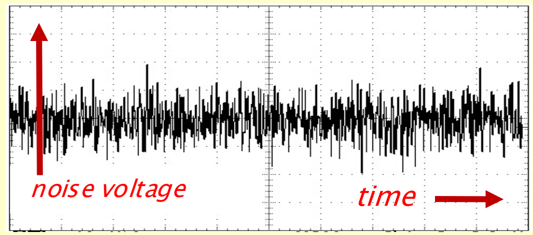
- other types – FET (field effect):
- A **FET** is another type of transistor, but it uses an **electric field** to control the flow of current.

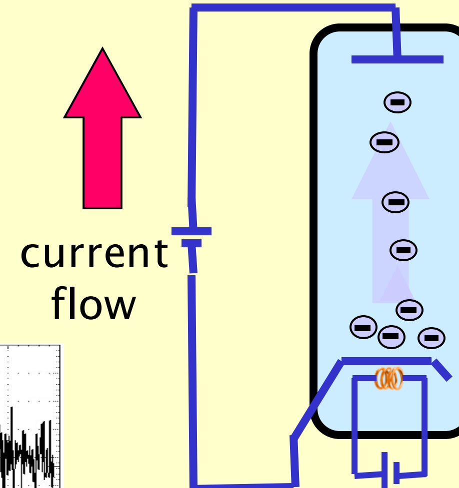
Noise – Causes

SHOT NOISE

- Random noise
- due to electron flow – i.e. discrete events
  … rather than a smooth flow
- also photons in an optical transmission
- significant problem with low signal levels
- generally across all frequencies







THERMAL NOISE (JOHNSON-NYQUIST NOISE)

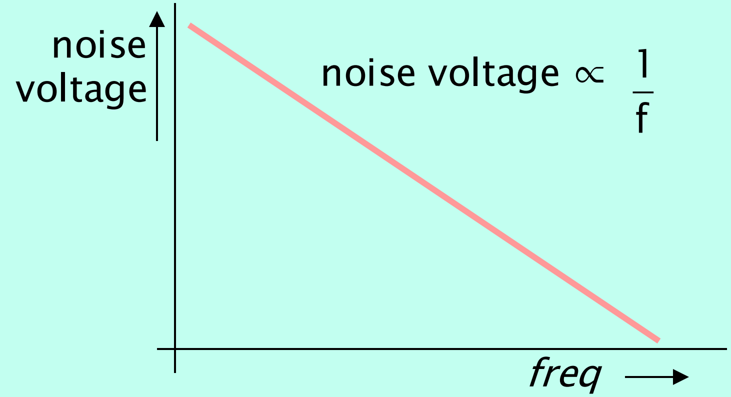
- thermal effects agitating the electrons in a circuit
- increases with temperature
  - Occurs in all resistors, capacitors, and other passive components.

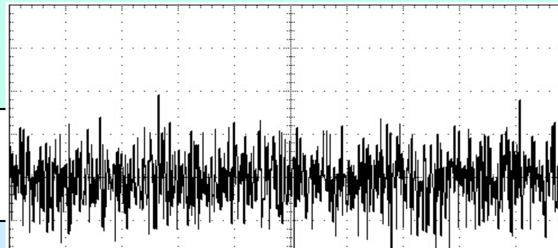
Power = $4kTRB$,
where:
- k = Boltzmann constant,
- T = temperature in Kelvin,
- R = resistance,
- B = bandwidth.

Noise – Causes

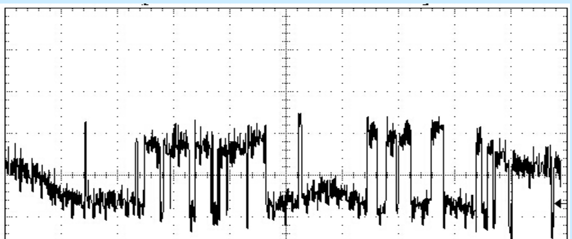
FLICKER NOISE

- **Cause**: Imperfections in materials and surfaces, often due to traps or defects.
- occurs in low frequency range
  - dropping off towards higher freqs with a 1/f characteristic (like Pink Noise)




Random white noise (e.g. shot noise) *(in time and freq)*

time →


Impulse (popcorn) noise

BURST NOISE; IMPULSE NOISE *("popcorn noise")*

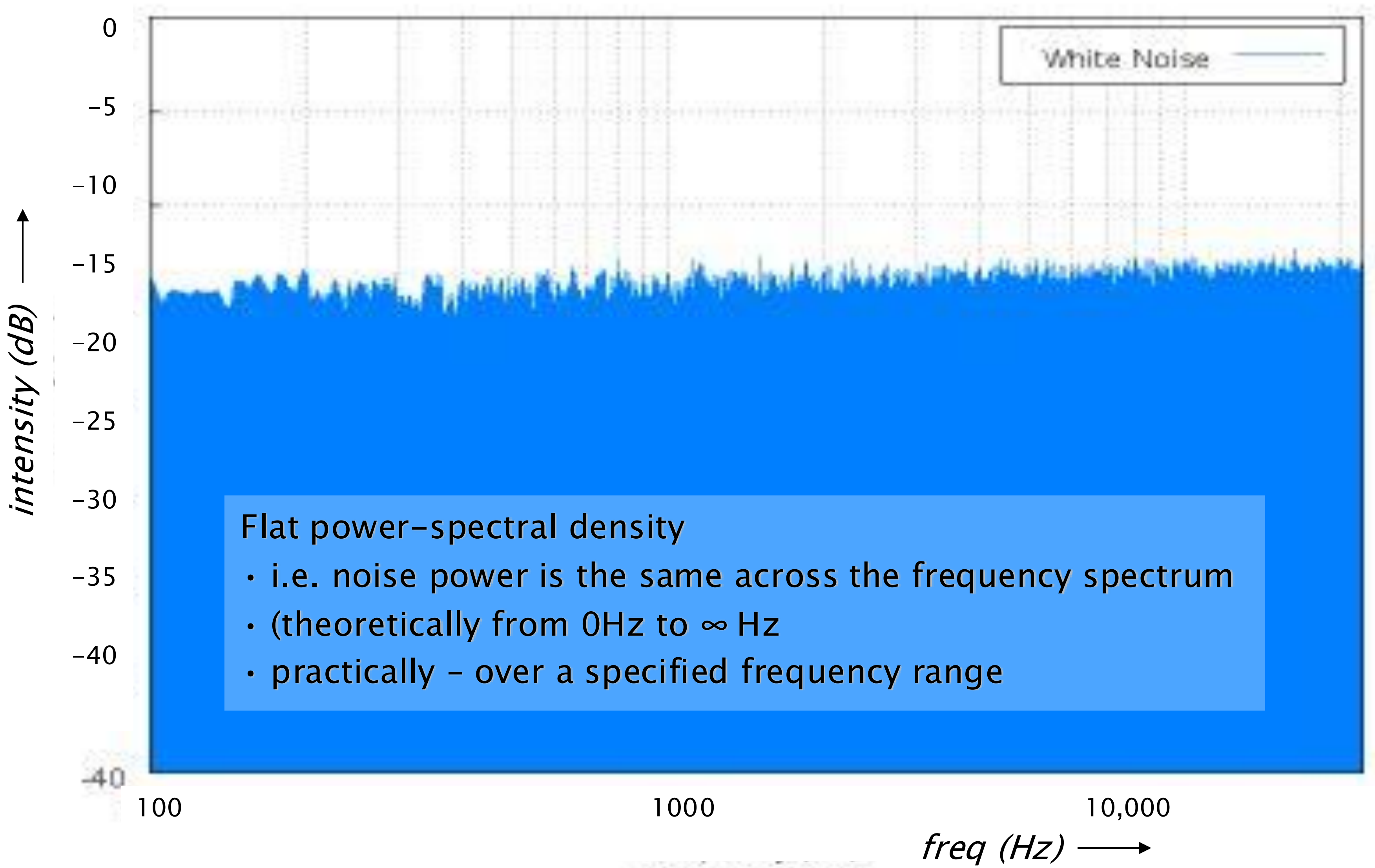
- noise spikes in the lower audio range
- different distinct levels of noise
- in audio circuits amplified into speakers, sounds like popcorn popping
- occurs in semiconductors
- due to imperfections in the semiconductor material

Noise – Other Causes

Several other causes of noise:

- nearby interference sources
- contact noise – oxidation on contacts
- avalanche noise – caused at a semiconductor junction
- partition noise – electrons flowing between a number of electrodes

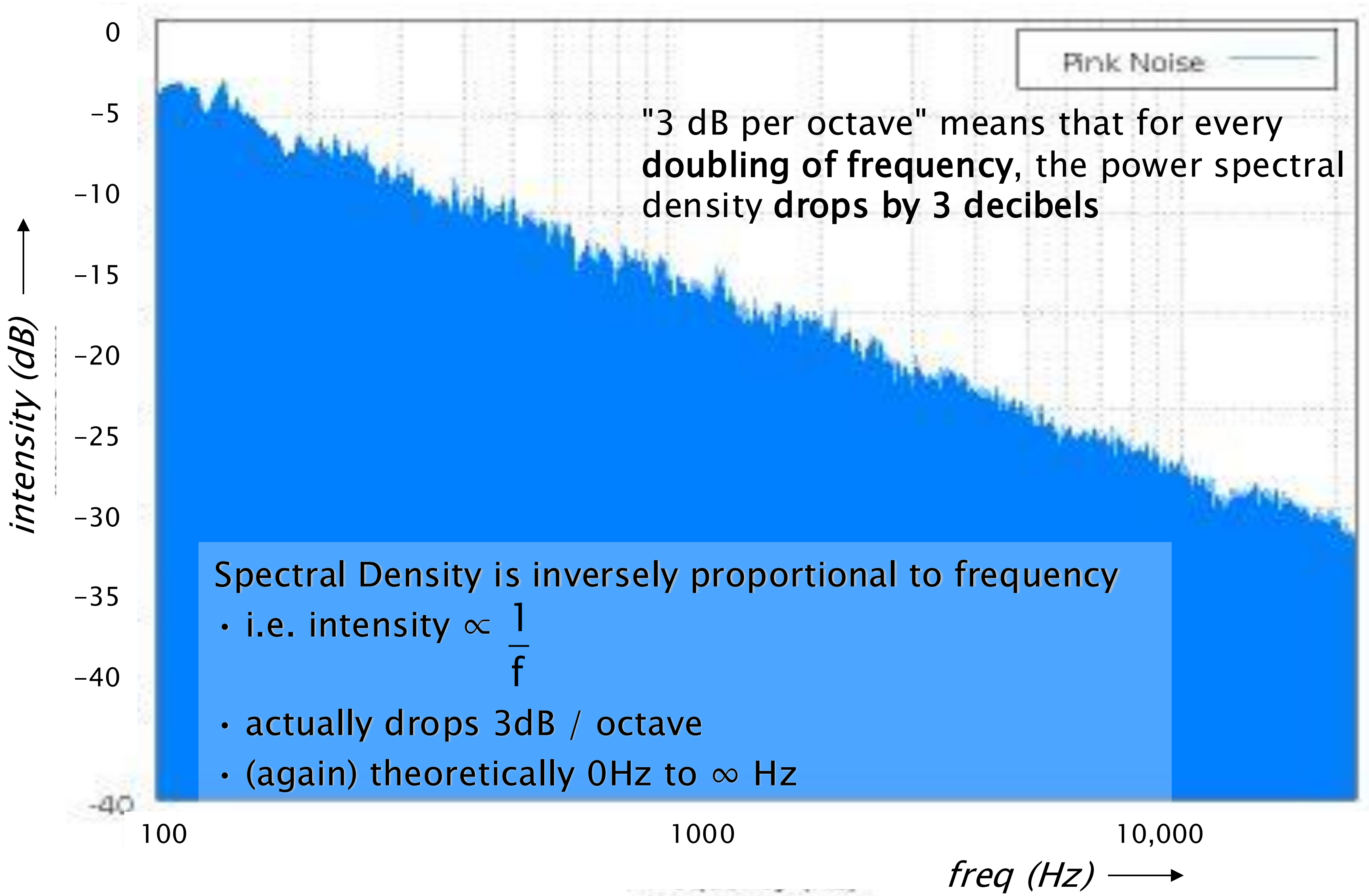
White Noise



Flat power-spectral density

- i.e. noise power is the same across the frequency spectrum
- (theoretically from 0Hz to ∞ Hz
- practically – over a specified frequency range

Pink Noise



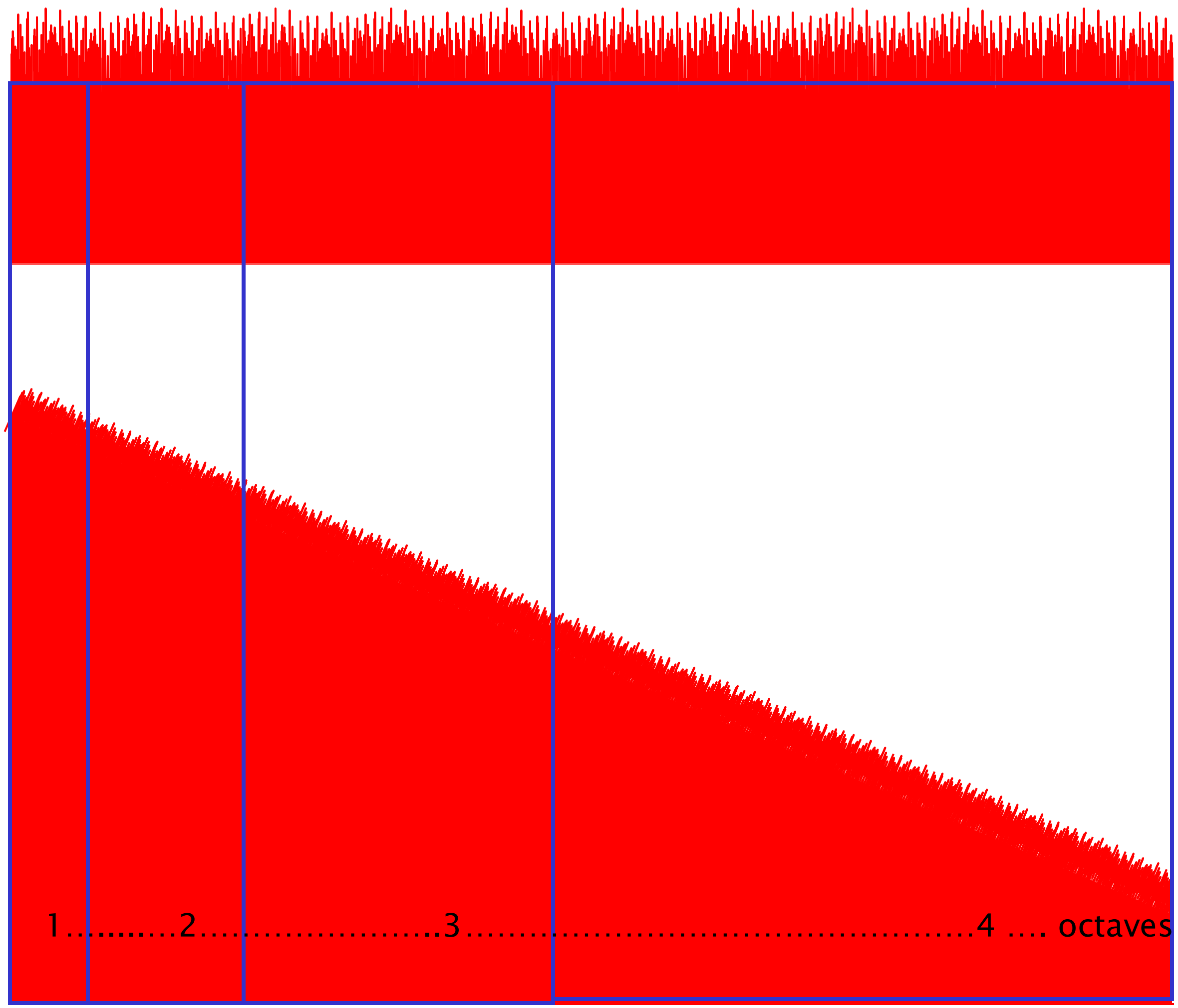
"3 dB per octave" means that for every **doubling of frequency**, the power spectral density **drops by 3 decibels**

Spectral Density is inversely proportional to frequency

- i.e. intensity $\propto \frac{1}{f}$
- actually drops 3dB / octave
- (again) theoretically 0Hz to ∞ Hz

Noise – White, Pink

Depending on frequency spectrum, i.e. the amplitude of the noise voltages across the freq range



WHITE NOISE

- has an increased intensity of 3dB for each octave increase (10log10(2))
- because each octave is twice the frequency range of the previous octave

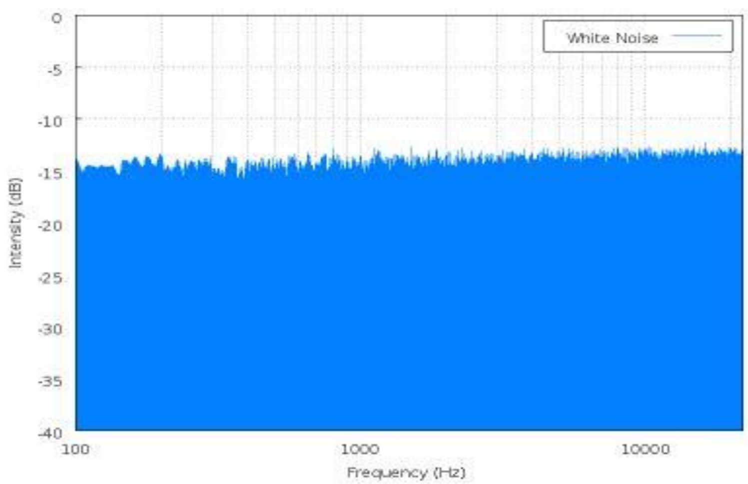
PINK NOISE

- power density drops 3dB intensity per octave
- therefore amount of power is the same in octave ranges:
- e.g. power in range 100 – 200 Hz
  = power in range 1KHz – 2 KHz
- therefore is useful for audio measurements

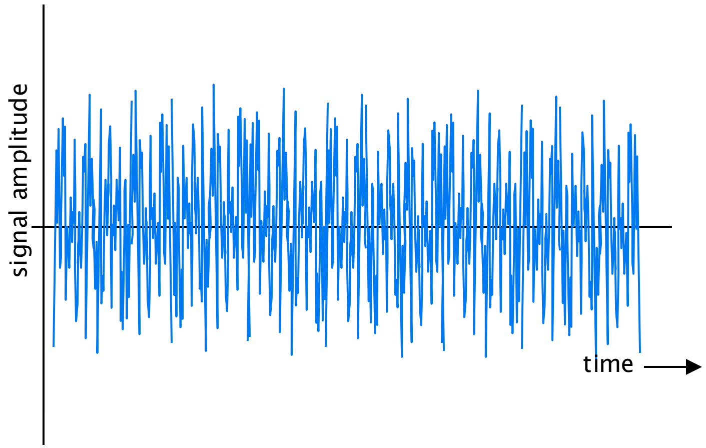
Frequency and Time Domains

This response (from the previous slide) is of intensity over the frequency range … i.e. observed in the frequency domain

Observing the noise-producing circuit over a period of time, we see only the changes in signal amplitude (i.e. intensity)



Take FFT



This is a snapshot in time of the frequency domain

Use Cases of FFT on Noise:
- To analyse **noise floor** in a system.
- To measure **Spectral Density** of noise (e.g., in dBm/Hz).
- To design filters that **remove certain noise frequencies.**
- To evaluate **SNR** (signal-to-noise ratio).

Noise is Probabilistic

Because it's generated by **random events** at the atomic or electronic level — such as:

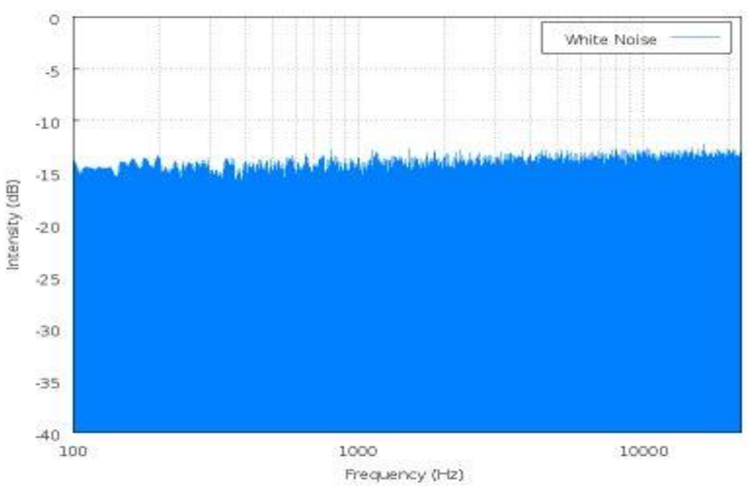
•**Thermal agitation** of electrons (→ *thermal noise*)

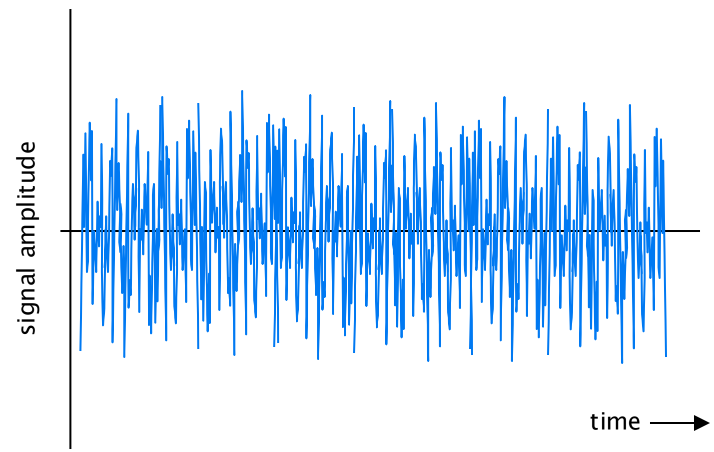
•**Quantum fluctuations** during current flow (→ *shot noise*)

•**Impurities** or defects in semiconductors (→ *flicker or burst noise*)

•**External EM sources** (→ *interference noise*)

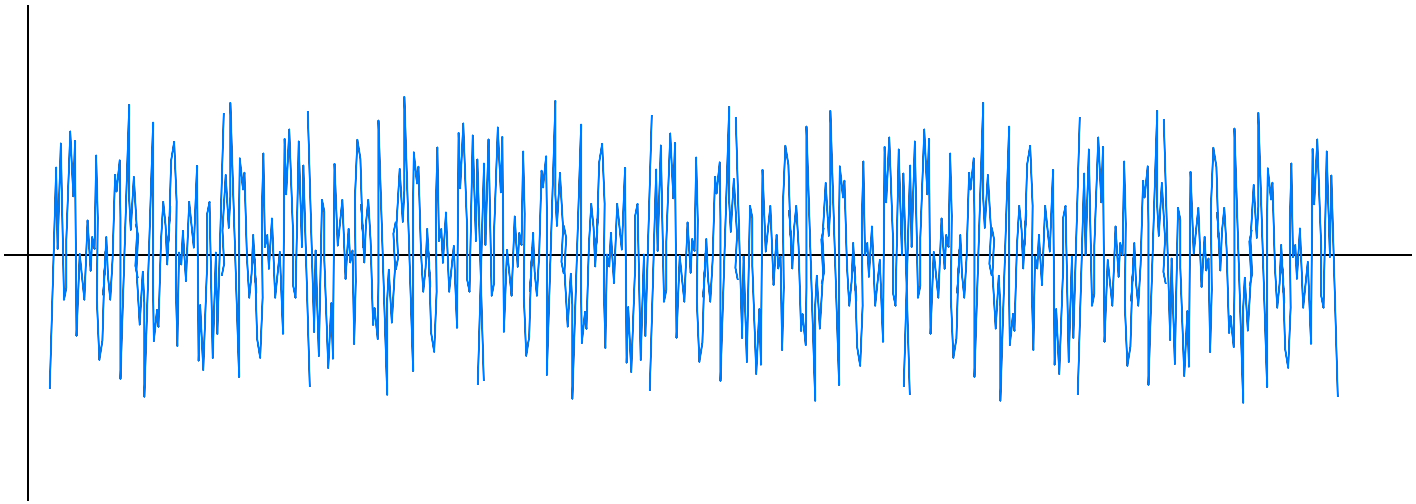
These events can’t be predicted exactly — only their **statistical behaviour** can be modelled.





… or that noise voltage amplitude = any specific value at any specific time

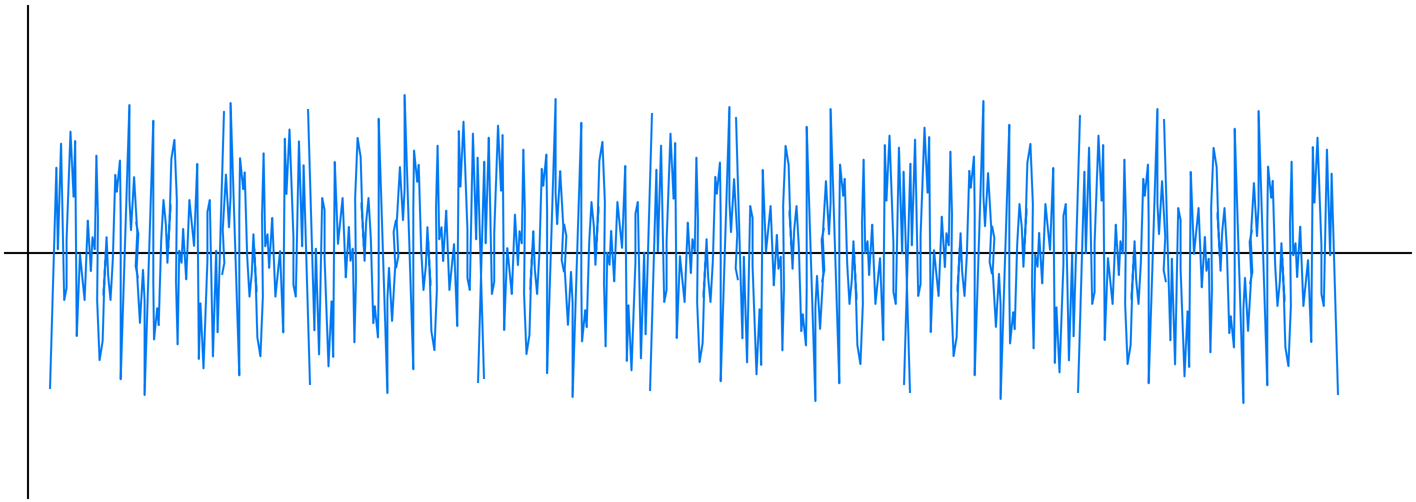
Noise is Probabilistic



Noise is typically treated as a **random signal**, with properties defined in **statistical** terms like:

Property	Description
Mean (μ)	Average value (usually 0 for AC noise)
Variance (σ^2)	Power of the noise
PDF (prob. density function)	e.g. Gaussian (normal), uniform, etc.
Autocorrelation	Measures similarity at time shifts
Power Spectral Density (PSD)	Shows how noise power is distributed over frequency

Noise is Probabilistic; Measuring Noise



As noise is random, it's mean value will be (usually) zero

∴ use mean square values for measurements of the noise power

- The effective noise power is measured in root mean square (RMS)
- Root Mean Square (RMS): RMS$=\sqrt{\langle x(t)^2\rangle}$,

  $\langle x(t)^2\rangle$ =mean of the square of the signal over time
- This gives an **effective value** — a single value representing the "equivalent DC power level" the noise would produce.

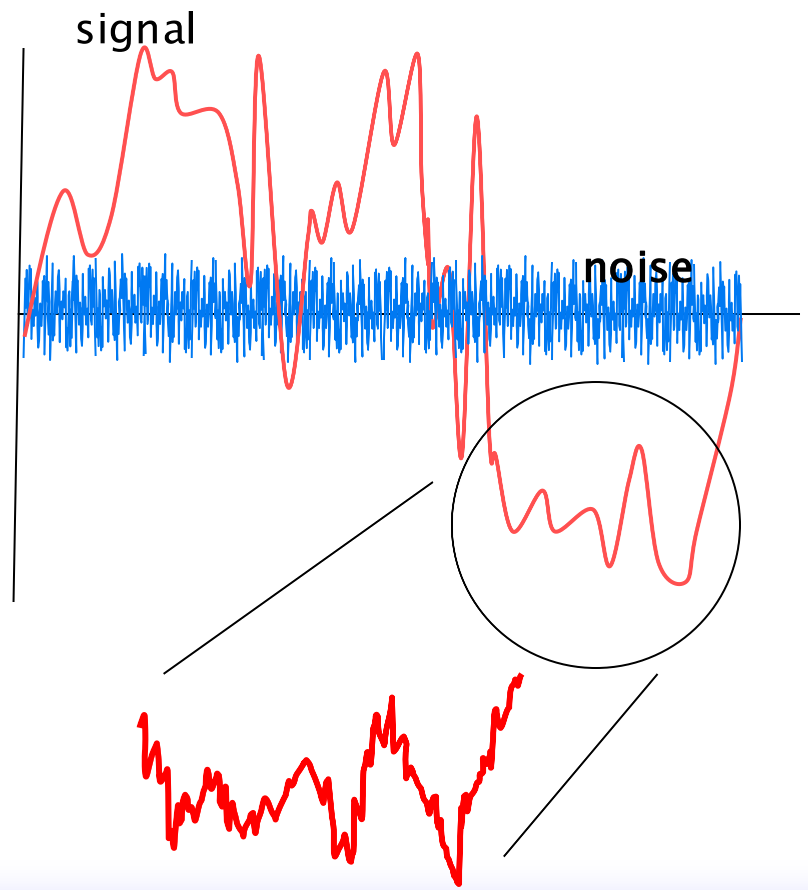
Measuring Noise: Signal to Noise Ratio (SNR)

Comparison of

- the power of the required signal
- against the power of the (usually) unwanted noise
- measured in decibels (dB)

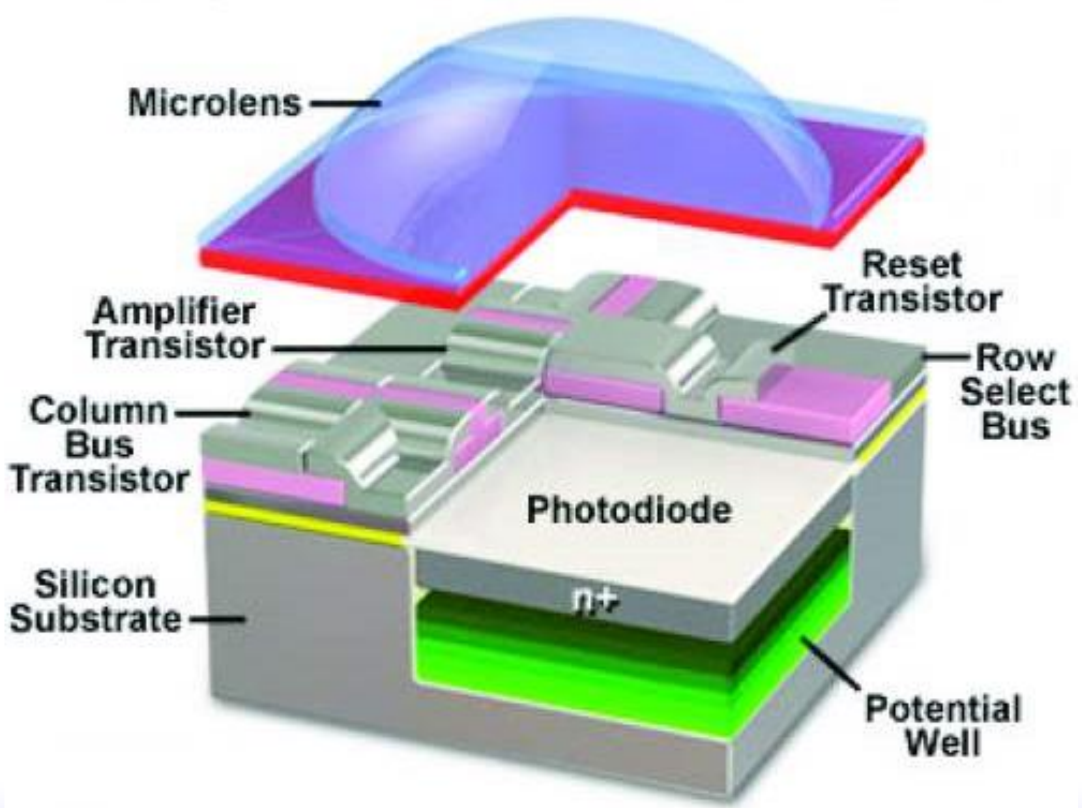
$$\text{SNR (dB)} = 10\log_{10}\frac{P_{signal}}{P_{noise}}$$

$$= 20\log_{10}\frac{\text{voltage}_{signal}}{\text{voltage}_{noise}}$$



Photoelectric effect

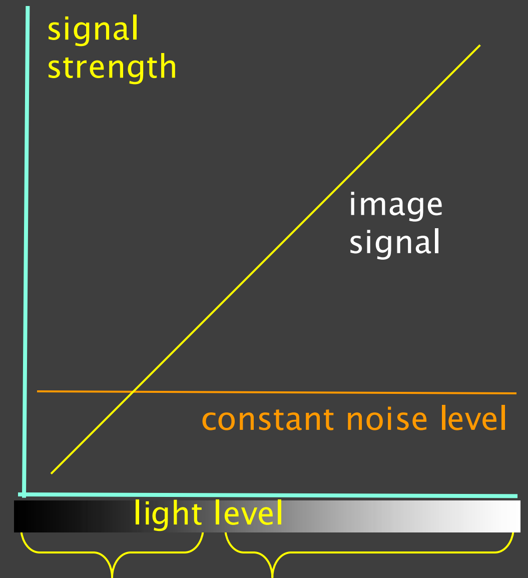
- photon (light particle) with energy *E* strikes the diode
- excites an electron
- creating a free electron and a (positively charged electron hole)
- The separation of charges creates a **photocurrent**, which flows through the external circuit.



Dynamic Range – Camera Sensor, Signal to Noise

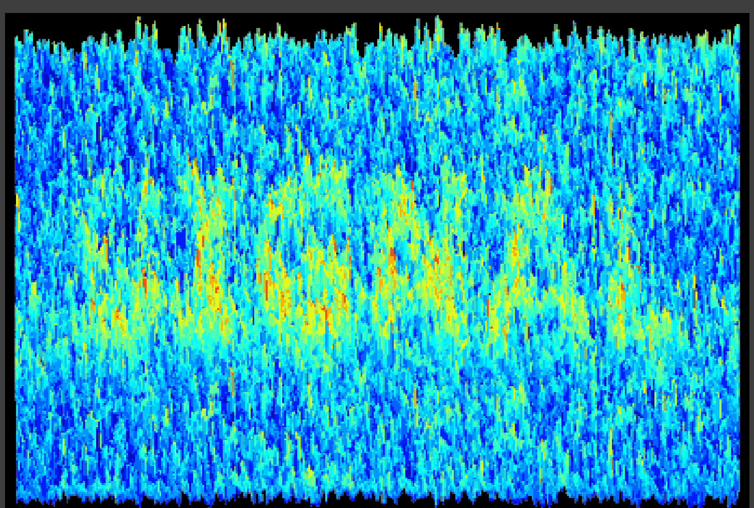
- signal: the light which hits the camera sensor
- noise is unavoidable
- can be small relative to the signal
- appears to be nonexistent

More light = more photons = more electron-hole pairs = more current.

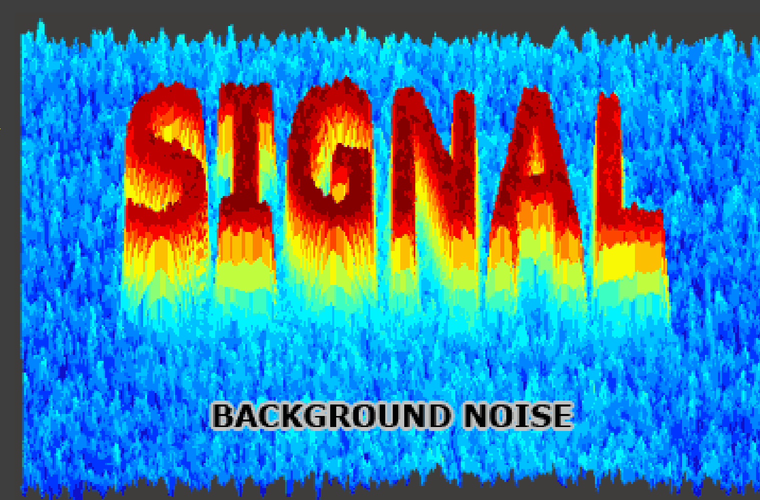


Signal to noise ratio (SNR)

- compare relative amounts of signal and noise
- used for any electronic system



noise low ratio = much noise



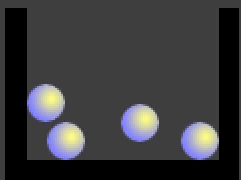
high ratios = very little visible noise

http://www.cambridgeincolour.com/

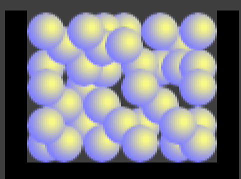
Dynamic Range – Camera Sensor, Photosites

Camera sensors – Each photosite works by capturing photons of light and converting them into an electrical charge (through the photoelectric effect).

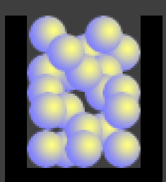
- Figures below – accumulation of charge is like filling buckets with balls (or water … whatever …); actually electrons



Black level – at low level, is *limited by the signal required to overcome the sensor noise*.



White is limited by the highest level of charge accommodated in the photosite



… Depends on the capacity of the photosite. A lower capacity element may accommodate less light before it is saturated

… smaller sensors usually have lower capacities

… compact cameras, smaller sensors

- Contrast ratio is the ratio between lowest level and highest level
- lower ISO levels usually have bigger dynamic ranges, since there is less noise at low ISO black levels
- Camera dynamic range usually measured in f-stops
- e.g. contrast ratio of $1024 \equiv 2^{10}$, i.e. 10 stops

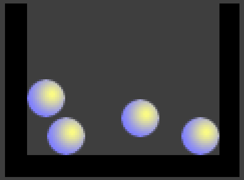
camera can record **1024 levels of brightness.**

ISO (International Organization for Standardization) in photography refers to the **sensitivity of a camera’s sensor** to light.

(state

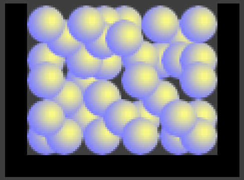
Dynamic Range – Camera Sensor, Photosites

$$\text{Dynamic Range} = \frac{\text{Full Well Capacity}}{\text{Camera Noise}}$$

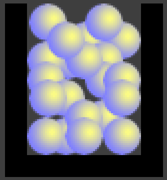


$$= 20\log_{10}\frac{\text{Full Well Capacity (stated as number of electrons)}}{\text{Camera Noise (stated as number of electrons)}}$$

… … stated in decibels (dB)



Dynamic range improves if the full-well capacity is higher
… and the camera noise is lower



Parameters that affect the dynamic range:

- Pixel size (Full-well capacity): Larger pixel size typically means a higher full-well capacity because the pixel can accumulate more charge
- Temperature (Dark noise): generated by the camera sensor itself when there is no light
- Readout rate (Readout noise): refers to the noise that occurs when the sensor data is transferred to the processor.

Noise Factor, F_n

specifies how much *additional noise* the device will *contribute* to the noise already from the source.

The noise factor of a system is

the ratio of output noise power (P_{no})

to input noise power (P_{ni}):

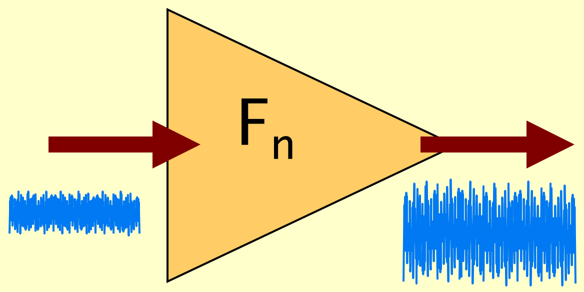
$$\boxed{F_n = \frac{P_{noise\_out}}{P_{noise\_in}}}$$

Ideally $F_n = 1$

(because noise out would = noise in)

- Noise sources are subject to temperature (thermal noise)

  ∴ needs to be measured at a standard temperature = T_0 = 20°C; = 293K



$$F_n = \frac{P_{noise\_out}}{P_{noise\_in}}$$

Noise Figure, NF	Noise Figure NF is the Noise Factor in dB $NF = 10 \text{ Log}_{10} F_n$

Signal Distortion

Frequency Distortion

- Frequency distortion occurs when certain frequencies are boosted or attenuated unevenly across the audio spectrum.

Phase Distortion

- When a signal passes through a system (like an amplifier, filter, or equalizer), the frequency components of the signal can experience different amounts of delay.
- This delay causes a shift in the phase of each frequency.

Total Harmonic Distortion (THD)

- Harmonics are multiples of the fundamental frequency (the main frequency of the signal).
- **THD is the ratio of the power of the harmonics to the power of the original signal.**
- Measured using Fourier Analysis

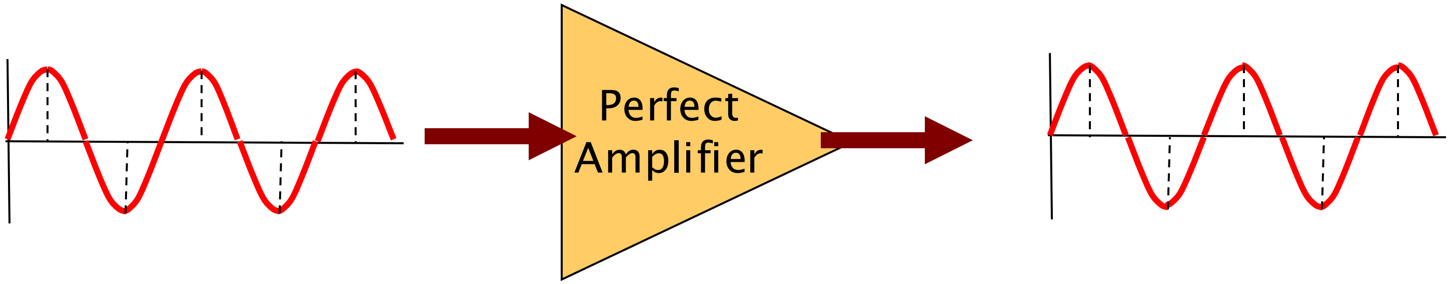
Intermodulation (IM) Distortion

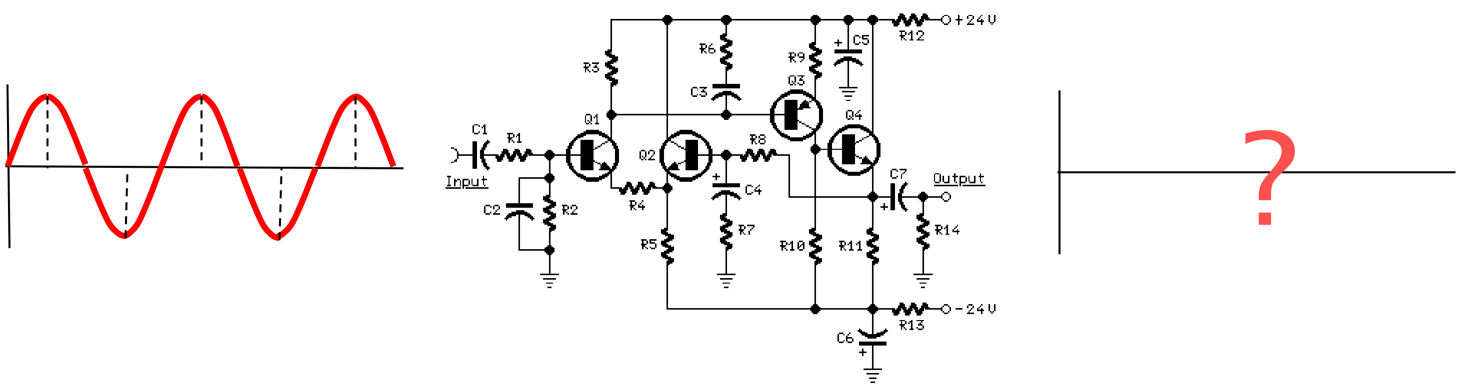
- IMD occurs when two or more signals interact in a non-linear system, creating unwanted additional frequencies.
- IMD affects audio quality by introducing new frequencies that interfere with the original signals.

Transient Intermodulation (TIM)

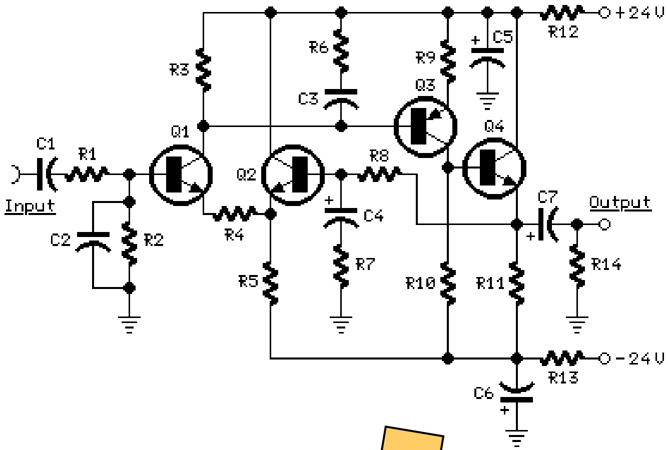
- TIM occurs when an amplifier cannot react quickly enough to sharp, high-frequency transients in the audio signal.
- It results in the generation of unwanted distortion products.

Signal Distortion





Signal Distortion

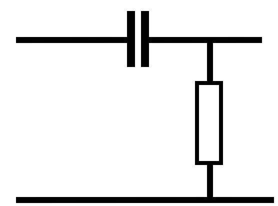


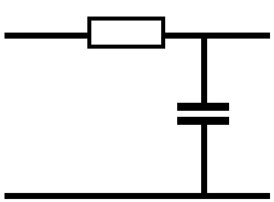
Unintended distortion

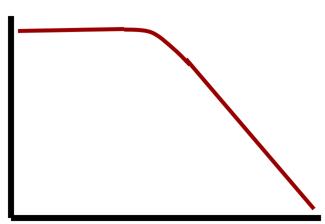
Frequency Response Limitations:

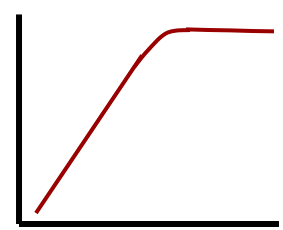
- Devices like amplifiers, filters, or transmission lines often have a limited bandwidth.

Capacitance between the **transistor casing** and **earth** can introduce filtering effects

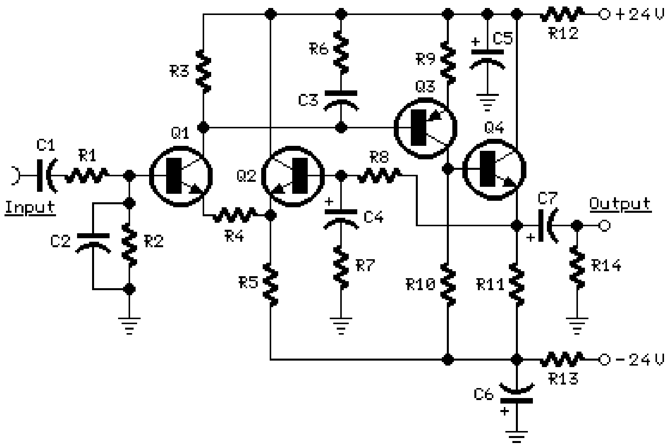






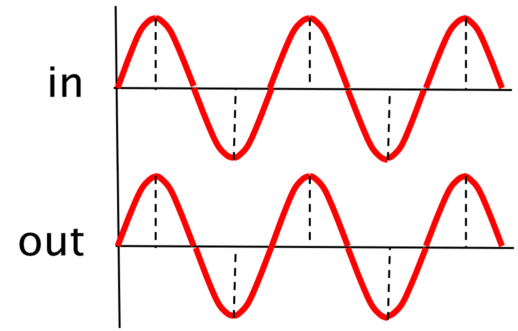


Signal Distortion



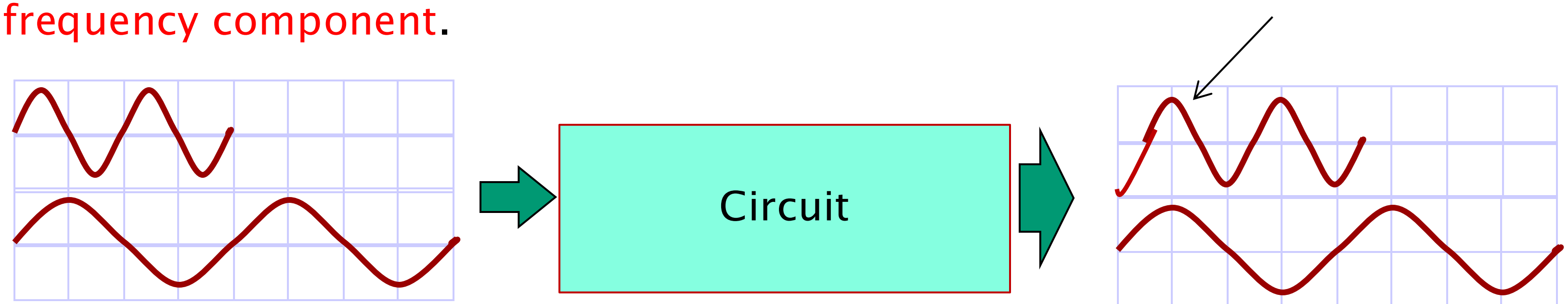
- If a system's frequency response is **flat**, all frequencies are amplified or attenuated by the same amount, and there is <u>no phase shift</u>.

Lower freqs: (assume) in phase

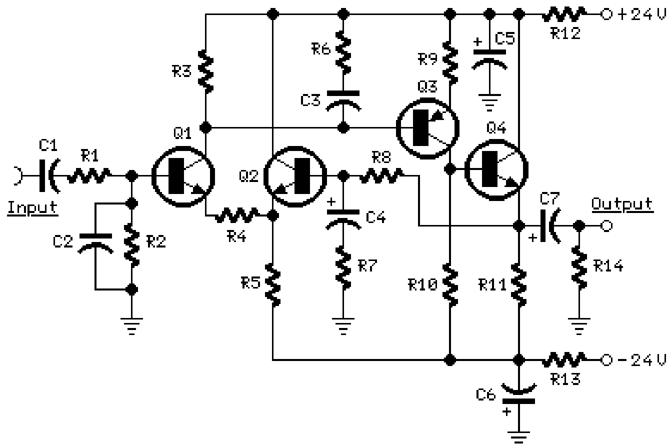


- If the frequency response is **uneven**, certain frequencies are amplified more or less than others. This unequal response leads to **<u>different phase shifts</u>** for each frequency component.

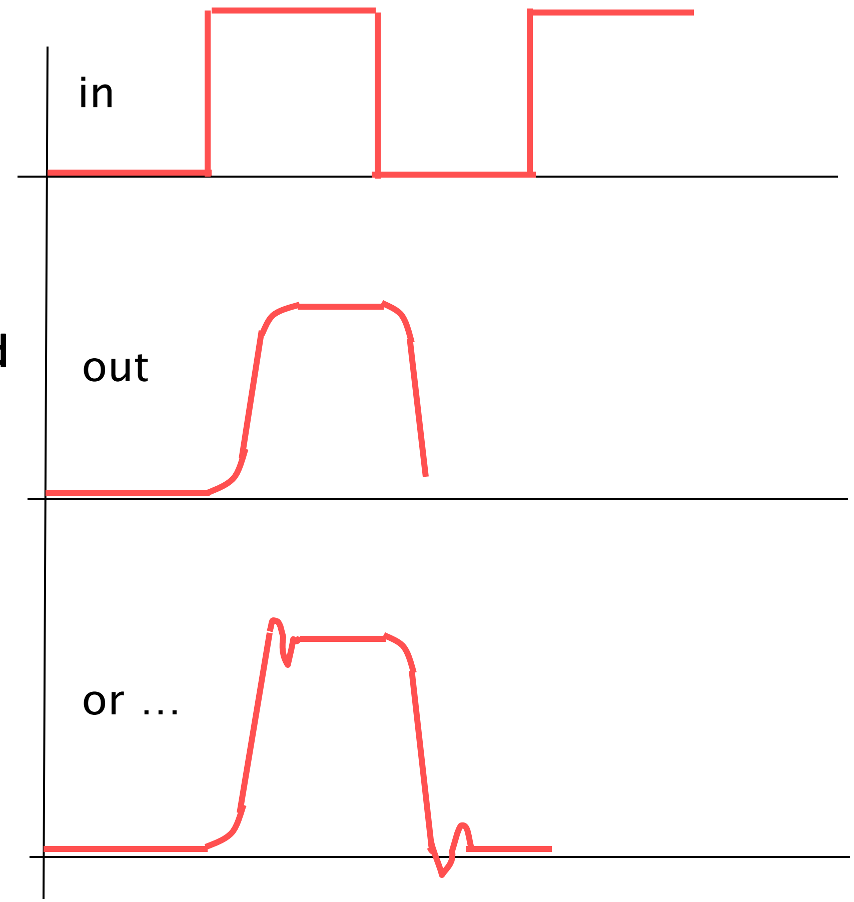
different frequencies have different phase responses



Signal Distortion



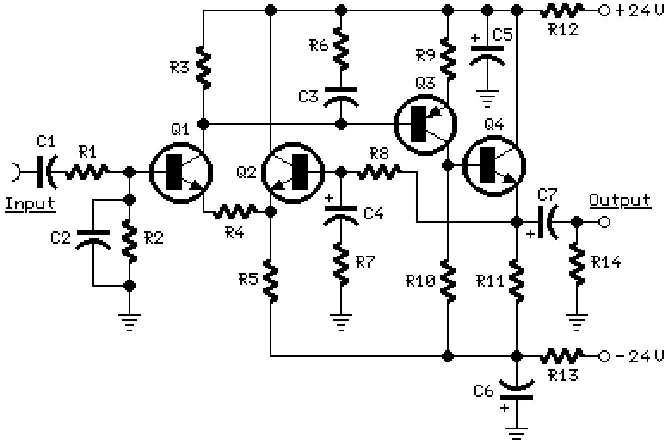
Transient response – circuit can't react immediately to changes in signal

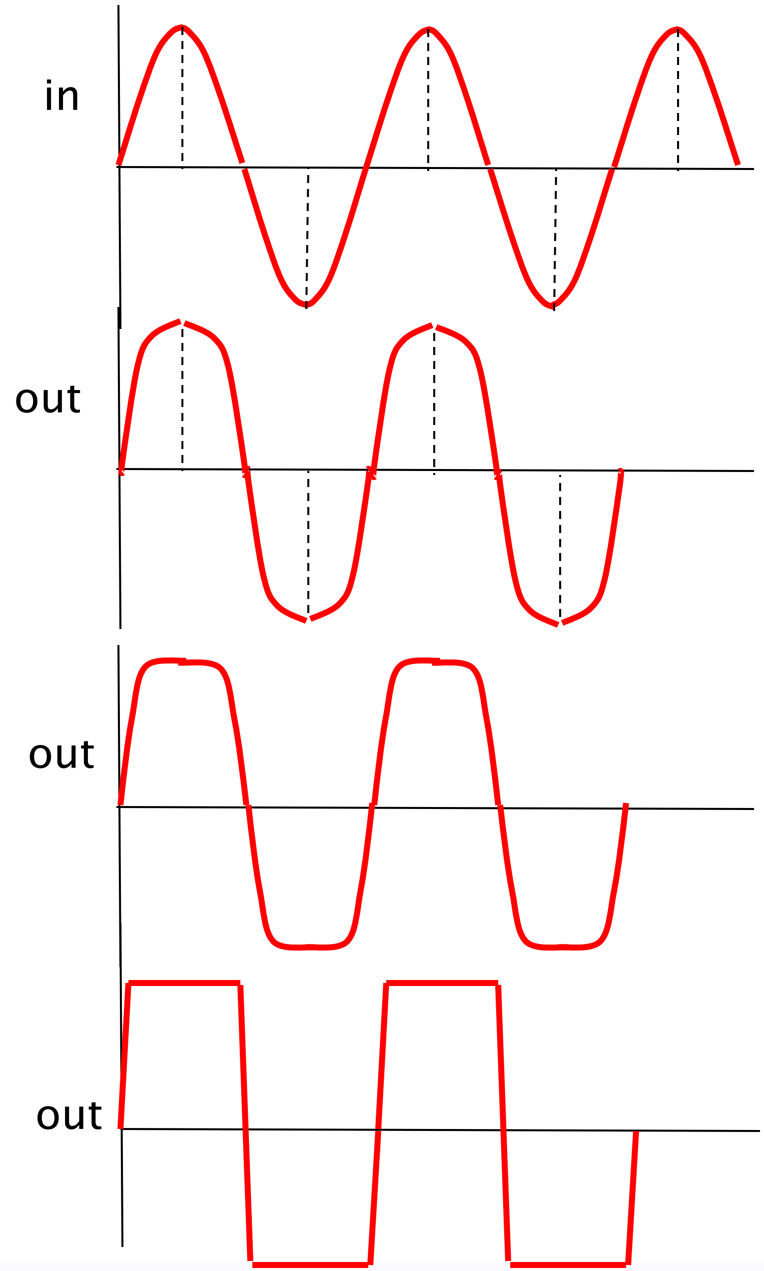


Practical systems are band limited:circuit produces rounding of edges due to **limited frequency response** (due to filtering) [caused by a **low-pass filter**, where higher frequencies that contribute to the sharpness of the signal's edges are attenuated.]

… or **Ringing** occurs when the system **over-shoots** the ideal response (the ideal rise and fall of the signal) before settling to its final value.

Signal Distortion

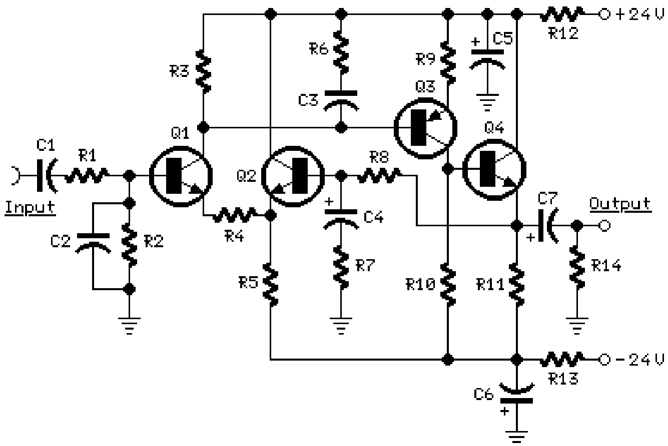


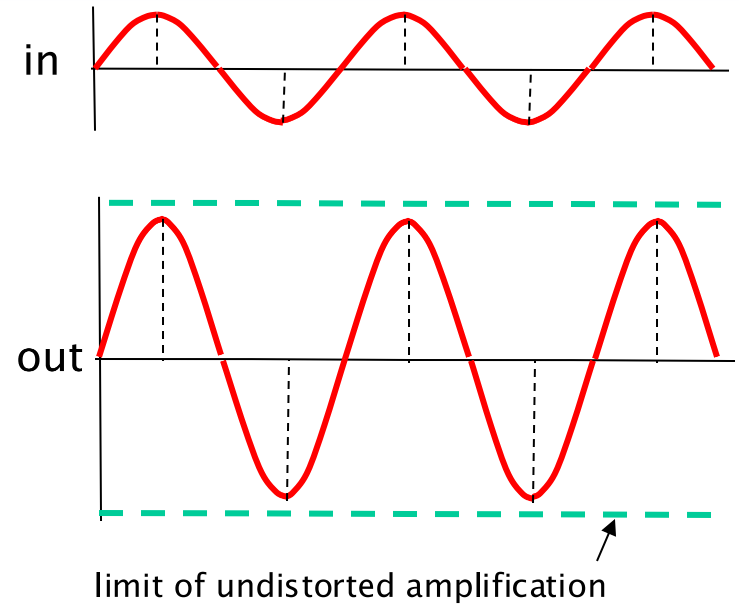


<u>Successive forms of compression, particularly clipping as a form of extreme compression:</u>

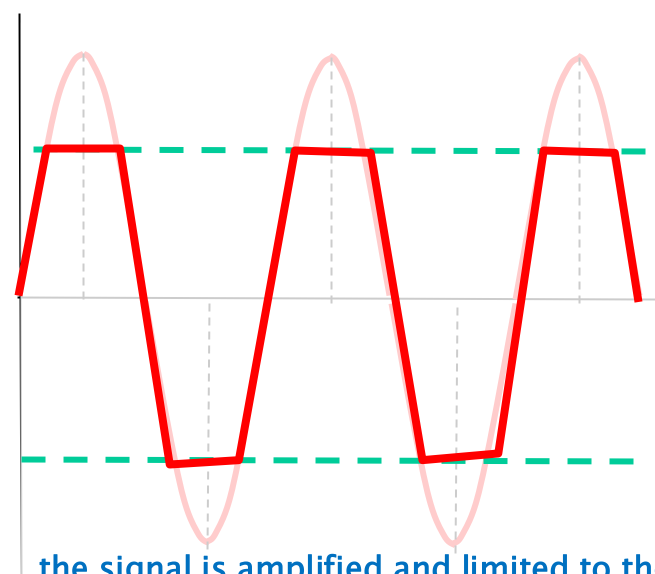
- clipping the waveform: when the amplitude of a signal exceeds the max limit of a system, causing the waveform to be "cut off" at the top and/or bottom.
- causes high freq harmonics
- (ref Fourier – many high freq harmonics required to turn a sine wave to square wave)
- According to Fourier analysis, any periodic waveform can be decomposed into a series of **sine waves** at various frequencies.

Signal Distortion





- when an amplifier gain is set too high
    - by mis-calculation or design for a specific effect
- the amplifier is trying to amplify the signal beyond the range of the supply voltage
- (simplified case):
    - e.g. an amplifier gain = 100 with a 0.5V input, but the power supply is 40V

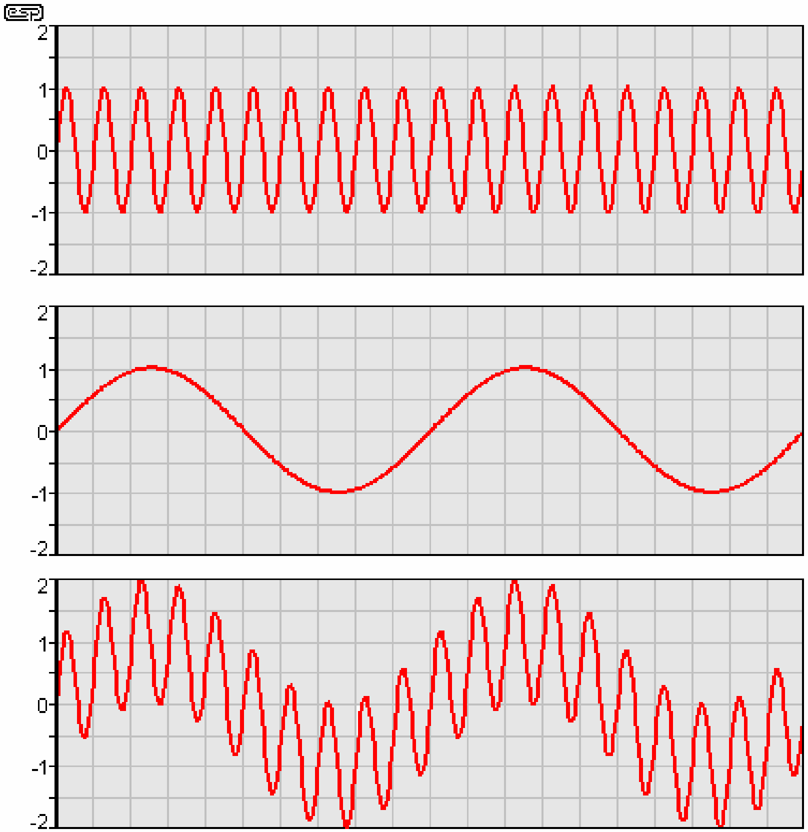


the signal is amplified and limited to the the power supply values

Intermodulation Distortion (IM)

Occurrence: two different frequencies are being amplified simultaneously

- one of frequency much lower than the other
- high frequency signal modulated by the low frequency
- different from the signals simply adding (as they are supposed to)



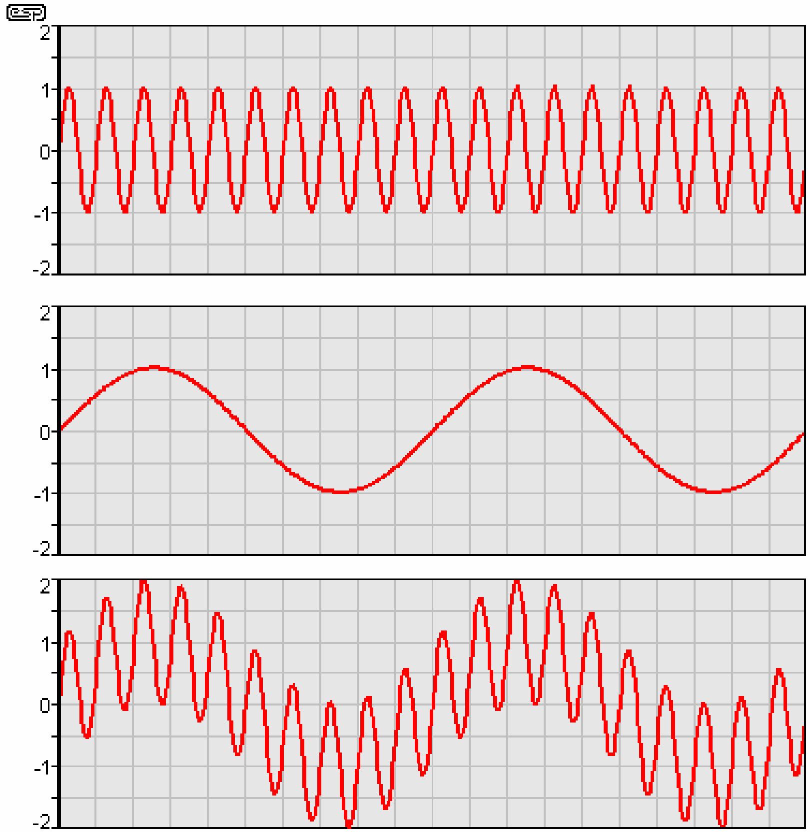
from ESP – http://sound.westhost.com

- effect (musically speaking) is that the sound is mixed
- Individual instruments become difficult to separate

Intermodulation Distortion (IM)

Signal includes components that are

- the sum of the two frequencies and
- the difference between the two frequencies



from ESP – http://sound.westhost.com

- usually a fraction of a percent of the original signal
- over 1% IM is noticeable
- over 3% is annoying

Harmonic Distortion

The production of harmonic frequencies by an electronic system when A signal is applied at the input

- when an input signal goes through nonlinear electronic circuitry, the output signal will include some harmonic distortion (or unwanted frequencies)
- the frequency of each unwanted frequency is a multiple of the centre frequency of the input signal.
  - If the input is at frequency ***f***, then the output might include: 2f, 3f, 4f, 5f, … (i.e., harmonic distortion)

Odd And Even-order Distortion

- if the frequencies of the distortion are 2, 4, 6, etc., times the centre frequency, it's even-order harmonics
- if the frequencies of the distortion are 3, 5, 7, etc., times the centre frequency, it's odd-order harmonics

Total Harmonic Distortion

The ratio of the powers of:

the fundamental (required) signal	:	sum of the power of all the harmonic components

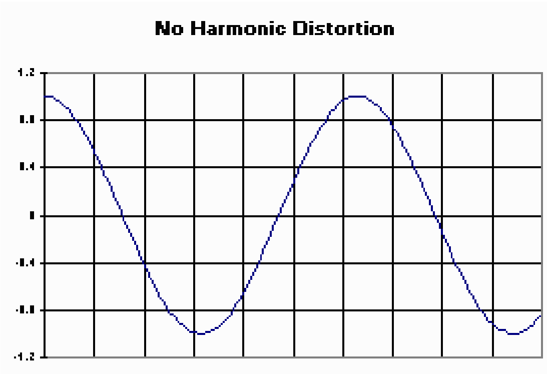
Usually expressed as a %

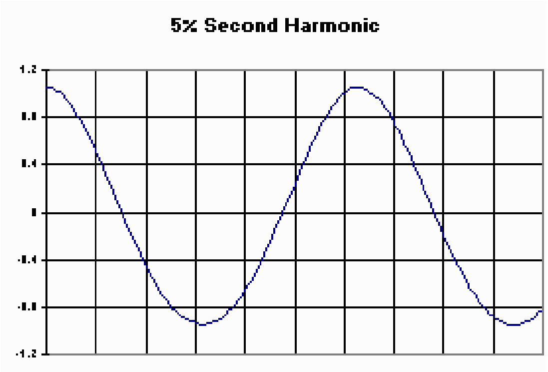
i.e. $\frac{\Sigma \text{ power of all harmonics}}{\text{power of fundamental freq}} = \text{THD\%}$

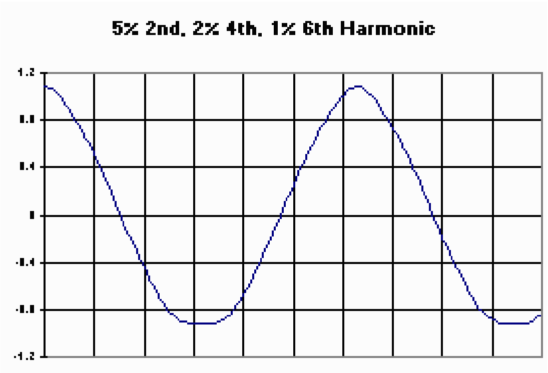
Total Harmonic Distortion

compare these even harmonic distortions

- Higher freq harmonics make the edges of the signal sharper.





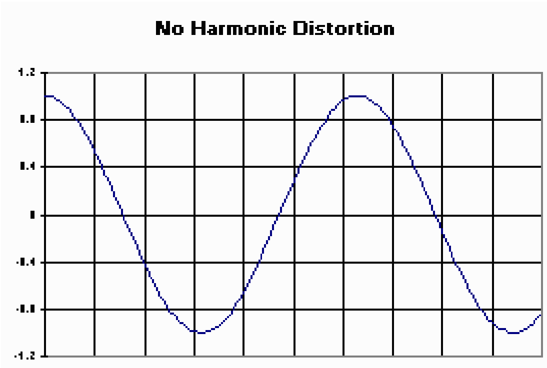


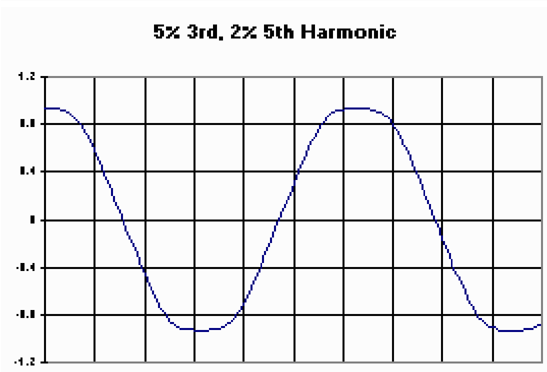
http://members.aol.com/sbench102/thd.html

Total Harmonic Distortion

•**Frequency and Shape**: Lower harmonics (e.g., 2nd) have longer wavelengths and contribute to broader, smoother changes in the waveform.

•Higher harmonics (e.g., 5th, 6th) have shorter wavelengths and introduce finer, sharper details, such as rapid oscillations or jagged edges.

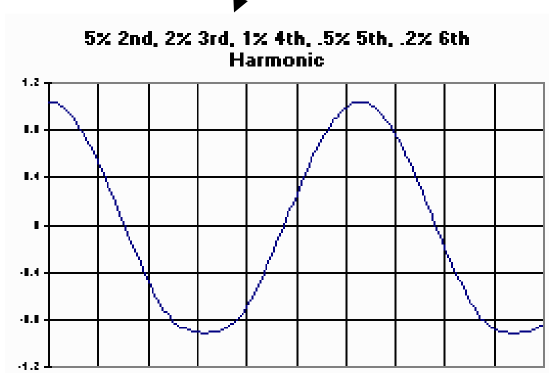


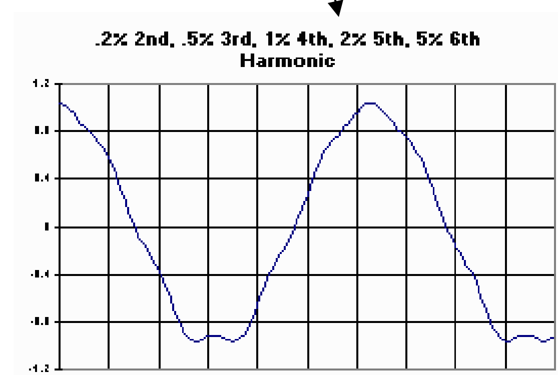


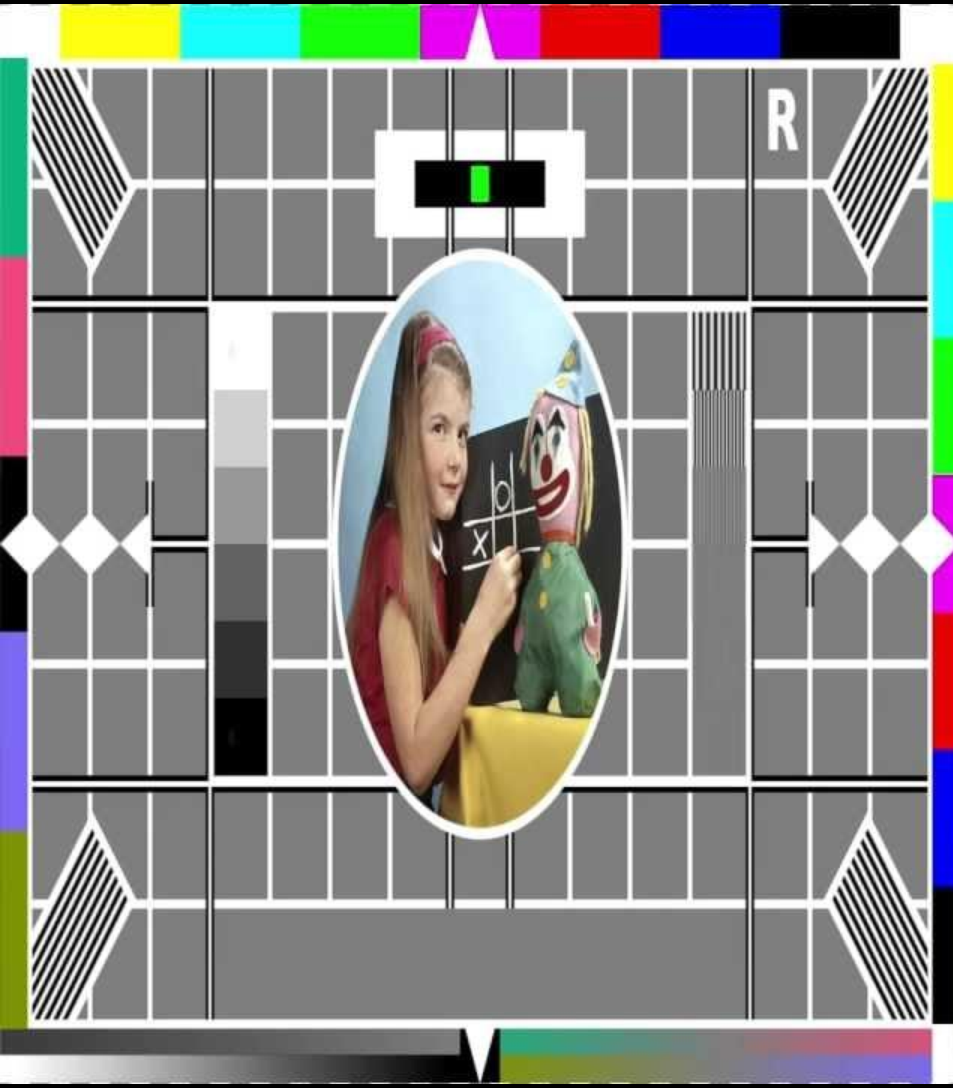
Odd harmonic distortions

compare the lower two:
this has higher THD in the 2nd and 5th harmonics

… this has higher THD in the 3rd and 6th harmonics



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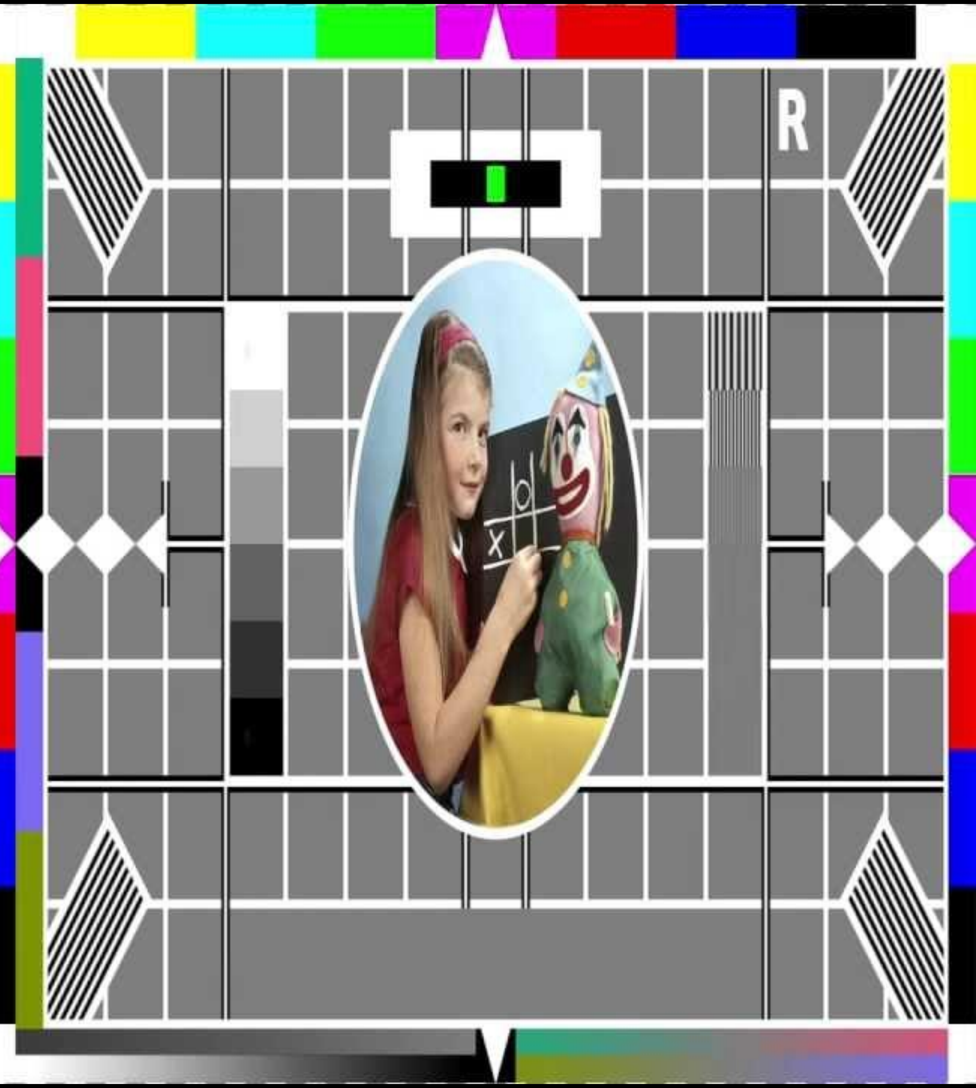
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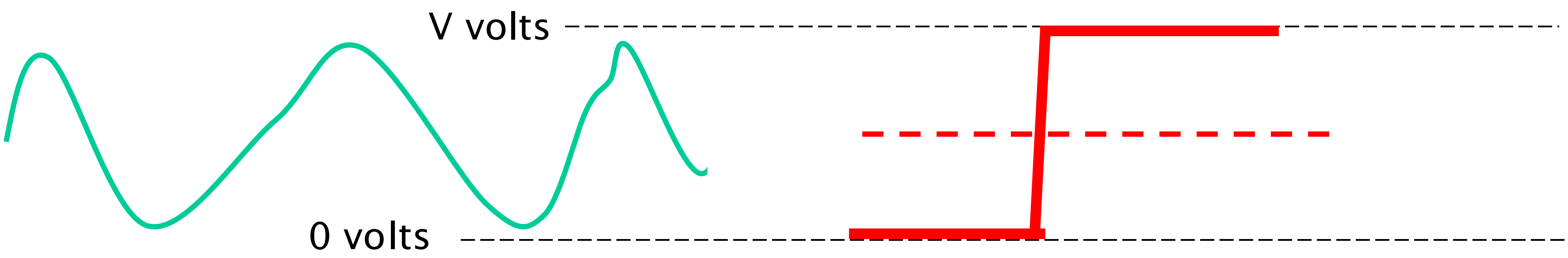
From Analogue to Digital

Analogue vs Digital Signals

Range of signal voltage:



Analogue

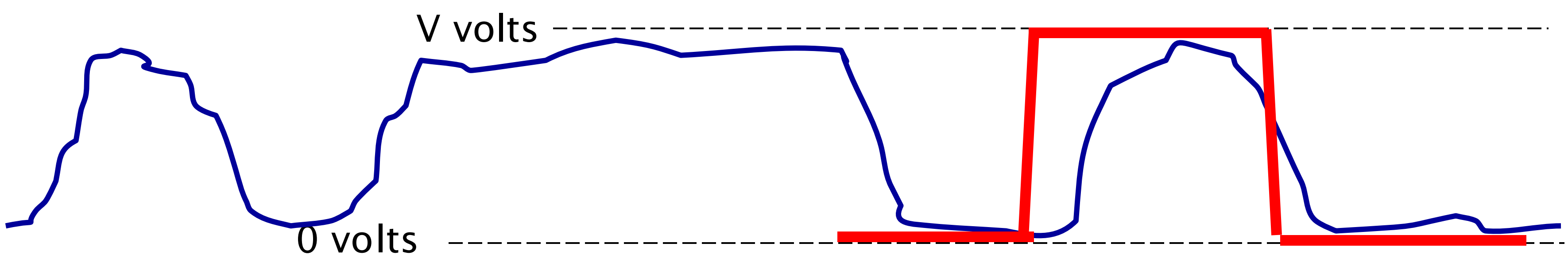
- offers an infinite range from 0 to V volts
- at any point the signal voltage should be *exactly* a specific voltage
- but there are many points in the chain where this *specific* voltage may be corrupted and subject to distortion (as previous slide)

Digital

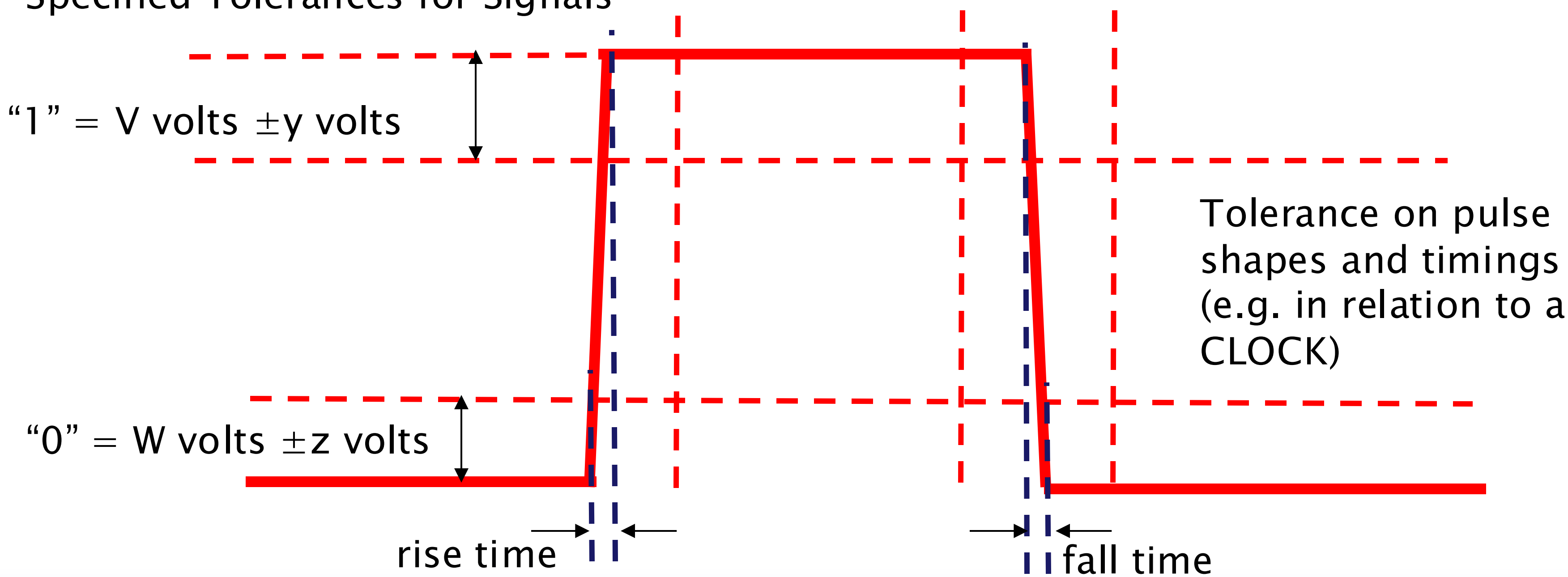
- offers a choice of only two values
  - Logical "0" = 0 volts
  - Logical "1" = V volts
- (simplisticly...) a decision point between 0 and V volts
  - above this = "1"
  - below this = "0"
- we can be (almost) certain that the signal is in either one state or the other
- allows the signal to suffer much distotion before it is mis-read

Distortion Affecting Digital Signals

Data: “1”........ “0”......... “1”.......... “1”.......... “1” .... “0” .... “1” .... “0” ....

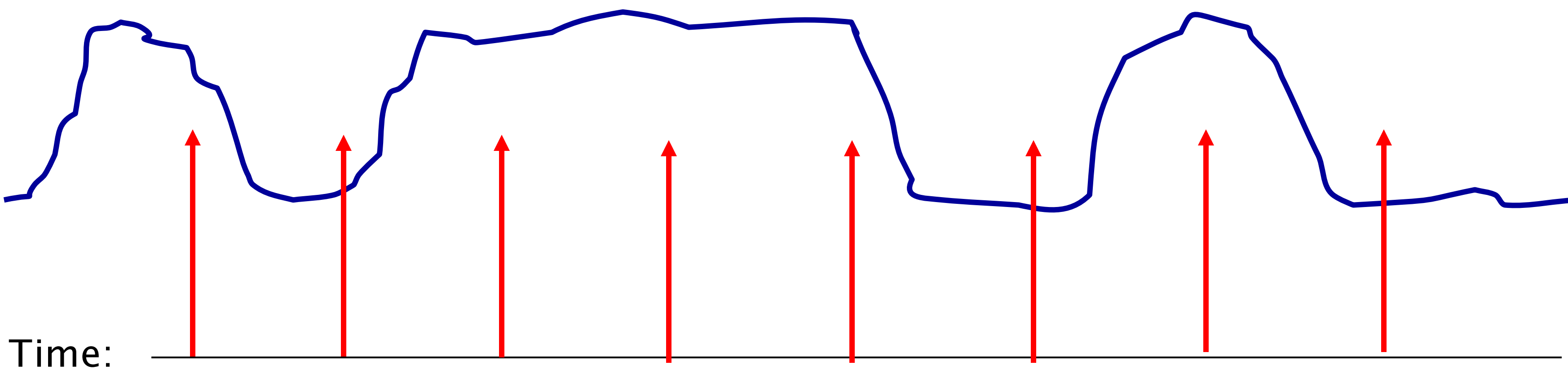


Specified Tolerances for Signals



Digital Signal Timing – Clocking

Data: “1”........ “0”......... “1”........ “1”......... “1” .... “0” .... “1” .... “0” ....



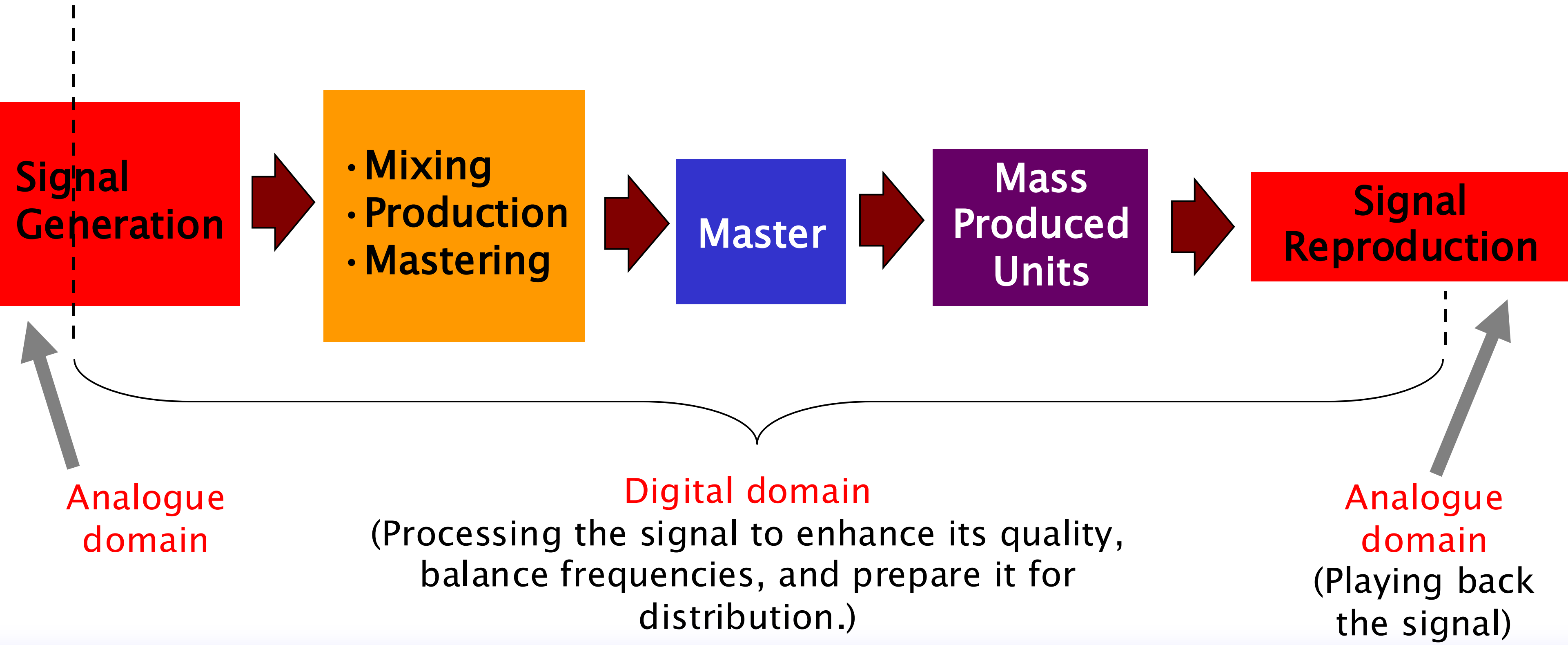
- Transmitting and receiving ends must agree on times at which data is read – the CLOCK frequencies
- Clocking information must also be inferred in recorded media (CDs, etc)

Go Digital!

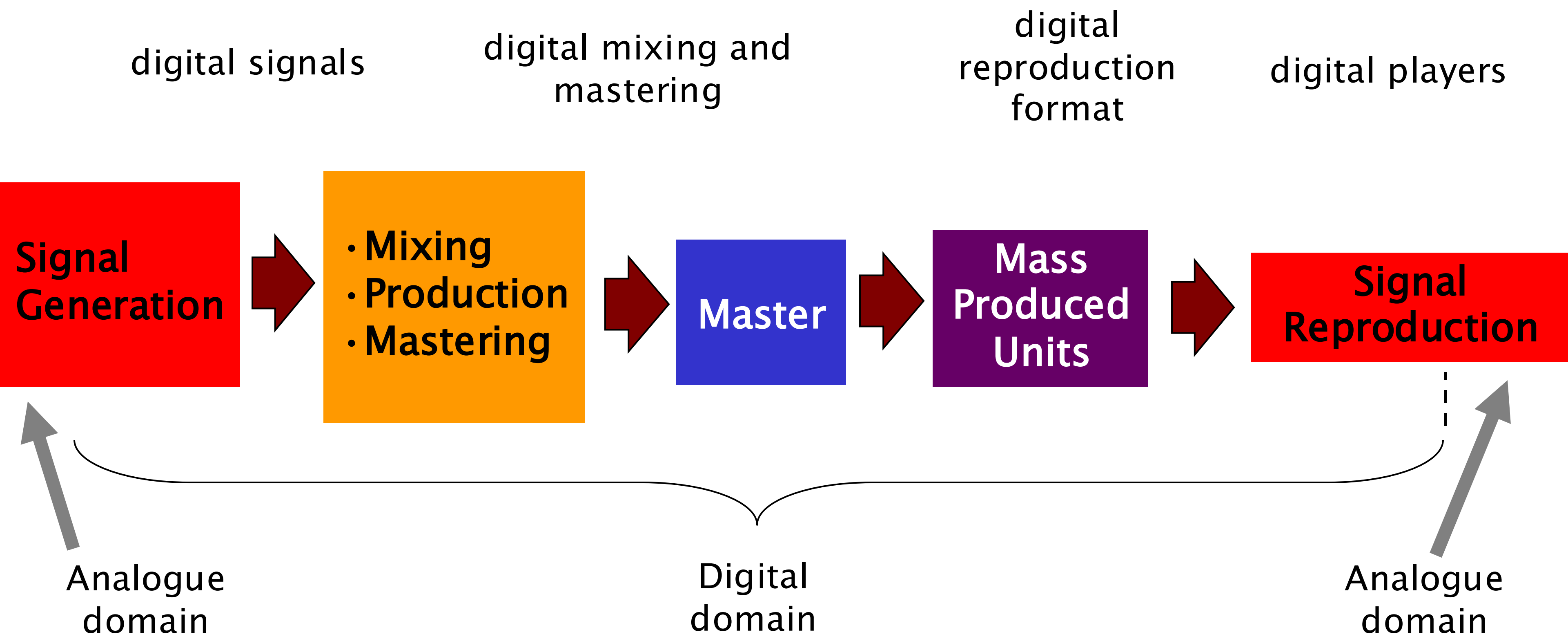
Digital Signals are either “1” or “0”

- Can add codings e.g. for error detection / correction
- if 1’s and 0’s can represent the analogue voltage, the produced / reproduced signal can be much more robust and faithful

Aim: to keep the process digital as much as possible



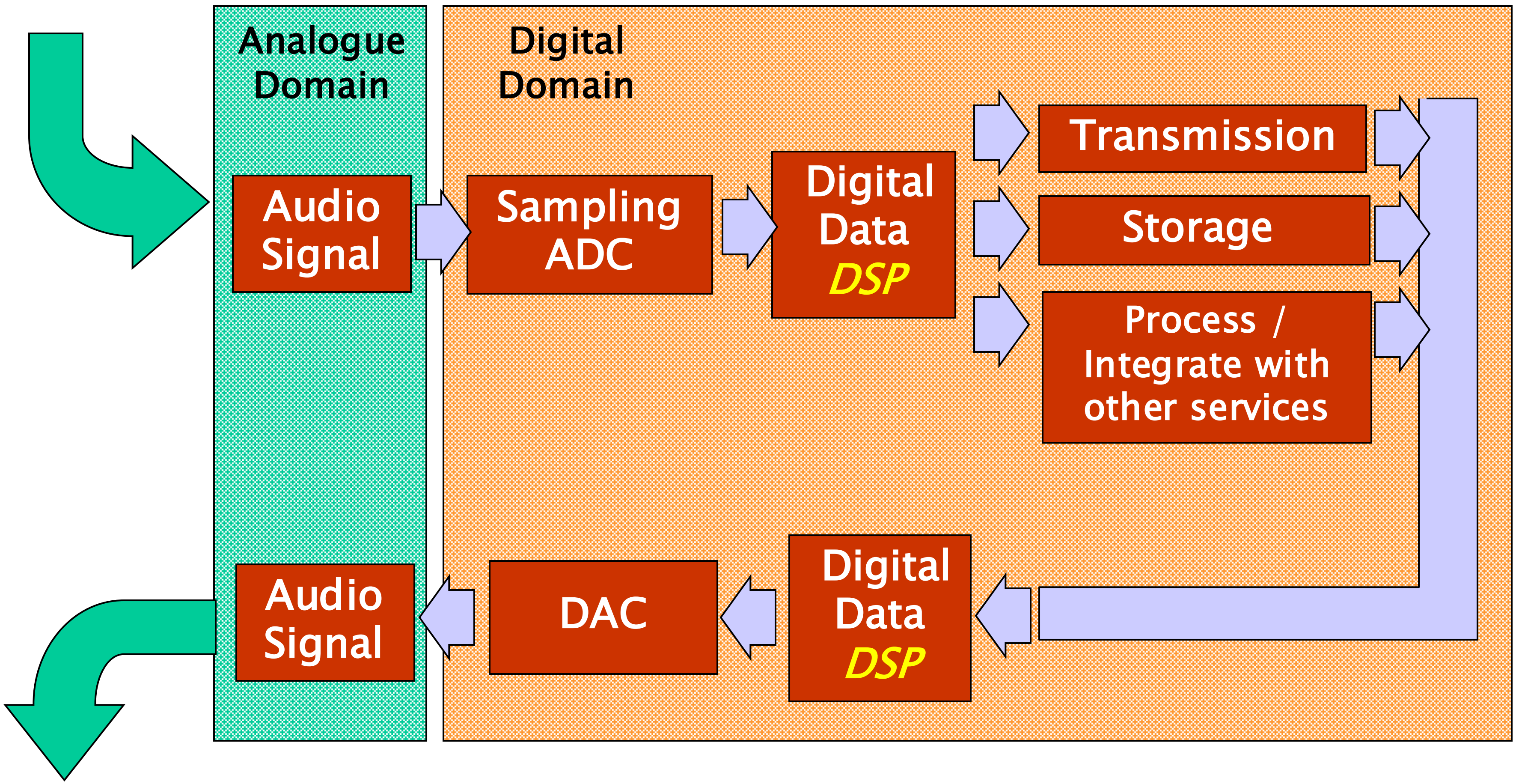
The Digital Chain of Production / Reproduction



ADC, DAC, Analogue and Digital Domains

ADC: Converts analogue signals into digital form so they can be processed, stored, or transmitted by digital systems.

Analogue signal in



Analogue signal out

DSP = Digital Signal Processing